GE Healthcare

MR Safety Guide

Operator Manual

MR Safety Guide
Operator Manual, English
2381696-100
Revision 10

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Medical Device Directive

These products conform with the requirements of council directive 93/42/EEC concerning medical devices, when they bear the following CE Mark of Conformity:

European Representative:

GE Medical Systems S.C.S
Quality Assurance Manager
283 rue de la Minière
78530 BUC France
Telephone: +33 1 30 70 40 40

This equipment generates, uses, and can radiate radio frequency energy. The equipment may cause radio frequency interference with other medical and non-medical devices and radio communications. To provide reasonable protection against such interference, the:

GE MR Systems

comply with emissions limits for (Group 2, Class A) Medical Devices as stated in EN 60601-1-2. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment is found to cause interference (which may be determined by turning the equipment on and off), the user (or qualified service personnel) should attempt to correct the problem by one or more of the following measures:

- reorient or relocate the affected devices;
- increase the separation between the equipment and the affected device;
- power the equipment from a source different from that of the affected device; and/or
- consult the point of purchase or service representative for further suggestions.

The manufacturer is not responsible for any interference caused by using interconnect cables that are not recommended or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the user's authority to operate the equipment.

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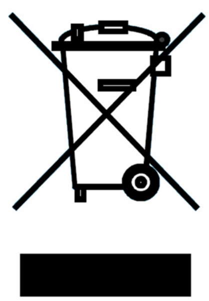
Do not use devices that transmit RF Signals (**cellular phones**, transceivers, or radio controlled products) in the vicinity of this equipment as they may cause performance outside the published specifications. Keep the power to these types of devices turned off when near this equipment.

The medical staff in charge of this equipment is required to instruct technicians, patients, and other people who may be around this equipment to fully comply with the above requirement.

Immunity/Emissions Exceptions: Note the exceptions from the EMC test results. Check with the business EMC engineer for this information.

In accordance with the international safety standard IEC 60601-1, this system is:

- a Class I device
- acceptable for Continuous Operation
- having ordinary protection against ingress of water (IPX0)
- type B and BF applied parts
- is not for use in the presence of flammable anesthetics.



CAUTION: This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.

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Chapter 1

About This Guide

Introduction

This chapter explains the purpose and design of this Learning and Reference Guide. It is an introduction to the guide, providing information on the purpose, prerequisite skills, guide organization, chapter format, and graphic conventions that identify the visual symbols used throughout the guide. This manual applies to all MR products except Profile and Ovation, which have separate safety guides.

Purpose of This Guide

This guide is written for health care professionals (namely, the MR technologist) to provide the necessary information relating to the proper operation of this system. The guide is intended to teach you the system components and features necessary to use it to its maximum potential. It is not intended to teach magnetic resonance imaging or to make any type of clinical diagnosis.

The MR Safety Guide contains information applicable to several MR system configurations. A topic heading, a note, or other wording indicates information that is applicable to a specific system configuration.

Prerequisite Skills

This guide is not intended to teach magnetic resonance imaging. It is necessary for you to have sufficient knowledge to competently perform the various diagnostic imaging procedures within your modality. This knowledge is gained through a variety of educational methods including clinical working experience, hospital based programs, and as part of many college and university Radiologic Technology programs.

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Chapter Format

Each chapter contains a consistent format. This consistency provides uniformity for content delivery and a better learning environment for you. Listed below are the components for each chapter.

Introduction

The **Introduction** provides a short introduction to the chapter and a list of tasks to be presented.

What Do I Need to Know About…

The **What Do I Need to Know About…** section lists and explains concepts necessary to perform the tasks within the chapter.

How Do I…

The **How Do I…** section provides the detailed steps necessary to complete a given task. These detailed steps not only provide the steps to complete a task, but also provide additional information, as needed, related to a step.

Each task also includes a **Quick Steps** table. This Quick Steps table is intended to be used as a quick reference by the experienced technologist and provides only the steps necessary to complete a task. Be sure to read the detailed steps before using this table.

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Mouse Controls

The mouse (Figure 1-1) is a hand-operated device that you maneuver across the surface of a pad. As you do, the on-screen cursor mimics the movement of the mouse, allowing you to move among windows and menus. For instance, moving the mouse to the right causes the on-screen cursor to move to the right. The mouse is used to make selections by clicking the left, right, and middle buttons. Table 1-1 describes the terminology used for the various mouse functions.

Figure 1-1 The Mouse

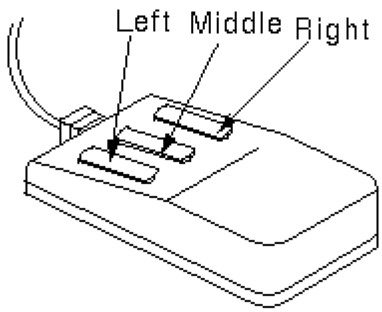

Table 1-1 Conventions for Mouse Actions

Mouse Action	Description
Click	Clicking the left mouse button to select a button or icon. The button can be pressed in, not pressed in, or popped in/out.
Right-click	Clicking the right mouse button.
Middle-click	Clicking the middle mouse button.
Click and drag	Clicking and holding the left mouse button down while dragging the cursor to the desired location.
Right-click and drag	Clicking and holding the right mouse button down while dragging the cursor to the desired location.
Middle-click and drag	Clicking and holding the middle mouse button down while dragging the cursor to the desired location.
Double-click	Clicking the left mouse button twice in rapid succession.
Triple-click	Clicking the left mouse button three times in rapid succession.

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Graphic Conventions and Legends

This guide uses special conventions for graphics and legends to make it easier for you to work with the information. Table 1-2 describes the conventions used when working with menus, buttons, text boxes and keyboard keys.

Table 1-2 Conventions for Menus, Buttons, Text Boxes, and Keyboard Keys

Example	Describes
Select	Selecting an option in a check box or radial button and selecting a tab.
Press **Enter**	Pressing a hard key on the keyboard.
Press and hold **Shift**	Pressing and holding down a hard key on the keyboard.
Press **Ctrl + Enter**	Pressing multiple hard keys on the keyboard simultaneously.
Click **[Viewer]**	A button label or Interface button name.
Click  (Exam prior)	Selecting an icon-based button.
In the **Matrix** text box,...	The name of text box in which you can select or type text.
Type ***supine*** in the Patient Position text box (different font and bold)	Text you enter into a text box.
Select **Sort** > **Sort by date**	The pathway of selecting option(s) in a pull-down menu.

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Safety Notices

The following safety notices are used to emphasize certain safety instructions. This guide uses the international symbol along with the danger, warning, or caution message. This section also describes the purpose of an Important notice and a Note.

DANGER: **Danger is used to identify conditions or actions for which a specific hazard is known to exist which <u>will</u> cause severe personal injury, death, or substantial property damage if the instructions are ignored.**

WARNING: **Warning is used to identify conditions or actions for which a specific hazard is known to exist which <u>may</u> cause severe personal injury, death, or substantial property damage if the instructions are ignored.**

CAUTION: **Caution is used to identify conditions or actions for which a potential hazard may exist which will or can cause minor personal injury or property damage if the instructions are ignored.**

CAUTION: **Coil Caution is used to identify conditions or actions for which a potential hazard of crossing or looping coil cables may exist which will or can cause minor personal injury or property damage if the instructions are ignored.**

IMPORTANT!: Important indicates information where adherence to procedures is crucial or where your comprehension is necessary to apply a concept or effectively use the product.

NOTE: A Note provides additional information that is helpful to you. It may emphasize certain information regarding special tools or techniques, items to check before proceeding, or factors to consider about a concept or task.

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Chapter 2

Working Safely

Introduction

The MR Safety Guide contains information applicable to several MR system configurations. A topic heading, a note, or other wording indicates information that is applicable to a specific system configuration.

This chapter focuses on the visible and invisible sources of hazard and concern in the magnetic resonance (MR) imaging environment and emphasizes the need to work safely. To ensure safe operation of your scanner, you must understand several components of your imaging system. This chapter provides brief guidelines for working in a magnetic field, key concepts regarding the patient alert system, as well as magnet, quench, radio frequency (RF), laser light, metal sliver, acoustic, peripheral nerve stimulation (PNS), and equipment hazards. It contains the step-by-step instructions to help you learn how to:

- Eliminate Magnet Hazards
- Respond to Emergencies
- Check the Cryogen Levels
- Handle Contact with Liquid Cryogens

IMPORTANT!: This chapter contains important safety information that you and the physician must understand thoroughly before using the system.

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What Do I Need to Know About...

This section presents the concepts necessary to successfully complete the working safely process. Specifically, you need to understand:

- Safety Information
- Magnetic Field Basics
- Static Magnetic Field
- Gradient Magnetic Fields
- Electromagnetic Fields
- Clinical Hazards
- Equipment Hazards
- Clinical Screening
- Patient Emergencies
- System Maintenance
- Cables and Equipment Connections
- Safety Review

Safety Information

The Magnetic Resonance Imaging (MRI) system uses a magnet, which can have a field strength several thousand times greater than that of the earth's magnetic field. The magnetic field surrounding the magnet may present a hazard to personnel and equipment within the immediate area. Therefore, the magnetic field safety information described in this chapter is very important. You and your physician must understand it thoroughly before you begin to use the system. You can find additional safety information throughout your Operator Manual and Learning and Reference Guide CD-ROM. If you need additional training, seek assistance from qualified General Electric (GE) Healthcare personnel.

Make sure your training guides are readily available at all times. Review the procedures and safety precautions periodically. Through Magnetic Resonance (MR) safety education, careful planning, and diligent upkeep of your MR facility, a safe environment can be provided for both patients and personnel.

For any hazardous incident or system malfunction related to the use of the GE Healthcare MR Scanner please use the following contact methods:

If Serviced by GE Healhtcare: Please contact your Field Service Engineer to report out on the incident.

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If third party serviced: Please conatct your third party Field Service Engineer and have them send a manufacturers notice to:

Regulatory Affairs Department
GE Healthcare
3114 N Grandview Blvd W-500
Waukesha, WI 53188

If the user is self servicing the GE Healthcare MR Scanner please provide the following information:

- System type
- System ID
- Date of incident
- Description of incident
- Contact Information (facility, address, contact name, title, and telephone numbers)

Locate the contact number on your scanner or visit GE Healthcare on the web http://www.gehealthcare.com/contact/contact_details.html and locate the appropriate telephone number for your location. In the US please us: 1-800-437-1171.

User Training

GE Healthcare provides a purchasable training program in Milwaukee for new system operators. You may arrange to participate through your local sales representative. In addition, GE Healthcare provides purchasable on-site training by an MR Applications Specialist. GE Healthcare advises that anyone who operates the system should attend this session after reading the Operator Manual, Learning and Reference Guide, and related training materials.

GE Healthcare strongly recommends that physicians who prescribe studies and review images on the MR system, attend at least two full days of professional meetings dealing with MR imaging each year. Such meetings include the Radiological Society of North America (RSNA), the Society for Magnetic Resonance in Medicine (SMRM) and the American Roentgen Ray Society (ARRS). In addition, MR system user groups present symposia and workshops throughout the year that provide additional learning opportunities.

The healthcare facility is responsible for training outside emergency personnel (e.g., fire department and other outside emergency personnel) not to bring any ferrous fire-fighting equipment, including axes, ferrous stretchers, or oxygen tanks into the magnet room. Be sure to show such outside emergency personnel where the Emergency Magnet Rundown switch is located.

Product Identification Labels

Product identification labels (ratings) can be found on the tops and sides of the cabinets, the rear of monitors, and other exterior surfaces on the equipment. Such product labels alert you to specific hazards and the level of hazard importance. The labels may also contain

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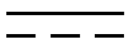
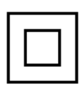
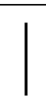
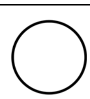
messages that communicate the specific hazard, the probable consequence of involvement with the hazard, and how the hazard can be avoided. In the event you are unable to identify these labels, contact your service personnel.

One or more of the product identification labels in Table 2-1 may be on your system or peripheral equipment. Please familiarize yourself with the labels that apply to your particular system.

Table 2-1 Product Identification Labels

Label	Description (typical use)
	Alternating current (rating plate, terminals)
	Direct current (rating plate, terminals)
	Earth (ground terminals)
	Protective earth (ground terminals)
	Equipotentiality (terminals)
	Class II Equipment (double insulated) (ratings)
	Type B Applied Part (ratings, AP connections)
	Type BF Applied Part
	Non-ionizing electromagnetic radiation (ratings)
	Main Power On (main disconnect/power switch)
	Main Power Off (main disconnect/power switch)

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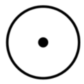
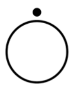
Label	Description (typical use) (Continued)
	Power On (only for a part of equipment)
	Power Off (only for a part of equipment)
	Dangerous voltage (components, points of entry)
	CAUTION – Dangerous voltage (barriers, points of entry)
	Emergency Stop
	Attention – Consult accompanying documents
	CAUTION
	WARNING
	DANGER
	CAUTION – Static Sensitive (Electrostatic discharge (ESD) susceptible parts)
	Laser Radiation (laser devices)
	Hazard: Hot Surface
	Hazard: Magnetic Field

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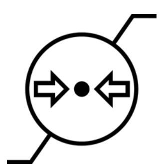
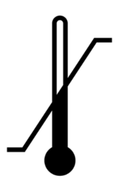
Label	Description (typical use) (Continued)
	Mandatory: Instruction Manual
	Prohibited: Do Not Touch - Hazardous Voltage
	Prohibited: No Metallic Articles
	Prohibited: No Metallic Implants
	Prohibited: No Pacemakers
	Mandatory: Maintenance Instructions
	Atmospheric Pressure Limitation
	Temperature Limit

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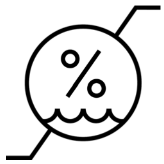
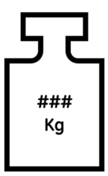
Label	Description (typical use) (Continued)
	Humidity Limitation
	Respiratory Bellows Port. Either label may be on your system.
	ECG Leads Port. Either label may be on your system.
	Peripheral Gating Port. Either label may be on your system.
	Patient Alert Port. Either label may be on your system.
	Table mass in Kg, without the patient

Indications for Use

GE MRI whole-body scanners are designed to support high resolution, high signal-to-noise ratio images, and short exam times. The scanner is indicated for use as a diagnostic imaging device to produce images in the axial, sagittal, coronal and oblique planes, spectroscopic images, parametric maps, spectra, and dynamic images of the structures and organs of the entire body, including, but not limited to, head, neck, TMJ, spine, breast, heart, abdomen, pelvis, prostate, blood vessels, and musculoskeletal regions of the body. The

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images produced by the scanner reflect the spatial distribution or molecular environment of nuclei exhibiting magnetic resonance. These images, when interpreted by a trained physician, yield information that may assist in diagnosis.

CAUTION: **These devices are limited by federal law to investigational use for indications not in the "Indications for Use" statement for a specific system type. Under federal law, these devices should only be used for the functions set forth in the "Indications for Use" statements.**

General system specifications can be found in the product data sheets. Contact your GE Healthcare Sales Representative for this information.

Restrictions on Use

CAUTION: **Federal law restricts the sale, distribution, and use of this device to or on the order of a physician.**

CAUTION: **Do not load non-system software onto the system computer.**

WARNING: **The MR system is not designed to provide information for clinical stereotactic use. The spatial accuracy obtainable with your MR system may not be adequate for stereotactic procedures and can vary depending on the patient, the pulse sequence used, and the system itself. It is therefore recommend that MR images not be used for stereotactic applications.**

WARNING: **Electrically conductive stereotactic devices may lead to high localized SAR. Excessive transmit power may result from interactions between the structure and the transmit coil. In addition, improper padding between the patient and any conductor may lead to excessive localized heating.**

NOTE: Clinical stereotactic use refers to being used in localization for surgical procedures.

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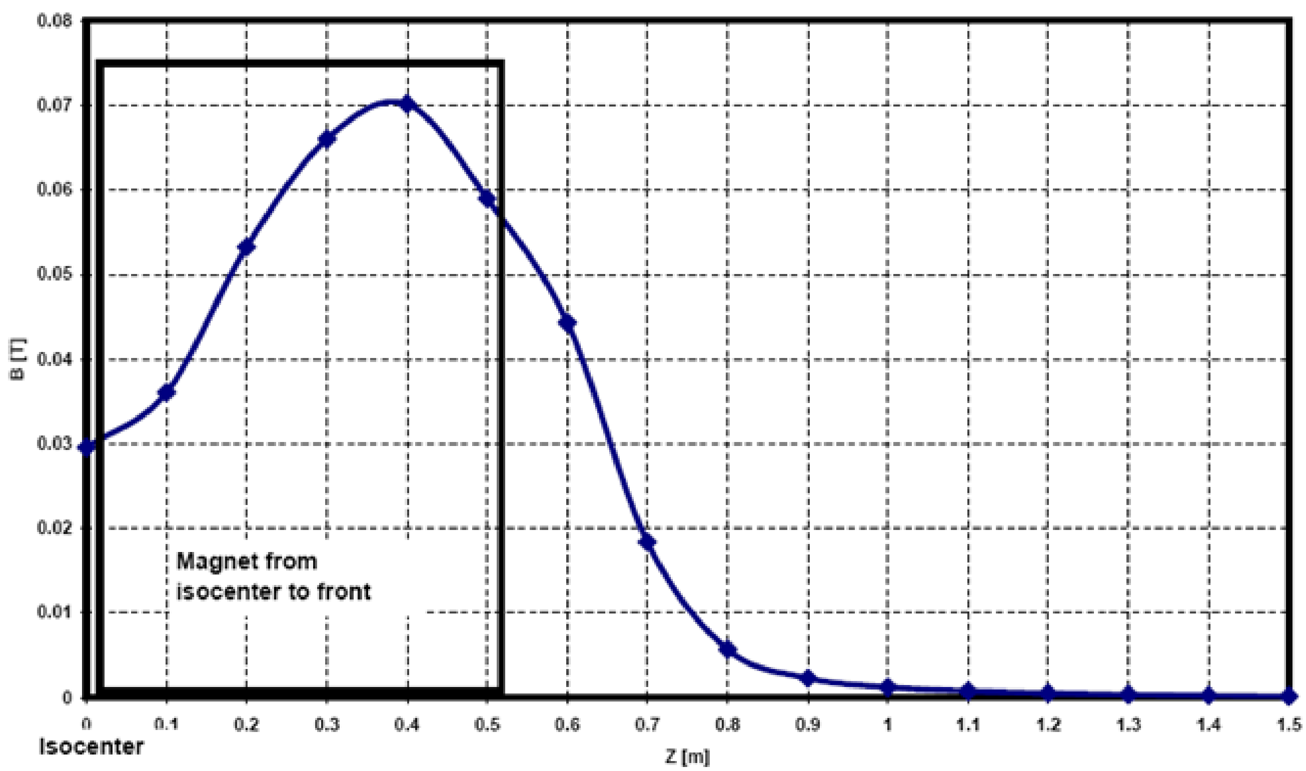
Instructions for Use

The 2nd amendment to IEC 60601-2-33 assumes that because no chronic effects from exposure to MR fields are known, worker safety limits are the same as for patients. However, it is prudent to minimize worker exposures.

Workers must prevent ferromagnetic materials from entering the magnet room. Ferrous projectile hazards are a major safety concern. Note that some materials that are initially non-magnetic may become magnetic when subjected to a static magnetic field over a period of time. Motion in static magnetic fields (especially near large spatial field gradients) may induce metallic tastes, vertigo, nausea, and possibly flashes of light (magneto-phosphenes). These motion effects are considered to be non-hazardous, provided they do not cause the worker to fall.

Time-varying gradient magnetic fields may induce peripheral nerve stimulation if the worker intercepts sufficient time-varying flux. Peripheral nerve stimulation is non-hazardous unless it causes the worker to injure himself when startled by the effect. Field plots of the maximum time-varying gradient |B| workers could experience outside the magnet bore is shown in Figure 2-1

Figure 2-1 Maximum Magnitude Gradient Magnetic Field from three Simultaneous Axes at the Patient Bore Radius (worker exposure is limited to these levels as a function of z).



Radio frequency fields at sufficiently high levels may cause heating. Outside the magnet bore the radio frequency fields rapidly decay. Let B1 be the magnetic field strength of the radio frequency magnetic field. A plot of the square B1normalized to its value at isocenter is

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shown in Figure 2-2. At most B1 at isocenter may produce the whole-body Specific Absorption Rate (SAR) limit. If as much of the body were exposed outside the bore then Figure 2-2 shows the scale factor for each Z location. This is a very conservative estimate of SAR since the total flux into the body is likely to be much smaller.

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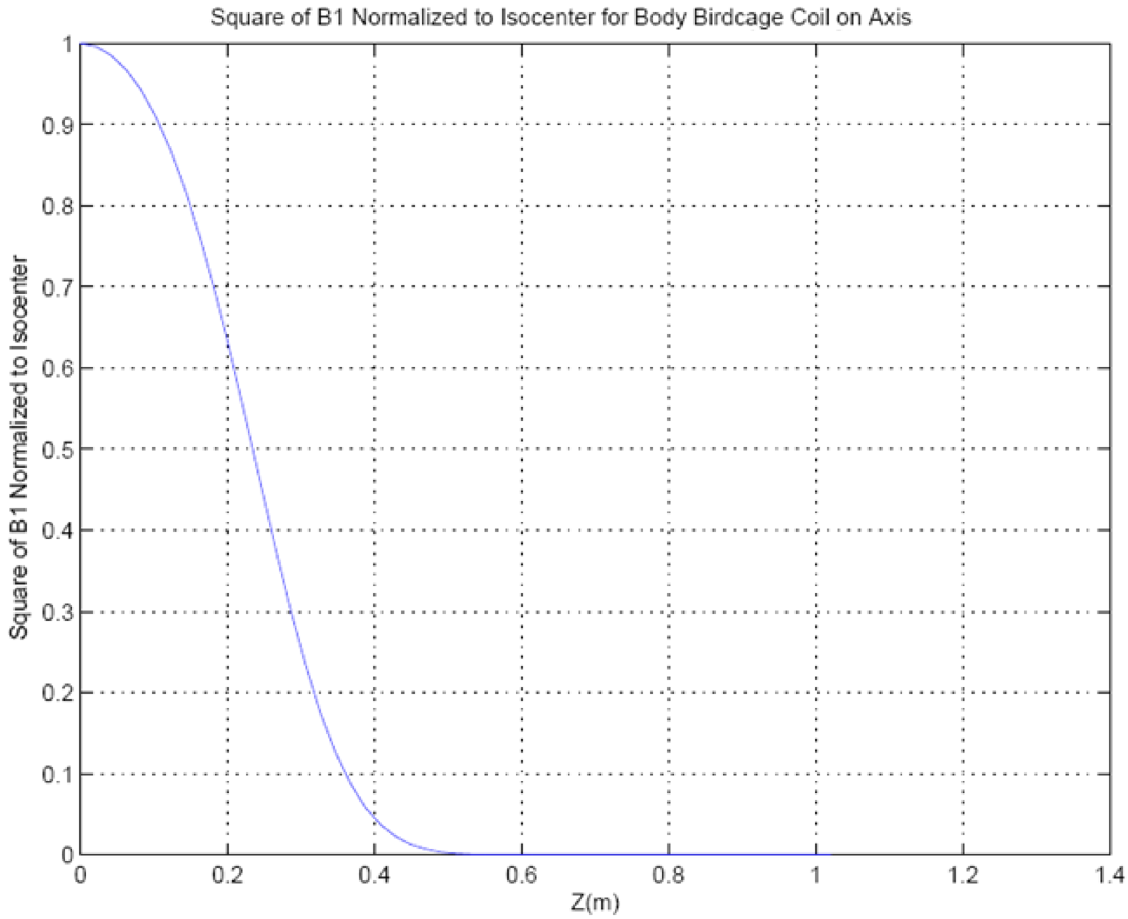
Figure 2-2 Plot of the Square of B1 Normalized to Isocenter for the Body Birdcage Coil on Axis.



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Contraindications for Use

Your MR system has a very strong magnetic field that may be hazardous to persons entering the environment or the system room if they have certain medical conditions or implanted devices. The use of your MR system is contraindicated (i.e., not advised) for patient and MR workers with any of the following:

- Electrically, magnetically, or mechanically activated implants (e.g., cardiac pacemakers and ferrous/electrically activated cardiac catheters) because the magnetic and electromagnetic fields produced by your MR system may interfere with the operation of these devices.
- Intracranial aneurysm clips.
- Non-compatible MR steriotactic frames.

WARNING: **The magnetic field of the MR system can cause a ferrous implant (e.g., surgical clip, cochlear implant, intracranial aneurysm clip etc.) or prosthesis to move or be displaced, resulting in serious injury. Patients and MR workers should be screened for implants and those individuals with implants are not to be scanned or allowed to enter the magnet room. Prostheses should be removed before scanning to help prevent injury.**

WARNING: **Induced electrical currents and heating can occur in the region of the metallic implants. Patients or workers with implants are not to be scanned or allowed into the magnet room.**

MR Safety Standards

In most countries the MR safety standard IEC 60601-2-33 provides safety limits for MR exams, for ventilation, and for occupational exposure of MR workers. The International Electrotechnical Commission (IEC) developed a widely-used MR safety standard. The IEC MR safety standard is three-tiered. The NORMAL OPERATING MODE is for routine scanning of patients. The operator must take a deliberate action (usually an ACCEPT button) to enter the FIRST CONTROLLED OPERATING MODE. This mode provides higher scanner performance, but requires monitoring of the patient. Finally, there is a SECOND CONTROLLED OPERATING MODE used only for research purposes under limits controlled by an Investigational Review Board (IRB).

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Table 2-2 IEC Safety limits

Operating mode	Whole body SAR (W/Kg)	Head SAR (W/Kg)	Partial body SAR (W/Kg)	Local head/ trunk SAR (W/Kg)	Local extremity SAR (W/Kg)	Short term SAR (W/Kg)	dB/dt (% mean PNS)
IEC Normal Mode	2	3.2	$= \left(10 - \frac{8M_{exposed}}{M_{patient}}\right)$	10	20	3 x long term	80% PNS
IEC 1st Controlled Operating Mode	4	3.2	$= \left(10 - \frac{6M_{exposed}}{M_{patient}}\right)$	10	20	3 x long term	100% PNS
IEX 2nd Controlled Operating Mode	IRB limit	IRB limit	IRB limit	IRB limit	IRB limit	IRB limit	IRB limit

Local SAR is averaged over the worst-case 10 g. Short term SAR is averaged over 10 s. SAR limits are reduced if temperature can exceeds 24 degrees C or if humidity exceeds 60%. Hearing protection (only earplugs have been validated) with NRR >/= 29 dB to reduce the A-weighted root-mean-squared sound pressure level below 99 dB(A) shall be used. IEC 60601-1 limits surface contact temperatures to 41 degrees C.

Magnetic Field Basics

Though it is generally accepted that no published evidence exists supporting cumulative or long-term negative effects of EMF exposure, it is advisable for pregnant MR workers to exercise extra precaution in limiting their exposure as much as possible. The existence of local regulations establishing upper limits for MR workers may not apply to pregnant MR workers, although no epidemiological evidence exists supporting negative effects of EMF exposure on the health of a pregnant worker or her fetus. The User is responsible for determining whether local or country legislation may exist establishing occupational limits for exposure to EMF. If such limits exist it is the User’s responsibility to ensure they are being observed.

To ensure safe operation of your system, for both you and your patient, you must understand several components of your MR system. Your MR system includes the following magnetic fields:

- Static Magnetic Field (the magnet)
- Gradient Magnetic Fields (the gradients)
- Electromagnetic Fields (the RF)

The following definitions are used throughout Magnetic Field Basics section:

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- **Normal Operating Mode (Clinical Mode):** mode of operation of the MR equipment in which none of the outputs have a value that may cause physiological stress to patients.
- **First Level Controlled Operating Mode:** mode of operation of the MR equipment in which one or more outputs reach a value that may cause physiological stress to patients, which needs to be controlled by medical supervision.
- **Second Level Controlled Operating Mode:** mode of operation of the MR equipment in which one or more outputs reach a value that may produce significant risk for patients, for which explicit ethical approval is required (i.e., a human studies protocol approved to local requirements).

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Static Magnetic Field

The main magnet is a stable and very intense magnetic field.

IMPORTANT!: Note that the MR magnet is always on even when the system is not acquiring scan data. The only exception to this is if service has ramped down the magnet or it has been quenched.

The main safety issues regarding the static magnetic field include the potential for biological effects, the potential for attraction of ferromagnetic objects, and the potential for a quench of the cryogens.

The MR system static magnetic field may be classified under several modes:

- **Normal**: the normal operating mode, admissible for all individuals.
- **First Level**: controlled operating mode, admissible for patients on whom a medical decision was made ensuring they can handle the increased static magnetic field.
- **Second Level**: controlled operating mode, IRB or Human Ethical Committee required.

Table 2-3 Static Magnetic Field

Mode	System
</= 2T for Normal Mode	0.7T /1.5T
> 2T </= 4T for First Level Mode	3.0T
> 4T for Second Level Mode	N/A

A magnet produces invisible lines of force that extend beyond the magnet that are called the fringe field (Figure 2-3). The size of the fringe field depends on the strength of the magnet and whether or not it is shielded. Active and inactive shielding are used to reduce or tighten the fringe field.

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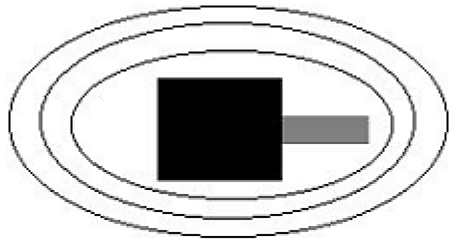
Figure 2-3 Fringe Field



CAUTION: **For some patients or MR workers, rapid movement of the head while in the magnetic field may cause dizziness, vertigo, or a metallic taste in their mouth. None of these motion effects are considered to be hazardous, provided they do not cause the worker to fall.**

It is recommended that the patient and the MR worker endeavor to remain still while in the region of high static magnetic field. The MR worker should always vacate the area of the static magnetic field when duties do not require otherwise.

The tesla to gauss conversion is 1 tesla = 10,000 Gauss.

The magnetic field exerts force on susceptible materials and biomedical implants and can create hazards. There are two critical zones: the Security Zone and the Exclusion Zone. Each zone has specific restrictions regarding people and materials.

WARNING: **It is your responsibility to ensure the creation of the Security Zone and the Exclusion Zone. Ensure occupational exposure to static magnetic field complies with local requirements.**

Security zone

The Security Zone (Figure 2-4) is the magnet room and the walls of the magnet room.

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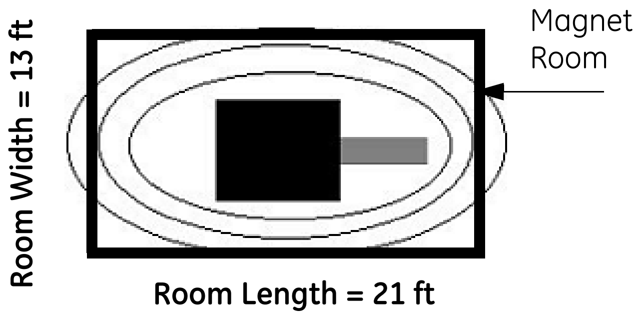
Figure 2-4 Security Zone



NOTE: Figure 2-4 states the approximate minimum room size for the 0.7T Open*Speed*, 1.5T EXCITE, and 3.0T systems, including Signa and MR750 systems. Consult your GE System Pre-Installation Manual for specific dimensions of your system and additional magnetic field plot information.

The 3.0T VH/i system approximate minimum room requirements are 17 feet x 28 feet. Consult your GE System Pre-Installation Manual for specific dimensions of your system and additional magnetic field plot information.

IMPORTANT!: You need to understand the meanings of ferromagnetic and ferrous substances or items:

- A substance that is ferromagnetic has a large positive magnetic susceptibility. (Example: Iron.)
- An item that is ferrous can posses intrinsic magnetic fields and react strongly in an applied magnetic field. (Examples: Iron, nickel, and cobalt.)

The attractive force of the magnetic field in the Security Zone can cause ferromagnetic items to become projectiles and contraindicated biomedical implants to fail. In short, ferromagnetic items and contraindicated biomedical implants are NOT allowed in the Security Zone.

The MR System operates with a highly sensitive RF receiving front end to be able to capture the signal of an object scanned. The Magnet Room part of the MR System installation provides the RF isolation to reduce the interference from electrical devices outside the shielded location.

It is possible that any device that functions with active electronic circuitry may potentially interfere the operation of the MR System if such device is introduced inside the Magnet Room even though the device does not have an intentional RF Transmitter. Extreme EMC measures must be taken into account in the design and manufacturing of an electrical device if such device is intended to operate inside the Magnet Room.

A device that may potentially interfere the MR System if introduced inside the Magnet Room are those containing active electronics. Some examples include: Switching Mode Power Supply (SMPS), microprocessor, Digital Signal Processors, analog to digital converters, LCD displays, keypad controllers, motors, battery operated devices.

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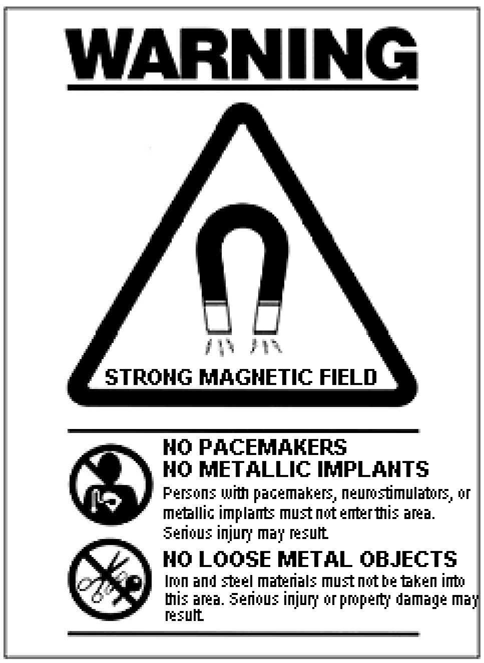
WARNING: The Security Zone warning sign (Figure 2-5) must be posted on the entrance to the magnet room to alert personnel to the high magnetic field and warn not to bring ferromagnetic objects into the magnet room.

WARNING: Ensure that the Security Zone complies with your local statutory requirements.

Figure 2-5 Security Zone Warning Sign



The Security Zone Warning Sign (Figure 2-5) alerts personnel and patients of the following:

- Strong magnetic field
- No pacemakers
- No metallic implants
- No neurostimulators
- No loose metal objects

3.0T EXCITE magnets have a specialized Security Zone warning sign (Figure 2-6) to distinguish the 3.0T magnet from a lower magnetic field.

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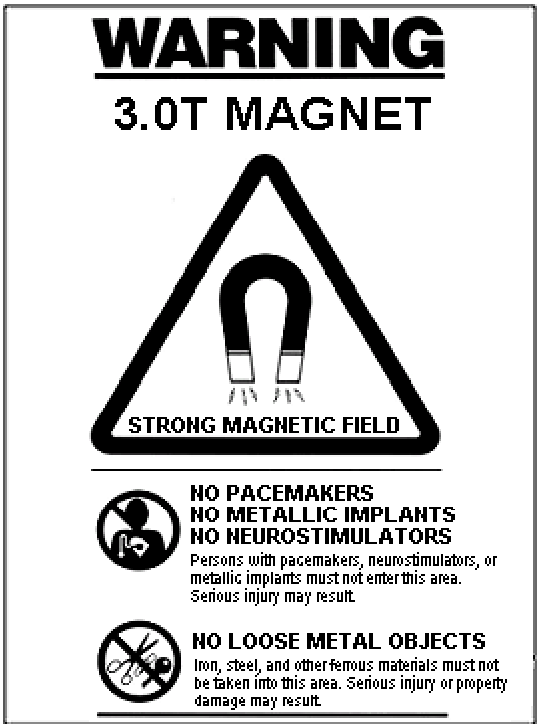
Figure 2-6 Security Zone Warning Signs for 3.0T Magnets



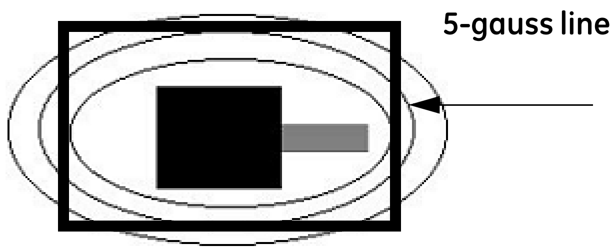
The Security Zone Warning Sign for 3.0T EXCITE magnets (Figure 2-6) alerts personnel and patients of the following:

- Strong magnetic field
- No pacemakers
- No metallic implants
- No neurostimulators
- No loose metal objects

Exclusion zone

The Exclusion Zone (Figure 2-7) begins at the 5-gauss line. Magnetic shielding may, however, restrict the 5-gauss line to the magnet room, making the security and the exclusion zone the same.

Figure 2-7 Exclusion Zone



All personnel should be aware of the gauss line and actively screen the changing conditions of the environment. There are gauss lines and equipment that must remain outside certain limits. Consult your GE Service Engineer to know where these gauss lines are located in your facility.

WARNING: The Exclusion Zone warning sign must be posted at the 5 gauss boundary. Locate and read the Exclusion Zone signs at your facility.

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Figure 2-8 Exclusion Zone Warning Sign

The Exclusion Zone Warning Sign (Figure 2-8) alerts personnel and patients of the following:

- Strong magnetic field
- No pacemakers
- No metallic implants
- No neurostimulators

WARNING: Ensure that the Exclusion Zone complies with your local statutory requirements.

Biological Effects

The static magnetic field strengths used by your MR system are within the guidelines provided by the United States Food and Drug Administration (FDA) for clinical imaging. However, there are several cautions that need to be understood:

CAUTION: Minimize the time spent near the magnet. Spend only the time necessary to attend to the needs of the patient.

CAUTION: MR scanning has not been established as safe for imaging fetuses or infants. Carefully compare the benefits of MR versus alternative procedures before scanning to control risk to the patient. A physician needs to decide to scan pregnant or infant patients.

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Ferromagnetic Objects

Ferromagnetic objects brought within close proximity of the static magnetic field can become projectiles, which could cause harm to someone standing between the object and the magnet. The force of attraction between a magnet and a ferromagnetic object is determined by the magnetic field strength (fringe field), the magnetic susceptibility of the object, its mass, its distance from the magnet, and its orientation to the field.

Use only non-ferrous oxygen tanks, wheelchairs, gurneys, intravenous (IV) poles, ventilators, etc. in the magnet room. Be sure anyone who has access to the MR suite is aware that only non-ferrous items are allowed in the magnet room. Make them aware that policies and procedures are in place for bringing medical devices and other equipment into the magnet room.

In addition to the projectile hazard, the static magnetic field can cause ferromagnetic objects within the patient (e.g., surgical clips, prostheses) to move, thus possibly causing harm. Electrically, magnetically, or mechanically activated implants can become dysfunctional due to the static magnetic field. If these devices are life-supporting, harm could result.

WARNING: The attractive force of the magnetic field of the MR system can cause ferrous objects to become projectiles that can cause serious injury. Post the security zone warning sign on the entrance to the magnet room and keep all hazardous objects out of the magnet room. If a ferromagnetic object has become attached to the magnet, contact GE Service for assistance.

WARNING: To help prevent patient or operator injury, do not bring ferrous materials such as battery operated devices into the magnet room.

WARNING: To help prevent patient or operator injury, do not bring ferrous oxygen bottles into the magnet room.

CAUTION: Common hospital equipment, which often have ferrous battery packs, such as patient monitoring, and life supporting devices, may be adversely affected when in proximity to the magnetic field or image quality may be affected by the presence of this equipment.

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CAUTION: The only GE supplied tools recommended for use inside the Security zone are the phantoms supplied with your system.

WARNING: Electrical discharges between conductive devices and the MR coils can startle or burn the patient and possibly cause the patient to injure himself/herself. To help avoid such reactions, do not place metal objects (e.g., limb braces, traction mechanisms, stereotactic devices, etc.) in the MR magnet.

WARNING: The fringe field can cause injury by interfering with the normal operation of biomedical devices.

Cryogen and Quench Concerns

With superconductive MR systems, another concern related to the static magnetic field is a quench of the cryogens. A superconductive magnet uses cryogens to super-cool the electrical conductor that creates the magnetic field. Temperatures as low as -269°C (-452°F) are achieved to create the proper environment within the magnet. A quench, which is a sudden boil-off of the entire volume of cryogenic liquid, causes a rapid loss of the static magnetic field.

Liquid Cryogen Hazards

Cryogens come in large vacuum containers called dewars. Liquid helium is generally used for cooling purposes, although some service procedures also require liquid nitrogen. Nitrogen dewars weigh from 400 to 500 pounds when full. Helium dewars weigh from 700 to 800 pounds. In addition to large dewars, there may be smaller helium gas cylinders present. This helium gas is used to fill the magnet to proper cryogen levels. Special considerations should be observed when dealing with cryogens.

CAUTION: Leaking helium or nitrogen gas will displace the oxygen. An ambient air oxygen concentration of less than 17% to 18% is not sufficient for human respiration. The limit of the air oxygen concentration should meet national laws or regulations.

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CAUTION: The following information defines the proper handling of cryogens.

- **Dewars and cylinders should not be tipped or heated, nor should the valves be tampered with.**
- **The cryogens boil off as they cool the magnet wires and must be replenished periodically by qualified personnel. The rate of boil-off should be monitored by checking the cryogen level meter found on the system cabinet.**
- **Contact with the cryogenic liquids or gas could result in severe frostbite; care should be taken when in proximity to these substances. The wearing of protective clothing is essential during all work in conjunction with liquefied cryogens. Such clothing consists of:**
  - **Safety gloves**
  - **Work gloves**
  - **Face shield**
  - **Laboratory coat or overalls (cotton or linen)**
  - **Non-magnetic safety shoes**
- **Dewars should be stored in a well-ventilated area. Cryogens could be accidentally released in gaseous form, resulting in an asphyxiation hazard.**
- **All dewars and gas cylinders must be non-magnetic.**
- **Gas cylinders should be stored upright and secured to the wall with a chain with the metal protective top in place. (If a cylinder falls over or the valve is knocked off, the container may act like a rocket; a full cylinder has enough power to penetrate walls.)**
- **Because the cylinder's metal cap may be magnetic, the cap should always be removed before bringing the cylinder into the magnet room.**
- **If possible, all personnel should stay out of the magnet room when a qualified service engineer is filling cryogens in the magnet. If personnel must be present, they must wear proper gloves, a face shield, and ear protectors.**
- **A qualified service engineer should be present any time cryogens are transported within the hospital or added to the magnet.**
- **It is crucial that ventilation and cryogenic systems be kept in good repair and checked regularly to ensure proper functionality.**
- **Flammable materials must not be brought near the cryogen containers.**
- **You are responsible for establishing and following a procedure, in accordance with your local and federal requirements (in the US: OSHA 29 CFR 1910.36), that includes possible evacuation of the MRI area, if flammable materials are identified near cryogenic gases. If grease, oil, or other combustible material is present in the vicinity of the containers, the escape of cryogenic gases can lead to the formation of a potentially combustible liquid due to liquefaction of air and concentration of oxygen.**

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Quench Vent Failure Hazards

Quenches are indicated by a loud noise, warning message, or the tilting of an image on the display screen. A quench is a hazard only if the vent fails, which would result in the release of white clouds of cryogen vapor into the magnet room. This would present a potential asphyxiation hazard to both the patient and personnel. It is critical to have a well-planned method to quickly evacuate the patient and personnel from the magnet room should a quench occur.

WARNING: **In the unlikely event of a quench and vent failure, a procedure needs to be in place to evacuate the patient and all personnel from the magnet room. Failure to follow these precautions can result in serious injury (e.g., asphyxiation, frostbite, or injuries due to panic).**

Table 2-4 lists the decay time for the system to reach 20 mT in the case of a quench or the Emergency Magnet Rundown switch is activated.

Table 2-4 Examples of system decay time to reach 20 mT

System	Decay Time (seconds)
0.7T	120
1.5T	120
3.0T VH/i	30
3.0T EXCITE	100

Gradient Magnetic Fields

Gradient magnetic fields produce rapidly changing magnetic fields during scanning. Gradients turn on and off very rapidly to spatially encode the MR signal during a scan. High frequency gradient amplitudes switched very quickly (high dB/dt) may cause nerve stimulation at periphery of body due to induced currents in nerves. Because a current can be induced in any conductive material lying on or near the patient's body, a potential biological hazard exists. The greater the rate of change of the magnetic field (dB/dt, slew rate), the larger the induced current. The muscles, nerves, and blood vessels of the human body are all conductive materials.

WARNING: **Ensure occupational exposure to time varying magnetic field caused by the gradients complies with local requirements.**

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Calculate Maximum Gradient Output

Let d be the total duration of the gradient ramp (minimum to maximum) or the period of a sinusoidal waveform divided by π. Let f be a fraction (0.8 for Normal Mode or 1 for First Mode). Let c be the chronaxie time in microseconds. Let $d|B|/dt\_0$ be the rheobase (infinite ramp duration) value of the time varying gradient magnetic field. Gradient output may be expressed by the following equation;

Maximum gradient output may be found from equation and Table 2-5. f is the fraction of the mean stimulation threshold, rb is the infinite ramp duration mean PNS threshold, c is the rheobase value (ramp duration where the mean threshold is 2*rb), and d is the gradient ramp duration.

$$\left(\frac{d|B|}{dt}\right)_{limit} = f(rb)\left(1+\frac{c}{d}\right)$$

Table 2-5 Rheobase and chronaxie constants for various gradient coils

	rb (T/s)	c (µs)
BRM	23.7	370
CRM	25.2	558
TRM Zoom	29.1	354
TRM WB	23.7	370
XRMB	23.4	334.3
HDw	20 T/s	360

Gradient Output Limits

Safety parameter levels approved for the Second Controlled Operating Mode are set by the Investigational Review Board (IRB) or other human studies boards. For Gradient Output these levels are measured as a percentage of the mean peripheral nerve stimulation threshold (PNS).

CAUTION: **Continuous patient observation and contact are required in all modes of operation. Medical Supervision is required in the First or Second Level controlled operating modes.**

The MR system gradients are capable of operating under several modes:

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- **Normal**: the normal operating mode, admissible for all individuals.
- **First Level**: controlled operating mode, admissible for patients on whom a medical decision was made ensuring that they can handle the increased gradient output effects or increased SAR. Limits for increased gradient output and SAR are based on current scientific literature related to safety.
- **Second Level**: controlled operating mode, admissible for customers with Investigation (ethical studies) Review Board (IRB) clearance and with a proprietary license agreement with GE.

The appearance of the operating interface changes on the basis of: whether the system is operating in clinical or research mode, the governing standard (typically the International Electrotechnical Commission (IEC)). Table 2-6 lists system operating modes and associated threshold limits.

CAUTION: Gradient output may be controlled by Local Approval.

Table 2-6 IEC Gradient Output Limits

Operating Mode	PNS Limit
Normal	80% mean nerve stimulation threshold
First Level (controlled mode)	100% of the mean nerve stimulation threshold. Requires you to click the **[Accept]** button to proceed when the Normal mode dB/dt or SAR limits are exceeded, but Second Level mode has not yet been reached.
Second Level (controlled mode)	Requires research key and IRB or Human Ethical Committee approval of the research to be conducted

Peripheral Nerve Stimulation

The concern from time-varying gradient fields is to prevent cardiac stimulation and ventricular fibrillation. Cardiac stimulation in the most sensitive population percentile requires at least 39 times as much energy as is produced by a 0.05 T/m, SR200 scanner.

Regulatory bodies use avoidance of painful nerve stimulation to limit gradient output with an adequate safety margin. Painful nerve stimulation typically occurs at approximately double the mean stimulation perception threshold. Some discomfort is experienced about 1.5 times the mean PNS threshold (Figure 2-10).

Peripheral nerve stimulation (PNS) problems are not of concern on systems compliant with IEC 60601-2-33 Normal or First Controlled Operating Modes. The IEC limits PNS to 80% of the mean threshold in Normal Mode (where stimulation should be rare) and 100% for the First Mode (where non-painful stimulation is expected in about half the patients).

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If the patient can not tolerate PNS, change to Normal Mode to eliminate the problem. Otherwise change to a lower slew rate pulse sequence to continue scanning the patient. MR workers may experience similar sensations if they are in or very near gradient coils during active scanning.

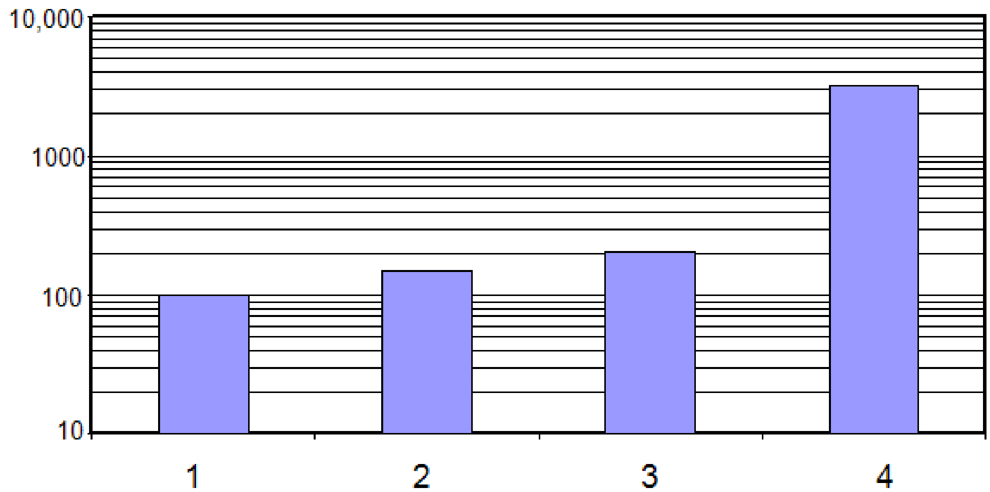
The Anterior/Posterior (A/P) (=Y) gradient axis typically has the lowest stimulation threshold. So it is prudent to keep the gradient waveform most likely to stimulate (the gradient waveform with the highest slew rate for the longest total ramp time) on a physical axis other than Y.

You should remain constant contact with the patient, especially in the FIRST CONTROLLED MODE, to ensure that the patient does not feel painful stimulation (or high localized heating).

Figure 2-9 displays a graph of the relative mean threshold (vertical axis) and discomfort stimulation levels (horizontal axis) where relative means are for perception (100), discomfort (1000), and pain (10,000).

- 1 = threshold
- 2 = uncomfortable
- 3 = intolerable
- 4 = 1% cardiac

Figure 2-9 Relative mean threshold and discomfort stimulation levels



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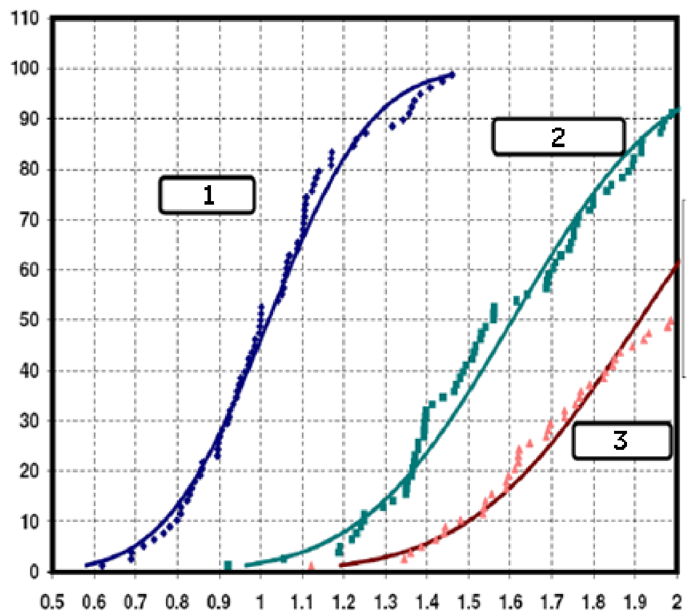
The distribution of those experiencing PNS is illustrated in Figure 2-10 three curves where the horizontal axis is the normalized level and the vertical axis is the% probability of PNS. The curves represent the following:

- 1 = threshold
- 2 = uncomfortable
- 3 = intolerable

Figure 2-10 PNS probability. X axis = Fraction of the Mean Threshold (100% PNS). Y axis = Population percentile.



CAUTION: **Due to the rapid rate of change of the magnetic fields (dB/dt) used during some scans, a percentage of patients may experience a non-hazardous tingling or touch sensation. Figure 2-10 indicates the type of sensations caused at different percentages of the mean nerve stimulation threshold. Note that stimulation is relatively rare in NORMAL MODE (x-axis=0.8), but occurs about 50% of the time in the FIRST MODE (x axis=1). If this sensation is bothersome or uncomfortable to the patient, stop the scan. Change to NORMAL MODE to continue scanning the patient. The MR worker may experience similar sensations if remaining within the gradient field during active scanning.**

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CAUTION: **To reduce the possibility of PNS, make sure the patient's hands are not clasped or touching and that their feet are not crossed. Either or both of which could form a conductive loop.**

CAUTION: **There is a possibility that mild peripheral nerve stimulation (PNS) may be induced in the MR worker when that person is exposed to the gradients when the system is operating in the First Level Controlled Operation Mode. The MR worker should remain outside the magnet room during scanning in this Mode except when circumstances dictate otherwise.**

CAUTION: **Peripheral nerve stimulation is not harmful. The potential for inducing peripheral nerve stimulation is kept within limitations. The MR system is limited from operating above 80% of the PNS threshold in the NORMAL Mode (100% of the mean PNS threshold in the First Mode) by the software (unless the system is in Second Controlled Mode). The point at which 50% of a population experiences PNS is the PNS threshold. PNS has been described as a light "touching" sensation felt on various areas of the skin surface. These areas vary depending upon which gradient axis is in use. Some common areas for the sensations are the bridge of the nose, arms, chest, and upper buttock/abdomen. Hands clasped together increase the potential for stimulation by approximately 65%. The potential for PNS is low, but it exists for all sequences in all gradient configurations.**

CAUTION: **for all sequences in all gradient configurations**

NOTE: Please report all complaints of patient discomfort that may be associated with PNS during MR examinations (e.g., muscle twitches, tingling sensations, or headaches) to GE Healthcare. See page 2-2 for contact information.

Acoustic Noise

Another potential safety issue associated with gradient switching is the loud noise. The rapid alternations of currents within the gradient coils cause the coil assemblies to vibrate against their mountings, thus generating a loud resonant noise. The acoustic noise produced during scanning can exceed 99 dBA in the bore.

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WARNING: The sound level at the operator's console should be limited to comply with local rules.

WARNING: Hearing protection is required for all people, including the MR worker, in the magnet room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dBA. Patient hearing protection with a noise reduction rating (NRR) of 29 dB or better is required to reduce acoustic level below 99dBA. The A-weighted RMS sound pressure level is measured according to section 26e and 26g of IEC 60601-2-33: 2002.

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CAUTION: **All personnel should be trained on the proper use of hearing protection.**

- **Special attention should be utilized to protect the hearing of neonates, premature infants, and any other condition that does not allow for hearing protection to be applied.**
- **Patients with increased anxiety may have a lower acceptance to sound pressure (e.g., young children, elderly, and pregnant women).**
- **Anesthetized patients have less then normal protection against high sound pressure. Hearing protection should not be omitted.**
- **Typical operator console noise levels are below 60 dBA, so hearing protection is generally not required at the operator console. However, it is important to ensure the sound level complies with all local regulations.**
- **In some countries, legislation exists that limits employee exposure to noise levels. Ensure compliance with your local regulations by providing additional hearing protection to MR workers for use in the magnet room where required.**
- **If a music sound system is in use by the patient during scanning, the music sound system must provide > 29dB NRR of attenuation. All hearing protection devices must provide > 29dB NRR of attenuation.**

Encourage the routine use of earplugs to prevent problems associated with acoustic noise during MR procedures. GE Healthcare offers disposable ear protection of various noise reduction ratings. These can be ordered through the GE accessories catalog. Table 2-7 describes the available types of disposable ear protection.

Table 2-7 Disposable Ear Protection

Description	dB
E8801BA EAR Disposable Foam Earplugs	29
E8801BB EAR Taperfit2 Foam Earplugs	32
E8801BC Max-Lite Foam Earplugs	30

IMPORTANT!: Since the acoustic noise of the Open*Speed* system does not exceed 92.2 dBA, ear protection is not required, but is recommended.

Electromagnetic Fields

The Radio Frequency (RF) field is an oscillating electromagnetic field. Pulses of RF energy are used to generate the signal, which cause tissues to absorb RF power. Under certain conditions, this may cause tissue heating. The amount of heating depends on several factors, such as patient size and pulse sequence timing. RF heating of tissues is greatest at the periphery of the skin. The Specific Absorption Rate (SAR) is the estimated amount of heat dose received by the patient. This value is expressed as watts of power per kilogram of the patient’s body weight.

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Tissue Heating

Before the patient is scanned, the computer estimates the level of heating and compares it to the predetermined exposure limits. If the scan is expected to exceed these limits, the system then adjusts the scan parameters before starting the scan. The complete estimate is based in part on patient weight. Therefore, take care to enter the patient's weight correctly to prevent excessive RF exposure or scan abortion.

When patient temperature is not changing, typical skin temperatures are about 33 °C while core temperatures are about 37 °C. Patients dissipate metabolic heat at the same rate it is generated so there are no skin or core temperature changes. Humans subjected to significant radio frequency power deposition (i.e., significant SAR) will normally attempt to dissipate the additional heat load through vasodilatation of skin blood vessels permitting skin to approach core temperature. This action typically causes the skin to flush (turn red) and enables the body to dissipate heat more rapidly. This skin flushing is a normal response to significant radio frequency power deposition. Skin reddening or to a lesser degree the report of a warming sensation without reddening regardless of the method it was created (SAR, Conatct, Metal, etc) is not hazardous if it clears in a few hours.

Patient Comfort Module

A sensor located in the bore of the magnet monitors bore temperature. The sensor posts appropriate messages in the System Status Display area of the monitor (upper left portion of the monitor) that notify you when the magnet bore wall is becoming warm.

The temperature inside the magnet room should be set at less than 70°F and the bore fan should be turned on at all times to keep air flowing inside the bore of the magnet.

Thermal Hazards

The increase in tissue temperature caused by RF exposure depends on a variety of factors associated with the thermoregulatory system of the individual and the surrounding environment. Thermoregulatory is the ability of the body to maintain regulated heat capacity levels.

Observe the following warnings concerning tissue heating:

WARNING: RF power deposition can heat the patient's tissue if delivered faster than the patient's tissues can dissipate the generated heat. The amount of tissue heating depends on the patient's weight, type of pulse sequence, timing factors, number of slices, SAR, and the use of imaging options such as saturation. Power deposition will typically be lower when the NORMAL MODE is selected for SAR. FIRST MODE for SAR offers higher performance but also higher power deposition

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WARNING: **A rise in body temperature can be a hazard to a patient with reduced thermoregulatory capacity. Reduced thermoregulatory capacity can be caused by pre-existing conditions, such as cardiac impairment that has reduced circulatory function, hypertension, diabetes, old age, obesity, fever, or an impaired ability to perspire. A patient with these complications must be carefully monitored at all times. Consider scanning with NORMAL MODE for SAR for patients that may not tolerate the higher levels.**

CAUTION: **The MR worker who remains in the scan room during a study could be subject to tissue heating caused by RF energy exposure. Care should be taken to limit the time the MR worker remains in the scan room during a study.**

WARNING: **RF can also raise the magnet bore temperature and cause thermal stress; medical conditions can reduce a patient's ability to cope with external temperature increases. If the temperature continues to rise, the scan stops until temperature within the bore is lowered. When the sensor detects temperatures that may cause patient discomfort, the system posts the following messages on the screen:**

The patient comfort level is warmer than normal.

The patient may be uncomfortable during the scan.

The bore cooling system or magnet room temperature may not be normal.

Further increases in temperature will inhibit scanning.

When the temperature drops to a comfortable level, the message is cleared from the screen. If the temperature continues to rise, a second message appears on the screen: Scan inhibited. Patient comfort sensor trip.

To facilitate a return to scanning, make sure the patient fan is ON, room temperature is normal, 21°C (70°F), and air flow through the bore is unobstructed.

When the magnet opening temperature decreases, the system posts this message: New scans can be initiated, but the patient comfort level is still warmer than normal.

CAUTION: **All patients should be monitored for increased temperature during the scan acquisition. If the patient reports discomfort due to warming, stop the scan. Patients should be provided with the hand-held Patient Alert bulb prior to scanning. The patient should be instructed to communicate any concerns through the intercom or by activating the Patient Alert bulb.**

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CAUTION: RF heating can be caused by:

- **Damp clothing**
- **Contact of body or extremities against the RF transmit coil surface, contact with metal, tattoos or metallic eyeliner, contact with other body parts**
- **Formation of loops with RF receive coil cables and ECG leads**
- **The use of non MR-compatible ECG electrodes. Never use ECG electrodes past their expiration date.**
- **Scanning with a unconnected receive coil or other cables in the RF transmit coil during the examination.**
- **ECG leads not compatible with MR. MR leads have very high impedances that limit current to below the level of concern.**

CAUTION: Extra attention should be utilized when scanning patients who are unconscious, sedated, or may have loss of feeling in any body part (temporary or permanent paralysis). They may not be able to alert you to RF heating.

CAUTION: The coil selected should match the coil that is connected. When scanning with a transmit/receive only coil, DO NOT scan using the body coil (or use the Body coil configuration) at any time. Using the body coil can cause RF heating and could result in patient burns. In addition, scanning with the body coil can damage the transmit/receive only coil, requiring the coil to be unusable and returned to the factory for service.

Contact Point Heating

Patient positioning can affect the safety of the scan procedure. To help prevent a patient burn from closed loops formed by the following examples: clasped hands, by hands touching the body, from thighs contacting, the patient's breasts contacting the chest wall over a small area, etc., insert non-conducting pads at least 0.25 inches thick between touching parts and between the patient and the bore wall and the patient and any coils or conductors (Figure 2-11).

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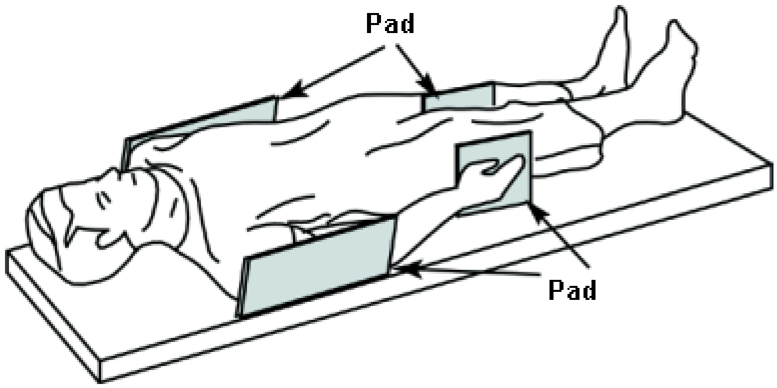
Figure 2-11 Patient Positioned with Non-conducting Pads



Observe the following warnings concerning contact point heating to protect patients from excessive heating or burns related to induced currents during MR procedures:

WARNING: RF can cause localized heating at contact points between the patient/bore and patient/RF coil resulting in discomfort or burns.

WARNING: RF can cause localized heating at contact points between adjacent body parts when a loop is formed. Such localized heating can result in discomfort, or burns. This could occur when a patient's hands are touching or when a female patient's breasts are compressed to her chest. Use pads between body parts to avoid creating a loop with adjacent body parts.

WARNING: Place appropriate non-conductive padding between the patient and the bore wherever a portion of the body may come into contact with the magnet opening.

WARNING: Always place appropriate non-conductive padding between the surface coil and the patient's skin to prevent burn injuries.

WARNING: For shoulder imaging, always place appropriate non-conductive padding between the patient's opposite shoulder or a portion of the patient's body and the bore wherever a portion of the body or opposite shoulder comes into contact with the bore.

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CAUTION: RF can cause localized heating at patient contact points. Wet diapers or incontinence products have the same electrical properties as human tissue. All patients with diapers, including adults, should have dry diapers on prior to the start of the scan. If the patient reports discomfort due to warming, stop the scan.

Metallic Heating

Another potential hazard related to the RF field is the heating of implants, devices, and objects of metallic compositions that may cause temperature changes during an MR examination. The induced currents in the metallic objects may cause them to get so hot that they can actually cause burns in adjacent tissues. Therefore, it is very important to determine each patient's work history and thoroughly screen for any accessories containing metal.

Observe the following warnings concerning metallic heating to protect patients from excessive heating or burns related to induced currents during MR procedures:

WARNING: Eye makeup that contains metal flakes can cause eye and skin irritation during MR scanning. Instruct patients to wash off removable makeup before the exam to avoid the risk of eye injury. Before scanning, warn patients with permanent eyeliner or other metallic ink tattoos about the risk of skin irritation and instruct them to get prompt medical attention if they experience severe discomfort following an MR exam.

WARNING: Metal fragments/slivers can deflect and/or heat in a magnetic field, damaging surrounding tissues. Patients thought to have metallic fragments in the eye should receive an eye exam to detect and remove any metal fragments that could deflect and damage the eye.

WARNING: Jewelry, even 14-karat gold, can heat and cause burns. RF can heat (even non-ferrous) metal and cause burns.

WARNING: Medicinal products in transdermal patches may cause burns to underlying skin.

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WARNING: **The use of MR non-compatible stereotactic frames and RF blankets is not recommended.**

Specific Absorption Rate (SAR) Limits

It is necessary to measure the RF absorption in tissues since RF exposure cannot be measured by the system. Energy dissipation through absorption by the body can be described in terms of Specific Absorption Rate (SAR). SAR is a rate, meaning it is a measure of the amount of RF power absorbed per unit of mass of an object in watts per kilogram (W/kg). There are several types of SAR, listed in Table 2-8, that must be understood.

Table 2-8 SAR Definitions

SAR Type	Definition
Whole Body	SAR averaged over total patient body mass over any six-minute period for Body and Receive Only surface coils, which use body transmit.
Partial Body	SAR averaged over the exposed mass in the coil averaged over any six-minute period.
Head	SAR averaged over the mass of the patient's head over any six-minute period.
Extremity	SAR averaged over the estimated mass of the patient's extremity over any six-minute period for small volume and Transmit/Receive coils.
Short Term	The SAR averaged over any 10-second period.

The MR system SAR operating modes:

- **Normal**: the normal operating mode, admissible for all individuals.
- **First Level**: controlled operating mode, admissible for patients on whom a medical decision was made ensuring they can handle the increased SAR effects. Increased SAR levels are based on current scientific literature related to safety.
- **Second Level**: controlled operating mode, admissible for customers with a propriety license agreement with GE and with Investigational Review Board clearance for the investigational protocol.

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CAUTION: When the system is operating in research mode, you are presented an interface to modify pulse sequence internal parameters. If one chooses to modify certain internal pulse sequence parameters like the pulse sequence repetition time, there is a risk of running the pulse sequence at a Specific Absorption Rate (SAR) higher than the regulatory limits. You are advised NOT to modify any internal pulse sequence parameters through Research Options, unless you are certain that by doing so, you are not violating or bypassing safety limits or other local requirements which are otherwise in place.

CAUTION: Continuous patient observation and contact is required in all modes of operation. Medical Supervision is required in the First or Second Level controlled operating modes.

CAUTION: SAR may be controlled by Local Approval.

WARNING: The magnet room temperature shall not be more than 21°C per the manufacturer's requirements and the relative humidity shall not be more than 60%. Temperatures above 21°C and humidity above 60% could result in lowering the system SAR limit.

The derating temperature is 25°C for relative humidity less than 60%. For each 10% increase of the relative humidity in excess of 60%, the temperature is reduced by 0.25°C, e.g., 24°C at 100% relative humidity.

For each degree of environmental temperature that exceeds the SAR-derating temperature, the whole-body SAR limit is reduced by 0.25 W/kg until the SAR is 2 W/kg or 0 W/kg for the First Level controlled operating mode or for the Normal mode, respectively.

WARNING: The RF power monitor and SAR limitations help prevent excessive RF exposure to the patient; SAR values are calculated based on the patient's weight. To help avoid injury, enter the patient's correct weight to set operating limits and prevent excessive RF exposure.

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WARNING: The SAR algorithms for the MR systems calculate SAR values and set a limit on the number of slices/echoes per second in order to limit RF power deposition. The power monitor and SAR algorithm limit SAR, regardless of the patient weight or pulse sequence used. SAR limits are conservatively estimated from worse-case patient positioning as a function of weight.

The legacy power monitor module limits the RF amplifier output power thus limiting the patient SAR in case of a catastrophic failure. This module monitors peak power based on the patient's weight, duty cycle, and pulse sequence parameters. The peak power limits prevent you from using incorrect patient weights.

WARNING: The average power monitor and SAR algorithm limit SAR based on patient weight and RF transmit coil used. SAR limits are conservatively estimated from worse case patient positioning as a function of weight. The power monitor limits the RF power, which in turn limits the patient SAR to within controlled limits over time.

Pulse sequence SAR predictions (estimated SAR) are based on patient weight at the worst-case landmark. To minimize nuisance power monitor trips caused by patient-to-patient variability, pulse sequence predicted SAR is the mean plus 1.96 standard deviations (typically the normalized standard deviation is about 18% at the worst-case landmark). The expected worst-case nuisance trip rate is approximately 2.5%. If you experience a significant number of power trips above the 2.5% frequency, please consult your local field service representative. The power monitor measures actual power and limits SAR appropriately. The power measuring accuracy of the power monitor is about +/-12%.

Errors in patient weight do not result in excessive SAR. Low patient weight entries result in power monitor trips below the SAR limit. High patient weight entries result in fewer slices/images per unit time than would have been permissible.

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WARNING: SAR Limits

The MR system's RF power monitor helps prevent excessive RF exposure due to equipment failure. Since the monitor protects the patient, it must be operational at all times, even when a scan is not in progress. If it detects an equipment failure, it immediately disables the RF system. This system must be repaired or adjusted by qualified service personnel.

Table 2-9 SAR Operating Limits

System	Normal Mode (W/kg)	First Level (W/kg)	Second Level (W/kg)
0.7T	Head = 3.2 Body = 2.0	N/A	N/A
1.5T	Head = 3.2 Body = 2.0	Head = 3.2 Body = 4.0	Head > 3.2 Body > 4.0
3.0T	Head = 3.2 Body = 2.0	Head = 3.2 Body = 4.0	Head > 3.2 Body > 4.0 *

IMPORTANT!: *When operating in the Second Level controlled operating mode, you need to click the **[Scan Modes]** button from the Rx Manager and click **[NO]** to monitor SAR. This is only available when Research mode is available. This disables the SAR slice limitations of the MR system, but not the slice limitations or SAR monitoring of the 3.0T system power monitor. When the Research mode is enabled, a **[Monitoring Parameters]** button is available. Clicking this button allows the researcher to enter an IRB approved SAR limit for the research scans to be performed. The system automatically returns to Normal operation for the next examination or can be manually returned to Normal operation by clicking the **[Go to Clinical]** button from within the **[Monitoring Parameters]** button.

The IEC also allows Short Term limits that allow a higher Short Term SAR for 10 second periods of time. These limits are 3 times higher than the 6 minute limitations per IEC guidelines. This allows short term bursts of RF energy for very short scans.

For example, the Body SAR limit is 2.0 W/kg over 6 minutes and the 10 second average is subsequently 6.0 W/kg averaged over 10 seconds. This would allow a scan of 4.0 W/kg to run for up to 3 minutes, since running at 2 times the long term limit can only run for half the time to equal the same average over the full 6 minutes.

NOTE: Patient acceptance of High SAR scanning can be increased by giving the patient breaks to cool down, providing light clothing, and limiting room temperature to 18 ± 3 °C, and by maximizing air flow.

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Clinical Hazards

Maintaining good patient contact and education can help reduce patient anxiety reactions and clinical scanning hazards in the MR environment and during procedures.

CAUTION: **Continuous patient observation and contact are required in all modes of operation.**

You need to be aware of the conditions and risks associated with the following:

- High-Risk Patients
- Scanning Hazards

High-Risk Patients

Several conditions are associated with high-risk patients, who may be at a greater risk of developing complications during an MR examination. Observe these warnings and be prepared to manage the needs of such patients during the examination.

WARNING: **Patients with the following conditions are at the greatest risk of complications during MR scanning:**

- **Patients likely to develop seizure or have claustrophobic reactions.**
- **Greater than normal potential for cardiac arrest.**
- **Patients who are unconscious, heavily sedated, or confused and patients with whom no reliable communication can be maintained.**

WARNING: **Patient screening is required for patients who are going to be imaged in the First or Second control modes.**

Some patients may experience feelings of fear or claustrophobia when undergoing an MR procedure. This could be related to the confining conditions of the magnet, the length of the examination, the acoustic noise, or the temperatures within the bore of the magnet. Discuss the procedure with the patient and be prepared to manage the needs of the patient during the examination.

CAUTION: **The confining conditions of the MR system can precipitate claustrophobia in some patients. To prevent injuries due to panic, provide instructions and comfort the patient as needed to alleviate anxiety.**

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WARNING: Since direct observation from the operator's console can be partially obscured by the magnet enclosure, be sure to more closely monitor these types patients at all times to quickly identify and respond to medical emergencies. In some cases, emergency personnel should remain with the patient or be on standby alert to help prevent serious complications or death.

Scanning Hazards

During scan set-up, acquisition, and conclusion, be aware of the following scanning hazards:

WARNING: Do not use Projection Images for localization.

WARNING: Do not use 3D views only to perform voxel value, distance, angle, or area measurements. Always refer to 2D baseline views.

CAUTION: Measurements are more reliable when done on 2D views. Always check on the 2D reformatted views where exactly the points have been deposited.

CAUTION: Most multiple-channel receive only coils are designed to function best with adult patients. For smaller patients using the multiple-channel receive only coil the patient positioning is critical for optimal image quality. For small patients use appropriate non-conductive padding to place patient anatomy of interest in the center of the coil.

For example, the HDHead coil is a multiple-channel receive only coil. Use appropriate non-conductive padding to place the patient's head in the center of the coil.

CAUTION: Make sure the patient connected IV lines, oxygen tubing, urinary catheters, and any other tubing and cables are long enough to allow full travel of the system and will not become entangled, pinched, or pulled.

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CAUTION: **Following the exam, your patient may need assistance when getting off the table. After lying in a prone position for a length of time, your patient may experience light-headedness upon sitting up.**

CAUTION: **If the magnet room door is open, the scan cannot be started. If the scan is already in process and the door is opened, the scan will pause. Close the door and press resume.**

If the magnet room door is opened when attempting to start a scan, close the door and try again.

International regulations require the system to function in this manner.

Equipment Hazards

There are also general equipment concerns in the MR environment. Make sure you are familiar with your MR equipment and the accessory equipment manufacturer's guidelines and precautions. Specifically, you need to be aware of the hazards associated with the following MR equipment:

- Laser Alignment Lights
- Cables and Equipment Connections

Also observe the following general equipment hazards:

CAUTION: **Using equipment that is damaged or has been compromised, can put the patient and/or operator at risk of injury.**

CAUTION: **The MR system applications run on equipment that includes one or more hard disk drives, which may hold medical data related to patients. In some countries, such equipment may be subject to regulations concerning the processing of personal data and the free circulation of such data. It is strongly recommended that access to patient files be protected from all persons not in medical attendance.**

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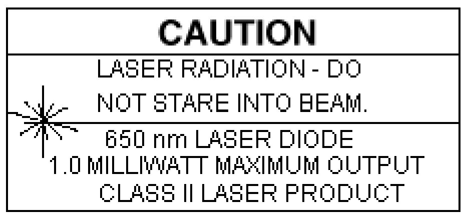
CAUTION: **Any application of physiological monitoring and sensing devices to the patient shall be made under the clinical staffs direction and is the clinical staff's responsibility. Use only MR-safe devices. Devices with conductors or ferromagnetic parts may introduce safety concerns.**

Laser Alignment Lights

The MR systems use semiconductor laser alignment lights for patient landmarking. This type of alignment light casts a thin red light on the patient for the purpose of positioning and landmarking. The laser lights can cause eye injury.

Figure 2-12 displays the warning label for laser products and is located on the bottom of the magnet front cover. Labels such as this provide warnings about laser radiation.

Figure 2-12 Laser Product Warning Label



The eyes must be protected from laser radiation. The patient needs to be instructed to close their eyes when landmarking and the laser light is turned on. Exposing eyes to the laser alignment lights may result in eye injury.

CAUTION: **Do not leave the laser light on after you position the patient.**

Figure 2-13 displays the laser aperture label that is located in the patient bore adjacent to the windows where the laser beams exit.

Figure 2-13 Laser Aperture Warning Label

NOTE: The 3.0T VH/i imaging systems do not contain laser lights. Therefore, these hazards do not apply.

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Cables and Equipment Connections

Various equipment and accessories are used in the MR environment for specific types of examinations that include cables and require connections to the MR system or the patient.

WARNING: **The following general warnings should be followed when using cables and accessory connection equipment:**

- **Use only GE or GE-authorized accessory coils, cables, monitoring and gating equipment that is labeled as compatible with MR equipment. Failure to restrict the use of such equipment not labeled for MR applications may result in patient burns or other injuries.**
- **Use only accessories, coils, and cables that are in good condition. If you suspect that an accessory is not in good condition, discontinue its use and contact your GE Service Engineer.**
- **Auxiliary devices indicated as compatible with MR equipment may still cause patient injuries if the instructions for use are not explicitly followed. Never use equipment unless it is accompanied by the use instructions.**
- **Remove unplugged surface coils or unused accessory devices from the magnet bore; a patient burn can result.**
- **RF can heat non-compatible surface coils/gating cables, damaged surface coils/gating cables, surface coils that are not properly plugged in, and improperly routed cables can cause burns.**
- **The use of cable-connected surface coils, the original peripheral gating probe (consult your GE Service Engineer for questions), or electrocardiogram (ECG) gating accessories for patient scanning can result in localized heating, leading to a burn or fire if proper scan preparation is not followed. The cables often extend into the high intensity region of the RF field and it is possible that induced electrical currents in the cables may cause arcing.**
- **Always bring the cable directly out of the magnet bore with no slack. Place cables under the cushion whenever possible to separate the cable from the patient.**
- **Keep the length of cable in the bore to a minimum. Avoid bending the cable 180° and route the cables out of the bore in the most direct way.**
- **Route cables through the center of the magnet bore. Place cables under the cushion whenever possible to separate the cable from the patient. Routing near the sides of the bore increases the likelihood of cable heating (from induced currents).**
- **Do not cross or loop cables. Arcing and patient burns could result.**

In addition to the warnings above, there are specific warnings related to Cardiac Gating Equipment and Accessory Coils you need to understand to maintain a safe MR environment.

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Cardiac Gating Equipment

Electrocardiogram (ECG) gating may be performed in an MR environment to monitor critically ill patients or for an MR triggering technique to synchronize the MR scan acquisition with the patient's heart beat. Only MR-safe electrodes should be used in the MR environment to ensure patient safety. The ECG triggering feature on the system should only to be used for cardiac gating and must not be used for patient monitoring. When using an ECG triggered MR technique, it is important to use only GE recommended disposable electrodes and compatible leads.

WARNING: Observe the following warnings when using ECG or peripheral gating:

- **The MR cardiac gating feature is intended for use solely in acquiring MR images using cardiac gating/triggering, not for physiological monitoring.**
- **Do not use monitoring equipment when conductors are in the bore and touching the patient; burns can result.**
- **Do not use leads with broken shields or exposed conductors. Only use accessories in good condition. If you suspect that an accessory is not in good condition, discontinue its use and contact your GE Service Engineer.**
- **Check to see that the cardiac or peripheral gating cable does not pass under or near the surface coil or surface coil cable.**
- **Check to see that only the peripheral gating sensor touches the patient. Keep cables from coming in contact with the patient.**
- **Do not use equipment that has not been specifically tested and approved for use in the environment of a MR system.**
- **Physiological monitoring and sensing devices should be used solely under the operators direction and it is their responsibility to ensure patient safety.**

Accessory Coils

It is important you familiarize yourself with the operating instructions for each accessory coil used in your MR environment. Follow the recommended guidelines and precautions by the manufacturer.

WARNING: Observe the following warnings when using surface coils:

- **Do not use surface coils with exposed coils or damaged insulation. Skin contact with metal conductors can cause burns.**
- **Do not allow the surface coil cable to touch the patient; patient burns can result. Use a thermal resistant material or pad to keep the cable from touching the patient.**
- **When using the 1.5T Breast Coil, make sure the patient's back and arms do not touch the magnet bore. Use thermal resistant material or padding between the patient**

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and the magnet to prevent burns that could be caused by patient contact with the interior of the magnet bore.

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Clinical Screening

To avoid potential health hazards in the MR environment, personnel and patient screening procedures should be established in your imaging facility. Every person working or entering the magnet room or adjacent rooms with a magnetic field needs to be instructed about the dangers. This should include all MR workers, maintenance, service, and cleaning personnel, as well as the local fire station team.

All MR workers need careful assessment prior to engaging in operation of the MR system. Additionally, all patients undergoing the MR examination need careful assessment prior to the procedure. Screening helps identify anything that might create a health risk or interfere with MR imaging. It also assists you in determining if the patient has any specific needs or limitations. Additionally, if another person accompanies the patient undergoing the MR examination, they should be screened and managed in the same manner as the patient. The aim of screening is to safely obtain high-quality images so an accurate diagnosis can be made. Maintaining a controlled and safe environment can be achieved by careful questioning patients, visitors/family members and all personnel.

IMPORTANT!: Screen each patient thoroughly for pertinent medical history and conditions that contraindicate scanning before initializing an examination. If proper scanning can not be performed, postpone MR examinations until the screening can be completed.

A documented screening procedure should be followed by a review of the completed form and a verbal conversation to verify the form information and provide the patient time to express his or her questions or concerns. The review and discussion should be conducted by MR safety-trained personnel to ensure there is no miscommunication about the MR safety issues.

A written screening form must be completed each time a patient is to have an MR examination. Even if the patient undergoing previous MR examinations and/or has completed the screening form previously, does not assure the patient another safe examination.

Screening Form

A comprehensive, printed screening form should be used to assess the patient and document the information. In the Safety Reference Forms appendix, an example Patient Screening Form is provided. This form can be customized for your MR suite. The example form consists of three sections:

- Section 1: General Information
- Section 2: Hazardous Items Checklist
- Section 3: Magnet Room Pre-Entry Checklist

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General Information

Section 1 of the patient screening form contains general information concerning patient demographics and the patient's medical and work history. Relevant patient-related information is valuable for obtaining current medical conditions and information on prior diagnostic studies that may be helpful in evaluating the patient's state.

Determining the patient's work history is important for those who work in machine shops or similar environments. These individuals may have small metal slivers or fragments of steel embedded in their eyes. If metal fragments are suspected, the patient should receive an eye examination to detect and remove hazardous materials before scanning.

IMPORTANT!: MR workers may be at risk if previous occupational, recreational or other life experiences have resulted in the accidental implantation of metallic substances, such as slivers or fragments. Page 2 of the patient screening form should be completed by each MR worker to ensure the magnetic field does not pose a hazard to their well-being.

This section also contains questions for female patients concerning matters that may affect the MR examination. Pregnant patients must be identified before they are permitted to undergo an MR procedure. A physician should carefully compare and discuss the risks and benefits of the MR examination versus alternative procedures before scanning to control risk to the patient.

Hazardous Items Checklist

The Hazardous Items Checklist, in section 2 of the patient screening form, is valuable in identifying various implants and devices that may be hazardous to the patient undergoing the MR examination. Items that may produce image artifacts, can also be detected through this section of the screening process.

Some patients, including those suffering from forms of dementia, may not be aware of having a pacemaker. On such patients, palpate the upper torso and abdomen to make sure no pacemaker is present. Pacemakers may be implanted in any of several locations in the chest and abdomen. Pacemaker lead wires are sometimes left in place after removal of the pacing mechanism. The wires can induce current, producing heat during the MR procedure. Therefore, checking for lead wires is also important.

The best way to ensure a metal-free environment is to have patients change into patient examination attire. The checklist also includes items the patient may externally possess. Do not limit your inspection to only ferrous objects alone. Even non-ferrous items, such as gold jewelry, can heat during a scan and burn a patient. Make sure the patient removes all of these objects. In addition, be sure to check small children for safety pins and snaps on diapers or undershirts.

Section 2 of the form also contains an anatomical figure of the human body for patients to mark the location of objects they have inside or on their body. This information can be useful in determining the approximate area of objects that may be hazardous or produce artifacts.

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Magnet Room Pre-Entry Checklist

The last section of the screening form lists metal or magnetic-sensitive items with which the patient can not enter the magnet room. Ensure the patient removes the checked items prior to magnet room entry. Also obtain signatures to document the person who completed and reviewed the screening form.

MR Compatibility

The potential magnetic force on an object must be evaluated before the object is brought into the magnet room. Familiarize yourself with the set of procedures and standards used to evaluate MR compatibility of hand-held, non-electronic equipment to be used in the magnet room found in the Appendix: MR Compatibility Guidelines and Tests.

There are three criteria by which a device can be considered compatible. A device shall be considered MR-compatible: it is MR-safe; the device is safe and effective in a magnetic field; and it is compatible with the scanner if and only if:

1. **MR-safe**: The presence of the device in the magnet room does not pose an increased safety risk to the patient or personnel from three areas of concern:
   - **Magnetic forces**: the device must not be attracted to the magnet with sufficient force as to be a projectile or sufficient torque as to cause personal injury.
   - **Heating from RF or gradient field induction**: the device must not be able to cause burns.
   - **Induced voltage from RF fields**: the device must not cause nerve stimulation.

WARNING: **The attractive force of the magnetic field of the MR system can cause ferrous objects to become projectiles, which can cause serious injury. Post the Security Zone warning sign on the entrance to the magnet room and keep all hazardous objects out of the magnet room.**

2. **Device Safety and Effectiveness**: The device cannot have its functions altered by the magnetic field nor can it cause a safety risk when in the magnetic field. The large static magnetic field, the rapidly changing gradient magnetic fields, and the RF energy may affect a device.

WARNING: **GE shall not be responsible for assessing the proper function of any device. The manufacturer of the device must make arrangements to test the device in the MRI environment. Written certification that the device functions as intended in the MRI environment shall be provided to the user before the device shall be considered MR-compatible.**

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3. **Scanner Compatibility**: The device cannot adversely affect the function of your MR system (SNR, artifacts, image distortion, etc.) in the following areas:
    - **Static magnetic field homogeneity**: the device must not cause significant changes in the shim.
    - **RF interference**: the device must not distort the RF transmitted by your MR system, nor can it generate RF in the sensitive, pick-up range of the MR coils.
    - **Dynamic field homogeneity**: the device must not create significant time-dependent field distortions.

The required degree of compatibility depends on the device's usage and the materials used in fabrication. For example, a scalpel made of a non-ferrous metal might be in Zone 3, while one made of ceramic material could be Zone 1. A device is classified as compatible in one of four zones. Refer to Table 2-10 for a description of the zones.

Table 2-10 Compatibility Device Zones

Zone	Description	Typical Devices
1	Suitable for use in the imaging volume, potentially in contact with the patient.	Biopsy needles Endoscopes
2	Suitable for use in the imaging volume, only if the device is at least 5 cm away from any part of the anatomy for which spatial accuracy is required.	Positioners Microscopes
3	Suitable for use in the imaging volume only when scanning is off. Spatial distortion and artifacts occur during scanning.	Majority of MR-safe hand-held surgical instruments
4	Suitable for use in the magnet room when kept more than 1.0 m from isocenter.	Furniture carts

NOTE: Refer to the MR Compatibility Test Guidelines chapter.

DANGER: **Devices compatible at one field strength, such as 1.5T, may not be compatible at another field strength, such as 3.0T. Prior to patient scanning, confirm with the device manufacturer that the device is compatible at the 3.0T field strength.**

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Patient Emergencies

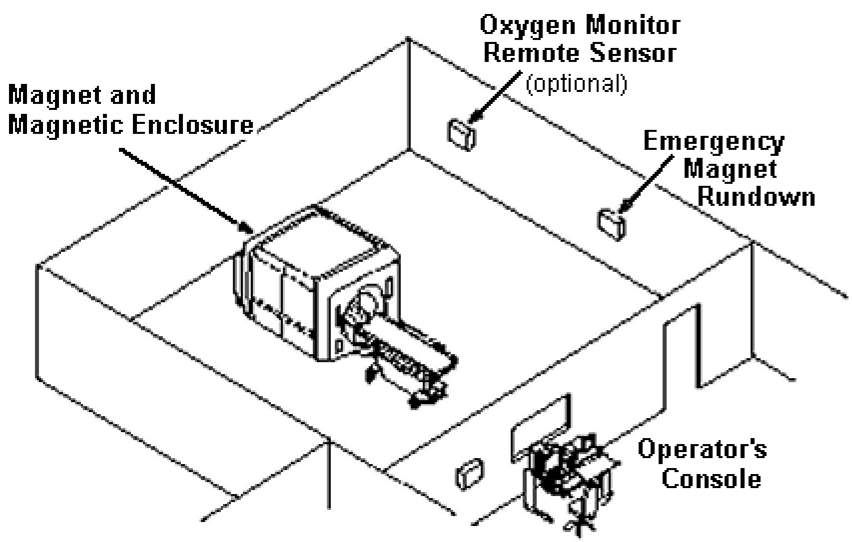
You must become very familiar with the location and proper use of certain emergency buttons and releases should an emergency occur in the MR environment. Advanced planning and being accustomed to your site's procedures and surroundings are necessary to ensure a safe environment.

Before you begin any scanning procedure, explain the use of the Patient Alert System to your patient. Make sure he or she understands its purpose and use. Remember that implants, pacemakers, and ferromagnetic life-support systems cannot be brought into the magnet room.

Be sure to closely monitor patients with a increased potential for cardiac arrest or claustrophobia, or patients who are unconscious or extremely ill. Always maintain visual contact with the patient. Be familiar with your site's predetermined location outside the magnet room where you can transfer patients if it becomes necessary for emergency personnel to intervene.

Figure 2-14 displays a general layout of an MR magnet room. You should always be able to maintain visual contact with your patient from the operator's console.

Figure 2-14 Magnet Room Layout



Patient Alert System

Your MR system has a Patient Alert system that enables the patient to alert you at the console by squeezing a bulb.

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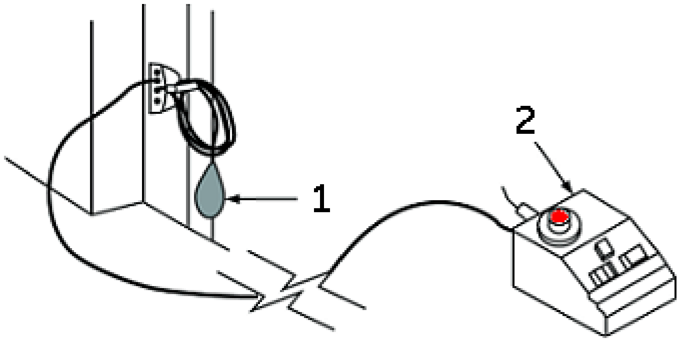
Figure 2-15 Patient Alert System, 1 = Patient Alert bulb, 2 = control box



Squeezing the Patient Alert bulb causes the control box to light up and emit an audible signal. A switch on the control box allows you to set the signal for intermittent or constant light and sound.

Your MR system also has an intercom system that enables you to maintain verbal contact with the patient throughout the examination.

CAUTION: **Provide all patients with the Patient Alert bulb. This can be especially important for procedures that require the concerted attention of the technologist/operator at the MR or Advantage Workstation (AW) operator console, e.g., BrainWave sequences.**

Latex

CAUTION: **THIS PRODUCT CONTAINS NATURAL RUBBER LATEX WHICH MAY CAUSE ALLERGIC REACTIONS.**

The black Patient Alert bulb and the Respiratory bellows contain latex. If the patient is aware of a sensitivity/allergy to latex or if the patient is unsure and concerned about the possibility of an allergic reaction, cover the bulb or the bellows with a towel, cloth, or plastic bag to shield the patient from the latex.

The gray Patient Alert bulb is made of PVC and does not contain natural rubber latex.

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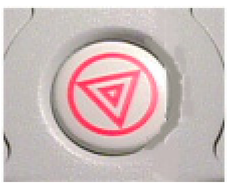
Emergency Stop

The **Emergency Stop** button (Figure 2-16) is located on the keyboard and on both the right and left sides of the magnet enclosure. This function cuts off electrical power from equipment located in the magnet room that may present a hazard to the patient in an emergency situation.

You can press the **Emergency Stop** button to stop a scan in a patient emergency situation. To quickly recover from an Emergency Stop situation, you can press the **Reset** button. You should not be afraid to press the **Emergency Stop** button because it may shut the system down for an extended length of time. This is not required to shut down the magnet coldhead.

Figure 2-16 Emergency Stop Button



The **Emergency Stop** button disables the following systems:

- RF
- Gradient Power Supply
- Magnet Room Unit
- Table and Patient Support Subsystem

WARNING: The Emergency Stop button does not remove the magnetic field, turn off the computer cabinets, operator's console, or camera.

Emergency Off

The **Emergency Off** button (Figure 2-17) is located on the wall next to all computer equipment and next to the MR magnet room doors. It removes ALL electrical power from ALL components of the system, including any power sources from uninterrupted power supply (UPS) devices.

The **Emergency Off** button not only stops a scan in a patient emergency, but also in the event of a serious equipment fault or hazards such as fire/water in the vicinity of the MR equipment. The entire MR system is to be turned OFF except for the static magnetic field and the magnet rundown unit used to shut down the magnetic field.

Figure 2-17 Emergency Off Button

Use this button only in a major emergency in the computer or MR magnet room. For example, use this button when you notice fire, sparks, or loud noises not associated with normal operation of the system.

NOTE: To restore power after emergency stop, the main circuit breaker must be reset before rebooting the system. Always contact a service engineer before restoring power.

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WARNING: The Emergency Off button does not turn off the magnetic field. To avoid personal injury or equipment damage, do not bring any ferromagnetic equipment into the magnet room. Assume that equipment is magnetic unless it is clearly labeled otherwise.

PDU Power Off

The PDU Power Off button turns off power to system components other than the Magnet Rundown Unit, cryogen compressor, scan room lights, chillers and magnet cryogen monitor. Typically, service uses this power off switch and the MR customer does not.

Figure 2-18 PDF power off button

Magnet Rundown

The **Magnet Rundown** (Figure 2-19) operates as follows and is located inside the magnet room:

- Rapid reduction of the magnetic field in about two minutes
- Boil-off of cryogens, accompanied by loud hissing sound
- Several days of down time to replace the cryogens

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Figure 2-19 Magnet Rundown Unit

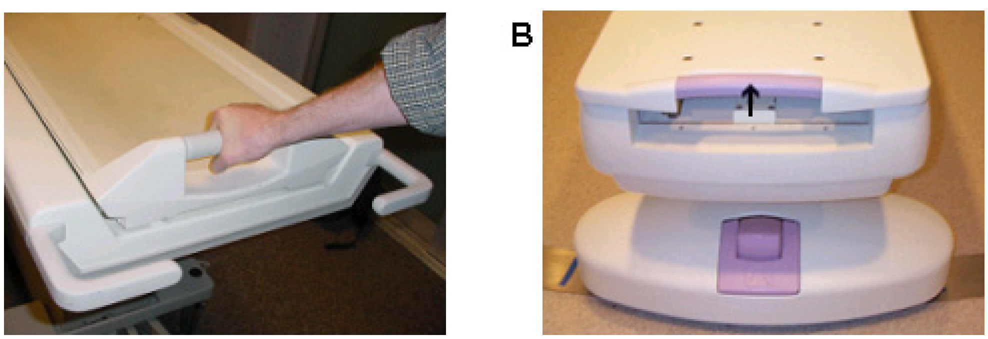
WARNING: The Magnet Rundown should only be used to free someone pinned to the magnet or to remove a large ferromagnetic object captured by the magnetic field when injury to persons is imminent. A controlled magnet rundown should be performed by a GE Service Engineer in non-emergency situations.

Table Transport Emergency Release

In an emergency, the patient cradle can be manually pulled out of the magnet. By squeezing the release handle and pulling the cradle, you can move the patient all the way out to the home position. Figure 2-20A displays the cradle release handle on movable 1.5T and 3.0T tables and Figure 2-20B displays the cradle release handle on 0.7T Open*Speed* tables.

Figure 2-20 Cradle Release Handle

A



B

IMPORTANT!: The patient table of the Open*Speed* system is permanently fixed to the magnet system and has lateral swing movement. A non-ferrous gurney should always be placed outside the magnet room for emergency patient transportation.

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CAUTION: **The Open*Speed* table should only be swung with the patient off the table. If you are pivoting the table for the examination, move the table prior to positioning your patient.**

The 1.5T and 3.0T patient tables can be detached from the magnet system. The tables can also be lowered and raised. In the event that a patient needs emergency medical attention during the scanning session, use the undock pedal (Figure 2-21 and Figure 2-22) or the emergency table lever release for quick transportation of patients outside the magnet room.

Figure 2-21 Undock Pedal at Foot of Table

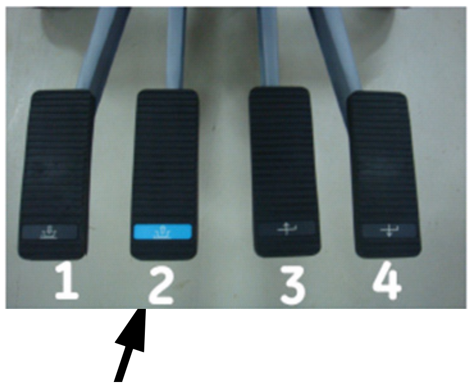

Figure 2-22 Discover undock pedal at foot of table

The Table Transport Emergency Release (Figure 2-23) is used in the event the undock pedal on the patient transport does not function.

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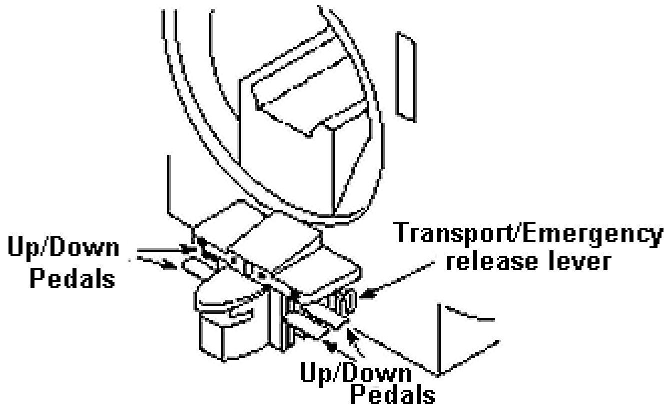
Figure 2-23 Table Transport Emergency Release



To release the transport with the emergency release:

- Make sure the cradle is fully retracted from the magnet bore at the home position.
- Grasp the handle on the red lever and pull straight out to release the table.

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System Maintenance

Maintaining a controlled environment also involves routine preventative maintenance checks by the service engineer and site personnel. Careful planning and diligent upkeep of an MR facility can provide a safe environment for both patients and employees. Your system requires maintenance at specific service intervals in which many of the maintenance checks should be performed by a qualified service engineer. There are several checks you can perform. Be aware of required maintenance and the personnel responsible for meeting each requirement.

GE makes available, on request, such information as circuit diagrams and component lists to assist your technical personnel in the repair of equipment classified by GE as repairable.

WARNING: **Electric shock hazard. No user serviceable parts. Refer service to qualified service personnel.**

General Cleaning

Background cleaning should be done by site personnel (e.g., technologists or housekeeping personnel) unless otherwise indicated in the following maintenance schedules.

IMPORTANT!: Inpsect pads for peeling or cracking. To prevent a biohazard, replace cracked or peeling pads before using.

Cleaning tips:

- To clean most accessories, use nothing stronger than alcohol or a mild soap-and-water solution.
- Use hydrogen peroxide to remove bloodstains.
- Open-cell sponges are coated with canvas to allow better durability and reliability. These sponge covers allow disinfection using only 5.25% sodium hypochlorite diluted between 1:10 and 1:100 with water and 10% bleach solution. Anything else may discolor the fabric.

CAUTION: **To avoid possible damage to equipment, do not use solutions containing amines, strong alkalis, esters, iodine, aromatic or chlorinated hydrocarbons, or ketones. Do not use autoclaves or the industrial washers and dryers found in most hospitals or professional laundry services.**

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Exhaust Fan System

The magnet (RF-shielded) room exhaust fan, vent, and duct system are intended to evacuate the magnet room of cryogenic gas at the MR product specified rate. Over time, the exhaust fan system may become blocked with lint, hair, and other air-borne particles. It is important for personnel safety reasons that the exhaust fan system (vent, exhaust fan, ducts, etc.) be kept clean to make sure the exhaust fan system operates properly and exhausts cryogenic gas to an outside area.

In the unlikely event of a magnet quench or a cryogen gas leak, it is important that this exhaust fan system performs at or above the specified airflow to remove the cryogen gas from the magnet room. The magnet room exhaust fan and air inlet must be sized for a minimum of 1200 CFM (34 m^3/minute) and minimum of room 12 air exchanges per hour. The minimum air flow and air exchange rate for mobile, transportable, and relocatable systems are different from those for fixed sites and varies depending on the type of site. Any blockage or obstruction could prevent the exhaust fan system from providing the required airflow. If the exhaust fan system fails to operate at or above specification, accumulation of dangerous levels of helium or nitrogen within the RF screen room could occur.

It is important that this exhaust system vent be cleaned regularly as part of the normal room cleaning. Regular customer inspection, cleaning, and testing of the exhaust fan system (vent, exhaust fan, ducts, etc.) are needed to make sure all equipment and parts of the system are always in good working order and able to perform to specification. It is recommended the exhaust fan system be cleaned and inspected annually to make sure the specified air flow rate can be met and thus ensures proper performance.

Maintenance Services

The planned maintenance (PM) services prescribed in the PM schedules represent the current manufacturer's recommendations. Specific customer requirements and/or your site environment may necessitate more or less frequent intervals for PM service. An agreement to perform PMs less frequently than these recommendations can be made with the understanding that a reduction of system performance may result.

The PM service schedules in the Maintenance Service Schedules of Appendix C list all the PM procedures and the frequency they should be completed by qualified service personnel. There are different schedules for each system type. There are different schedules for each system type.

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You should perform the maintenance services shown in Table 2-11.

Table 2-11 Operator Services

Item	Required Maintenance	Service Interval
General	Clean	4 months
	Check the table emergency release.	4 months
Patient Cradle and Pads	Check for cleanliness of pads and clean the inside of the cradle.	Daily
Patient Table	Check the table alignment and proper operation.	6 months
Coils, pads, and straps	Clean with non-abrasive cleanser. Clean coil anti-skid pads with water and mild detergent only.	Daily
Coils and coil cables	Check for defects or damage, worn cable or exposed wires.	Daily
Image Quality	Perform quality assurance and functional checks.	As recommended

Cables and Equipment Connections

To avoid trip hazards, you should install yellow and black trip hazard indicator covers over any cables routed on the ground of the equipment room.

Safety Review

Table 2-12 Safety Review

Situation	Procedure
Fire, sparks, a loud noise or other emergency condition in the magnet room not associated with normal operation of the system.	Press an **Emergency Off** button, either in the computer equipment room or at the magnet room door. Remove the patient from the magnet room.
Magnet quench, indicated by a loud noise, warning message, dense white vapor **with vent failure**, helium meter dropping considerably or the tilting of an image on the image screen.	Evacuate the patient and personnel from the magnet room and close the magnet room door. Follow the usual overnight procedure. All helium vapor should automatically be vented outside of the magnet room.

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Situation	Procedure
Magnetic-field emergency, e.g., a person pinned between the magnet and a ferromagnetic object.	Press the **Magnet Rundown** button in the magnet room. Remove the patient from the scan room.
Fire, sparks or a loud noise, indicating a severe system malfunction in the computer equipment room.	Press an **Emergency Off** button, either in the computer equipment room or at the magnet room door. Remove the patient from the magnet room.
Fire or severe condition relating to the power distribution unit (PDU) or service outlets.	Press an **Emergency Off** button, either in the computer equipment room or at the magnet room door. Remove the patient from the magnet room.
Overtemp indicator lights up at the remote power panel (RPP) or at the PDU, and an error message appears on the scan console's System Status Display area.	Remove the patient from the magnet room. Check the PDU vent for obstructions. If the vent is obstructed, or if the overtemp light or message remains on, perform a system shutdown and then press an **Emergency Off** button, either in the computer equipment room or at the magnet room door.
Patient needs medical attention.	Press the **Emergency Stop** button on the console or magnet and remove the patient from the magnet room.
Hydraulic failure of the table.	Make certain the cradle is fully retracted on the transport (the home position) before undocking the transport. Keep all personnel (including patients) away from any spill. Keep patients on the table until safe transfer is possible. Check for oil leaks and if any exist, clean them up to prevent anyone from slipping on the oil. If the table latch is stuck and the table cannot be removed, pull the **Table Transport Emergency Release**. Remove the table from clinical use until it is repaired.
Imaging functions are lost without warning.	Follow your facility's emergency procedures during this type of occurrence.

IMPORTANT!: In all cases, notify a GE Service Engineer as soon as possible.

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How Do I...

This section provides the step-by-step instructions for working safely in a magnetic field environment. Specifically, it describes how to:

- Eliminate Magnet Hazards
  - Protect the Security and Exclusion Zones
  - Screen Patients and Personnel
- Prepare the Patient
- Protect the Patient from RF Burns
- Protect the Patient's Eyes and Ears
- Respond to Emergencies
  - Patient Emergencies
  - Equipment or Environmental Emergencies
  - Magnet Emergencies
  - Quench with Vent Failure
- Check the Cryogen Levels
  - Systems with a Helium Level Meter
  - Systems with a Magnet Monitor Unit
- Handle Contact with Liquid Cryogens

IMPORTANT!: You should understand the following definitions before continuing:

- A substance that is ferromagnetic has a large positive magnetic susceptibility, meaning it is very easily magnetized (example: Iron).
- An item that is ferrous can posses intrinsic magnetic fields and become a projectile in an applied magnetic field (examples: Iron, nickel, and cobalt).

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Eliminate Magnet Hazards

Protect the Security and Exclusion Zones

It is vital to have supervised and controlled access within the MR environment to keep it safe from ferromagnetic items and to guard against accidents, injuries, or damage to MR systems. Keep in mind that even a paperclip inside the bore of the magnet can cause image artifacts or a patient burn. All personnel should be aware of the following important steps.

1. Keep the door to the MR environment and the magnet door closed.
   - The doors should not be held open for other people or propped open.
   - Only essential personnel should be allowed to enter the magnet room.
2. Limit and monitor access to the MR environment and magnet room.
   - Personnel trained in MR safety should be present at all times during the operation of your MR facility to ensure that no unaccompanied or unauthorized individuals are allowed to enter the MR environment or magnet room.
   - Personnel trained in MR safety are also responsible for performing thorough screening of patients and other individuals before allowing them to enter the magnet room.
3. Supervise non-MR personnel when working in the magnet room.
   - Everyone who needs to enter the MR environment on a regular or periodic basis should be educated regarding the potential hazards related to the magnetic field.
4. Prominently display the Security and Exclusion Zone warning signs to make all individuals and patients aware of the risks associated with the MR system.
   - The Security Zone sign must be posted on the entrance to the magnet room.
     - These signs warn patients about the strong magnetic field and stresses the presence that no pacemakers, metallic implants, neurostimulators, or loose objects are allowed.
   - The Exclusion Zone sign must be posted at the 5 gauss boundary.
     - This sign warns against the strong magnetic field and stresses the presence of no pacemakers, metallic implants, or neurostimulators.
5. Test all items for ferromagnetic properties before taking them into the magnet room.
   - Use a hand magnet to test items.
6. Remove ferrous items from the immediate vicinity of the magnet room.
   - This can reduce the chance that someone might carry a ferrous item into the magnet room.
   - Replace ferrous items that must remain in the vicinity of the magnet room with non-ferrous versions whenever possible.

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7. Tag ferrous items that remain at the facility so that all personnel know the item cannot be taken into the magnet room.
   - Tag all ferrous items with the same label to be consistent in identifying items that are not to be in the magnet room.
8. Do a pocket check before entering the magnet room.
   - Check for loose metal objects, such as keys, and remove.
9. Keep the magnet door in sight at all times.
   - When working in the magnet room, do not stand between the door and the magnet or turn your back to the door.
10. Do not turn your back on the patient or anyone else in the magnet room.

Quick Steps: Eliminate Magnet Hazards – Protect the Security and Exclusion Zones
1. Keep the door to the MR environment and the magnet door closed. 2. Limit and monitor access to the MR environment and magnet room. 3. Supervise non-MR personnel when working in the magnet room. 4. Prominently display the Security and Exclusion Zone warning signs to make all individuals and patients aware of the risks associated with the MR system. 5. Test all items for ferromagnetic properties before taking them into the magnet room. 6. Remove ferrous items from the immediate vicinity of the magnet room. 7. Tag ferrous items that remain at the facility so that all personnel know the item cannot be taken into the magnet room. 8. Do a pocket check before entering the magnet room. 9. Keep the magnet door in sight at all times. 10. Do not turn your back on the patient or anyone else in the magnet room. 11. Ensure all MR personnel are trained in the protection of the Safety Exclusion Zones.

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Eliminate Magnet Hazards

Screen Patients and Personnel

For your safety and the safety of the patient, an MR safety-trained health care worker at your facility should carefully screen for hazards before patients and personnel enter the Exclusion Zone. All personnel must be aware of and comply with your facility's screening procedure.

1. Use a Patient Screening form routinely before bringing patients or other personnel into the Exclusion Zone.
   - MR is contraindicated (not indicated for use) in certain patients, e.g., patients with cardiac pacemakers, electrically activated cardiac catheters, ferrous implants or intracranial aneurysm clips, etc.
   - Every patient, individual, and employee must be carefully screened prior to admission to the magnetic field. Refer to the Patient Screening Form.
2. Review the completed screening form and evaluate the individual prior to entry.
   - Identify circumstances that contraindicate admission to the Exclusion Zone or items that need to be removed before entering the Security Zone.
   - In addition to safety issues, metal objects or materials containing metal distort the magnetic field and detract from the image quality.
3. Discuss the items on the screening form with the patient or other individual.
   - Verbally interview the patient to verify the information on the form and ensure the patient understands each question he/she is answering.
   - Allow discussion of any question or concern that the patient may have.
4. Examine all patients with diapers or incontinence products, including adults, should have dry diapers on prior to the start of the scan.
5. Examine or X-Ray patients who are at risk for metal eye slivers.
   - Serious injury may occur as a result of movement or heating of the metallic foreign body as it is attracted by the magnetic field of the MR system.
   - Follow your departmental clinical screening policy.
6. Require that patients change clothes.
   - Provide clothes without metallic fasteners and pockets.
   - Patients should not wear shoes into the magnet room as they may have collected metal on the soles.
7. Instruct the patient to wash off non-permanent make-up.
   - Follow the precautions for patients with permanent make-up such as permanent eyeliner, which can cause tissue heating.

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8. Keep metal out of the bore.
   - A metal-free bore prevents burns and image artifacts.

Quick Steps: Eliminate Magnet Hazards – Screen Patients and Personnel
1. Use a Patient Screening form routinely before bringing patients or other personnel into the Exclusion Zone. 2. Review the completed screening form and evaluate the individual prior to entry. 3. Discuss the items on the screening form with the patient or other individual. 4. Examine all patients with diapers or incontinence products, including adults, should have dry diapers on prior to the start of the scan. 5. Examine or X-Ray patients who are at risk for metal eye slivers. 6. Require that patients change clothes. 7. Instruct the patient to wash off non-permanent make-up. 8. Keep metal out of the bore.

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Prepare the Patient

Some patients undergoing an MR procedure may experience feelings of fear, anxiety, or claustrophobia. The following techniques may help reduce or eliminate these sentiments. Use the following techniques for all patients, even if your patient does not exhibit signs of fear or claustrophobia.

Larger patients can feel anxious due to a confined feeling in the magnet. Degradation of image quality can also occur. Verify that the patient's weight does not exceed the table weight limit of 350 pounds or 500 pounds (weight limits vary dependent on system). Consult your system operator manual for details.

1. Patients must be thoroughly screened prior to preparing them for scanning to make sure they are MR-safe.
   - Screen for pertinent medical history and conditions that contraindicate scanning.
   - If proper screening cannot be performed, postpone the MR examination until screening can be done.
2. Determine scan protocol and enter the patient's information in advance.
   - This saves time during the preparation for the procedure so the patient is not left waiting for the examination to begin.
3. Provide the patient an information booklet to read.
   - Educating the patient concerning specific aspects of the MR examination is an effective way to prepare for the situation and explain what is about to happen.
4. Have the patient use the restroom prior to the examination.
   - Fewer interruptions during the scanning procedure can help you stay on schedule and keep the patient focused on holding still during the examination.
5. Examine all patients with diapers or incontinence products, including adults, to make sure the patient has dry diapers and dry clothing on prior to the start of the scan.
6. Discuss the procedure with the patient.
   - The length of the examination
   - What can be seen during the examination
   - What can be heard during the examination
   - What can be felt during the examination

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7. Transfer the patient to the MR table.
    - If the patient is using a wheelchair or gurney, lock the wheels.
    - If the patient transport table is undocked from the system, complete the following before you transfer the patient to the MR table:
        - For the Discovery MR system tables, press the table lock foot pedal to lock all wheels to prevent the table from moving during patient transfer. Once the patient is securely on the table with the rails up, depress the table foot lock pedal again, to disengage the locks and to move the table.
        - For all other transportable table configurations, **depress the pedal on *every* wheel to lock it so that the table does not move.** Once the patient is securely on the table with the rails up, **depress the pedal on each wheel again to release the lock.**
    - On either MR transportable table, engaging the steering lock can facilitate directing the table to the dock.
8. If the patient was transported into the magnet room via the MR table and the IV pole connected to the table is in use, once the table is docked, replace the MR table's IV pole with a non-ferrous free-standing IV pole.

CAUTION: Do not move the patient into the magnet with the MR table's IV pole in use. To avoid any pinch points from the MR table's IV pole, remove the IV pole from the table, store it, and use a non-ferrous free standing IV pole.

9. Let the patient see the MR system while you explain the features of the bore.
    - Soft lighting
    - Good ventilation
    - A microphone and speaker to enable the patient to hear and be heard at all times
10. Demonstrate the use and function of the Patient Alert System.
    - This system is patient-activated and allow the patient to signal for assistance during a scan.
11. Explain the use of straps.
    - Insert the straps into the mounting track on the table and wrap the straps around the patient. Straps are to stabilize, not restrain the patient.

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Figure 2-24 Strap inserted in table mounting track

- If you find it difficult to insert the strap into the track or move it in the track, then slightly bend the strap base on a hard surface.

Figure 2-25 Bend the strap base

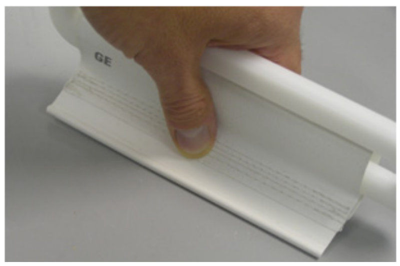

Figure 2-26 Bent strap that more easily slides in the mounting track

12. Ensure the patient is comfortable.
    - Use sponges and wedges to relieve pressure points and support the body in the correct position.
    - Ask if a blanket is needed while being aware that once the scan begins, a blanket may increase patient warming.
13. Explain the need for hearing protection.
    - Use recommended earplugs (>/= 29dB NRR) to minimize the noise from the gradient magnetic field.
    - Alternatively, use recommended MR-compatible headphones (>/= 29dB NRR) to provide relaxing music to the patient and minimize the noise.

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14. Stress the need for cooperation to attain a diagnostic study.
    - It is extremely important the patient not move during the examination.
15. Stay in constant oral and visual communication with the patient throughout the examination.
    - Some patients may require the physical presence of an family member or nurse in the magnet room.

Quick Steps: Prepare the Patient

1. Patients must be thoroughly screened prior to preparing them for scanning to make sure they are MR-safe.
2. Determine scan protocol and enter the patient's information in advance.
3. Provide the patient an information booklet to read.
4. Have the patient use the restroom prior to the examination.
5. Examine all patients with diapers or incontinence products, including adults, to make sure the patient has dry diapers on prior to the start of the scan.
6. Discuss the procedure with the patient.
7. If the patient was transported into the magnet room via the MR table and the IV pole connected to the table is in use, once the table is docked, replace the MR table's IV pole with a non-ferrous free-standing IV pole.
8. Demonstrate the use and function of the Patient Alert System.
9. Explain the use of straps.
10. Ensure the patient is comfortable.
11. Explain the need for hearing protection.
12. Stress the need for cooperation to attain a diagnostic study.
13. Stay in constant oral and visual communication with the patient throughout the examination.

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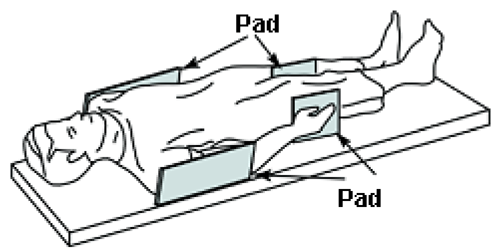
Protect the Patient from RF Burns

All personnel should be aware of several factors to protect a patient from burns and peripheral nerve stimulation. The following steps should be observed by all personnel that position patients for scanning.

1. Remove any accessory devices from the bore of the magnet that are not required for the procedure.
    - This includes any unused electrically conductive materials such as surface coils, cables, etc.
2. Examine all patients with diapers or incontinence products, including adults, to make sure the patient has dry diapers on prior to the start of the scan.
3. Position the patient to prevent direct contact between the patient's skin and the bore of the magnet or an RF surface coil.
    - Use additional pads to immobilize the patient and make them comfortable.



    - Preventing patient warming is one of the most important safety measures you must take into consideration as you prepare a patient for an MR exam. Appropriate RF padding and proper patient positioning are the most effective means of preventing injury related to RF heating. The following are a few golden rules to remember as you position and pad your patients:
        - Only use GE-approved RF padding.
        - Use non-conductive padding that is at least 0.25 inches (0.635 cm) thick between the patient's skin and the magnet bore.
        - Appropriate padding must be used *EVERY* time without exception
        - Sheets and gowns are not a substitute for approved RF padding.
        - Never allow your patient’s skin to come in direct contact with the scanner bore or any surface coil or cable.
        - Never allow skin-to-skin contact.
        - If a patient does not fit in the MR scanner bore with the required padding, another modality should be used to scan the patient.
    - While some of these rules may seem a little tough to follow at times, remember that RF injury, which can in extreme cases include burns such as the one you see below, can happen very quickly and your patient may not have time to warn you in time to prevent an injury

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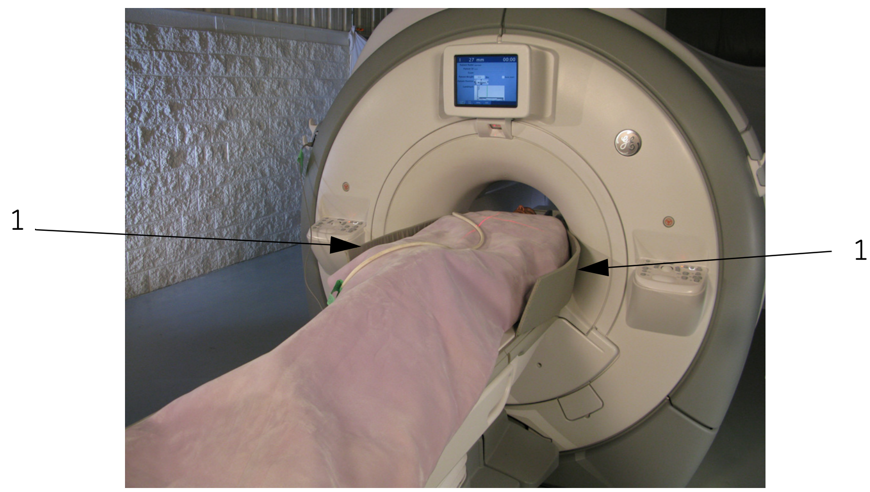
Figure 2-27 Elbow RF burn.

- The following are a tips that will assist you in properly positioning and applying RF padding to your patients. Should you need more information on prevention of patient warming than what is provided here, refer to your surface coil and refer to Tissue Heating in this manual. If you need help beyond the documentation please do not hesitate to reach out to your local Applications Specialists.
- **Whole body padding**
  - An important consideration when padding your patients is that you will need to double check the position of the pads once the patient is in the bore. Table movement may dislodge padding and expose skin to the scanner bore.

Figure 2-28 Padding between patient and bore. 1 = bore pads



  - Notice that padding is positioned not only at the patient's sides to prevent their arms from touching the bore, but that padding is also placed between the hands and thighs and between knees and ankles to prevent forming conductive loops.

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Figure 2-29 Patient padding

- **Surface coil padding.** Padding with a surface coil presents different challenges from a patient RF padding perspective.
  - First rule of thumb is to remember to use all manufacturer provided padding to prevent motion and the patient's skin from coming in contact with the coil, and to also use additional padding if appropriate to secure an opposing extremity to prevent contact with the coil which could also lead to burns or motion artifacts.
  - Just as with the whole body RF padding demonstration, you'll need to make certain that the patient's skin does not come into contact with the scanner bore and that padding is placed between the hands and thighs to prevent conductive loops.

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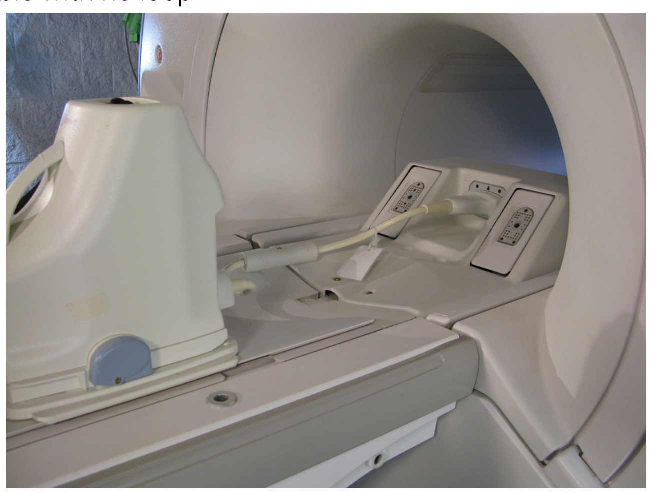
Figure 2-30 Extremity padding

- A final safety consideration for surface coils is to ensure that the patient does not come into contact with the coil cable, therefore you may need to use additional RF padding to protect the patient.
- Care should also be taken to ensure the cable is not looped in the bore and that it is routed down the center of the scanner bore.

Figure 2-31 Coil cable with no loop



- **Cardiac coil padding**. Follow your basic padding recommendations to prevent contact with the scanner bore and prevent conductive anatomical loops, but there are a couple of additional steps you'll need to take to ensure patient safety
  - The cardiac coil does not require additional RF padding to be placed between the patient and the anterior coil component, but you should use the manufacturers pad on the posterior component of the coil for patient RF protection. You should also cover the patient with their gown before placing the anterior component of the coil and make certain both the anterior and posterior elements are in alignment.

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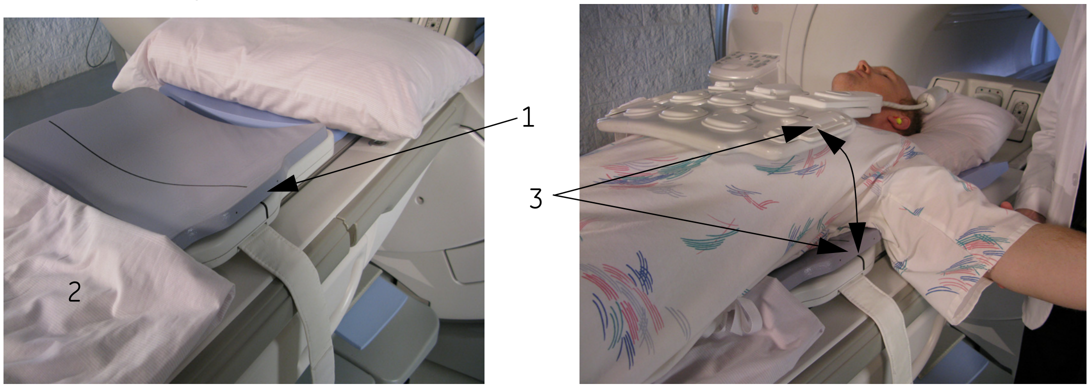
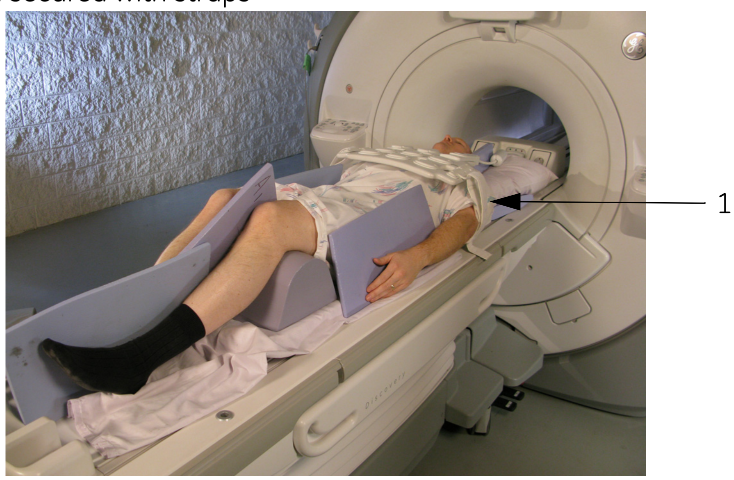
Figure 2-32 1 = coil pad aligned with coil, 2 = sheet to cover pad, 3 = anterior and posterior coil elements aligned



– Secure the coil snugly, but comfortably with the straps.

Figure 2-33 1 = Coil secured with straps



– As is the case of all surface coils ensure that the cables do not come in contact with the patient and that they are not looped and routed down the center of the bore. As you can see there is significantly more cable that we need to isolate from the patient, so be sure to use as much padding as needed.

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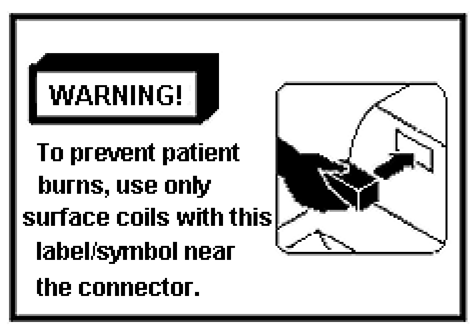
Figure 2-34 1 = Pad placed between cable and patient skin

- If you're using the cardiac coil, it's likely you're also using the ECG leads and cable. The rules for the ECG cable are the same as the coil cable. Route the ECG cable down the center of the bore, do not loop the ECG cable and do not allow it to come in contact with the coil cable.

Figure 2-35 1 = ECG cable with no loops

4. Only use approved RF coils that are not damaged.
   - Labels such as the one in the figure to the right, provide warnings about working with RF coils.
   - Before using the coil, check the integrity of the electrical insulation of the components or accessories of the device.



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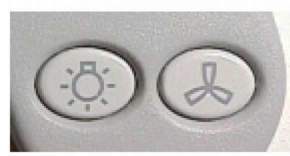
5. Keep electrically conductive material that must remain in the magnet bore from directly contacting the patient by placing insulation between the conductive material and the patient.
   - Place a clean cotton sheet over the coil and comfort pad so the patient's skin does not come in contact with the coil or the comfort pad.
6. Position RF cables down the center and directly out of the bore, without looping or crossing the cables.
   - Use the cable holders provided to route the cables so there are no loops in any cables in the magnet. Cable holders are located on both side of the cradle near the edges. Use the appropriate gating cable for surface coil imaging.
   - Use only MR system recommended monitoring equipment, ECG leads, wires, electrodes, and other components and accessories.
   - Follow all instructions for the proper operation of physiologic monitoring or other equipment provided by the manufacturer of the device.
7. Test the patient intercom.
   - Make sure that the patient can hear you and you can hear the patient.
8. Enter the correct patient weight.
   - Correct weight entries maximize performance and help prevent excessive RF exposure.
9. Turn on the bore light and fan.
   - Lights turned on inside the bore can help alleviate feelings of claustrophobia.
   - A fan inside the magnet bore provides adequate air movement for the patient. Keep the fan on at all times.



10. Show the patient how to use the Patient Alert System.
    - Patients experiencing uneasiness or concern can squeeze the Patient Alert bulb.
    - When the Patient Alert bulb is squeezed, an alarm emits a signal to you.
11. Respond to bore temperature direction messages throughout the procedure.
    - The temperature messages are located in the message window on your console.
    - If the patient reports feeling warm, discontinue the procedure.

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Quick Steps: Protect the Patient from RF Burns

1. Remove any accessory devices from the bore of the magnet that are not required for the procedure.
2. Examine all patients with diapers or incontinence products, including adults, to make sure the patient has dry diapers on prior to the start of the scan.
3. Position the patient to prevent direct contact between the patient's skin and the bore of the magnet or an RF surface coil.
4. Only use approved RF coils that are not damaged.
5. Keep electrically conductive material that must remain in the magnet bore from directly contacting the patient by placing insulation between the conductive material and the patient.
6. Position RF cables down the center and directly out of the bore, without looping or crossing the cables.
7. Test the patient intercom.
8. Enter the correct patient weight.
9. Turn on the bore light and fan.
10. Show the patient how to use the Patient Alert System.
11. Respond to bore temperature direction messages throughout the procedure.

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Protect the Patient's Eyes and Ears

During scanning gradients produce noise that can exceed 99 dBA in the bore. Hearing protection is required to prevent hearing impairment. Patients must close his or her eyes when the alignment light is on during positioning. Follow these guidelines to ensure proper eye and ear protection for your patient.

1. Provide the patient with hearing protection.
   - Earplugs or a headphone system with stereo music. For details, see Acoustic Noise
     - Earplugs reduce the intensity of the sound, while allowing your patient to hear normal conversations.
     - Headphone systems soften acoustic noise, but may impede verbal communication with patients while the system is operating.
   - Refer to Table 2-7 for available types of disposable ear protection.

WARNING: Hearing protection is required for all people in the magnet room during a scan to prevent hearing impairment. Acoustic levels may exceed 99 dB(A).

IMPORTANT!: Since the acoustic noise of the Open*Speed* system does not exceed 92.2 dBA, ear protection is not required, but is recommended.

2. Make sure that the hearing protection device is worn properly.
   - Earplugs should be comfortable for the patient and inserted fully. Pliable earplugs compress when they are rolled between the fingers and conform to the ear after they are inserted.
   - The headphone system should be audible and comfortable for the patient.
3. Instruct the patient to close his or her eyes when the alignment light is on.
   - The Laser Alignment Lights for patient positioning can cause eye injury.
   - 3.0T VH/i systems do not use semiconductor laser alignment lights for patient landmarking, therefore this only applies to 0.7T, 1.5T. and 3.0T EXCITE systems.

CAUTION: Turn off the laser light after positioning the patient.

Quick Steps: Protect the Patient's Eyes and Ears

1. Provide the patient with hearing protection.
2. Make sure that the hearing protection device is worn properly.
3. Instruct the patient to close his or her eyes when the alignment light is on.

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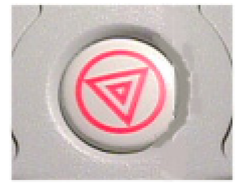
Respond to Emergencies

Patient Emergencies

Dealing with patient emergencies requires special planning in the MR environment because of the magnetic field. Certain equipment used for resuscitation does not function in a magnetic field, and ferrous items can become projectiles. If a patient needs emergency medical attention during the scanning session, follow these guidelines:

1. Press **Emergency Stop** on the operator's console or magnet enclosure.



   - The scan aborts.
   - The power disables the patient-handling and scan-related equipment.
2. Notify emergency personnel, if necessary.
   - Since ferromagnetic life support and related equipment cannot be brought into the magnet room, it must await the patient outside the magnet room.
3. Squeeze the cradle release handle and pull the cradle all the way out to home.
   - The cradle releases. If the cradle does not release, give it a strong tug.
4. If you have an Open*Speed* system, continue with step 5. Otherwise, move the side rails of the transport into the vertical position then, continue with step 7.
   - The side rails help prevent the patient from rolling off the side of the patient table and provide a handrail for the patient to hold.
   - The side rails also provide you a handrail as you guide the table during movement.
5. Move a **non-ferrous** gurney next to the magnet table.
6. Move the patient from the table to the gurney and continue with step 8.
7. Undock the transport.
   - Use the undock pedal at the foot of the table or emergency release handle at the head of the table.
8. Keep the patient on the gurney or transport and remove the patient from the magnet room as quickly as possible.
   - It is important to have an assigned emergency area outside of the magnet room where you can take a patient so that the emergency team can use the necessary equipment.
9. Follow your facility's emergency protocol.

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WARNING: The Emergency Stop button does not remove the magnetic field, turn off the computer cabinets, operator's console, or camera.

Quick Steps: Respond to Emergencies – Patient Emergencies

1. Press Emergency Stop on the operator's console or magnet enclosure.
2. Notify emergency personnel, if necessary.
3. Squeeze the cradle release handle and pull the cradle all the way out to home.
4. If you have an OpenSpeed system, continue with step 5. Otherwise, move the side rails of the transport into the vertical position then, continue with step 7.
5. Move a non-ferrous gurney next to the magnet table.
6. Move the patient from the table to the gurney and continue with step 8.
7. Undock the transport.
8. Keep the patient on the gurney or transport and remove the patient from the magnet room as quickly as possible.
9. Follow your facility's emergency protocol.

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Respond to Emergencies

Equipment or Environmental Emergencies

If you experience a serious equipment fault (system overheating, smoke, or odor associated with the system) or hazards such as fire/water in the vicinity of the MR equipment, you may need to perform a system shutdown with the **Emergency Off** button. The entire MR system is turns OFF with this button, except for the static magnetic field and the Magnet Rundown Unit used to shutdown the magnetic field.

Use this procedure to perform an emergency shutdown on your system during an equipment or environmental emergency.

1. Press the **Emergency Off** button located on the wall next to the computer equipment or next to the magnet room door.
   - This stops power to the magnet room, removing all electrical power from all components of the system.
   - This button also removes any power sources from UPS devices.
2. Evacuate the patient.
   - Follow the guidelines in the Patient Emergencies procedure.
3. Evacuate the MR suite.
   - Follow your facility's emergency protocol.
4. Call the fire department, if appropriate.
   - When the fire department arrives, evaluate the need for an emergency MRI magnet quench.
   - If the firefighters need to take ferromagnetic equipment into the MRI magnet room, quench the magnet.
5. Contact a service engineer before restoring power.
   - To restore power after an **Emergency Off**, the main circuit breaker must be reset before rebooting the system.
6. After service has examined the system, document the correct cause of the emergency.
   - Keeping the events documented allows you to reference this information in the future and may help prevent similar incidents.

WARNING: **The Emergency Off button does not turn off the magnetic field. To avoid personal injury or equipment damage, do not bring any ferromagnetic equipment into the magnet room. Assume that equipment is magnetic unless it is clearly labeled otherwise.**

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Quick Steps: Respond to Emergencies – Equipment or Environmental Emergencies
1. Press the Emergency Off button located on the wall next to the computer equipment or next to the magnet room door. 2. Evacuate the patient. 3. Evacuate the MR suite. 4. Call the fire department, if appropriate. 5. Contact a service engineer before restoring power. 6. After service has examined the system, document the correct cause of the emergency.

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Respond to Emergencies

Magnet Emergencies

In addition to patient and equipment emergencies, magnetic field emergencies can also occur. Examples of magnetic field emergencies include instances where the presence of the magnetic field may cause injury or harm, if someone is pinned between the magnet and ferromagnetic object. All personnel should be familiar with how to respond to magnet emergencies.

A magnet rundown results in several days of downtime and may jeopardize your magnet. Your facility needs to define the specific circumstances that would require a magnet rundown so that no one makes an expensive mistake.

Use this procedure to perform an emergency magnet rundown on your system during a magnetic field emergency.

1. Press the **Magnet Rundown** button located in the magnet room.
   - This results in a rapid reduction of the magnetic field in about two minutes.
   - There is a boil-off of cryogens, accompanied by loud crackling and hissing sounds.
   - Expect several days of downtime to replace the cryogens.
2. Evacuate the patient and all other personnel from the magnet room.
   - Follow the guidelines in the Patient Emergencies procedure.
3. Evacuate the magnet room.
   - Follow your facility's emergency protocol.

IMPORTANT!: Plan and rehearse for a magnet rundown that results in venting of cryogen vapor into the magnet room.

Do not activate the Magnet Rundown switch during practice.

WARNING: **The Magnet Rundown should only be used to free someone pinned to the magnet or to remove a large ferromagnetic object captured by the magnetic field when injury to persons is imminent. A controlled magnet rundown should be performed by a GE Service Engineer in non-emergency situations.**

Quick Steps: Respond to Emergencies – Magnet Emergencies
1. Press the Magnet Rundown button located in the magnet room. 2. Evacuate the patient and all other personnel from the magnet room. 3. Evacuate the magnet room.

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Respond to Emergencies

Quench with Vent Failure

A magnet quench can result in the release of cryogen vapor into the magnet room <u>if the vent fails</u>; white clouds of vapor appear in the magnet room. Cryogens released during a quench can cause asphyxiation, frostbite, or injuries due to panic. Magnet quenches are indicated by a loud noise, warning message, or the tilting of an image on the image screen. It is critical to have a well-planned method to quickly remove the patient and all personnel from the magnet room if a quench should occur.

Use this procedure in case of a sudden cryogen release into the magnet room.

1. Do not panic.
   - Staying calm helps you remain focused so you are able to safely remember and follow your planned method of action.
2. Using the intercom, tell the patient to stay calm and remain on the table.
   - Tell the patient that someone will be in shortly to offer assistance.
3. Turn on the magnet room exhaust fan.
   - Some systems vent automatically and there is no fan to turn on.
4. Prop open the door between the operator room and hallway or if in a mobile unit, open the door to the outside.
   - This promotes air circulation.
5. Prop open the door to the magnet room.
   - If helium is venting in the room, the magnet room door may not open.
   - If the door cannot be opened, slide open the window between the console and magnet rooms, or if needed, break the window to the magnet room to relieve pressure.
6. Enter the magnet room and help the patient exit.
   - If a gurney or wheelchair is needed to remove the patient, make sure it is a non-ferrous type.
   - When exiting, stay near the floor where the oxygen will be and immediately exit the magnet room.
7. Evacuate all personnel from the area until the air is restored to normal.

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Quick Steps: Respond to Emergencies – Quench with Vent Failure
1. Do not panic. 2. Using the intercom, tell the patient to stay calm and remain on the table. 3. Turn on the magnet room exhaust fan. 4. Prop open the door between the operator room and hallway or if in a mobile unit, open the door to the outside. 5. Prop open the door to the magnet room. 6. Enter the magnet room and help the patient exit. 7. Evacuate all personnel from the area until the air is restored to normal.

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Check the Cryogen Levels

Systems with a Helium Level Meter

It is very important to check and record the helium level. Sufficient helium is necessary to avoid accidental quench. If the helium levels fall below 60%, or the level your service engineer says is acceptable, contact a service engineer immediately.

Use this procedure to check cryogen levels.

1. Turn on the **Helium Meter** power switch.
   - This switch is located at the system cabinet.
2. Press **Update** on the system cabinet.
   - An updated reading of the helium level posts.
3. Record the percent of helium reading.
   - Keep a logbook notate readings daily.
   - It is crucial that cryogenic systems be checked regularly to be sure they are properly functioning.
4. Turn off the **Helium Meter** power switch when complete.

Quick Steps: Check the Cryogen Levels – Systems with a Helium Level Meter
1. Turn on the Helium Meter power switch. 2. Press Update on the system cabinet. 3. Record the percent of helium reading. 4. Turn off the Helium Meter power switch when complete.

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Check the Cryogen Levels

Systems with a Magnet Monitor Unit

It is very important to check and record the helium level. Sufficient helium is necessary to avoid accidental quench. If the helium falls below 60%, or the level your service engineer says is acceptable, contact a service engineer immediately.

Use this procedure to check cryogen levels of a system with a Magnet Monitor Unit.

1. Locate the magnet monitor, which is typically in the console room. Push the button on the Magnet Monitor Unit and hold it for approximately 10 seconds.
    - An updated reading of the helium level posts.
2. Record the displayed value when the LHe% light is illuminated.
    - Keep a logbook notate readings daily.
    - It is crucial that cryogenic systems be checked regularly to be sure they are properly functioning.

Quick Steps: Check the Cryogen Levels – Systems with a Magnet Monitor Unit
1. Locate the magnet monitor, which is typically in the console room. Push the button on the Magnet Monitor Unit and hold it for approximately 10 seconds. 2. Record the displayed value when the LHe% light is illuminated.

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Handle Contact with Liquid Cryogens

Since the cryogen gasses are odorless, tasteless, and colorless, it is particularly important to have specific procedures in place to avoid frostbite if there is ever an accident in which the liquid or gaseous cryogen contacts human skin. Such procedures should be established and made readily available to all personnel.

Use this procedure immediately in the unlikely event that someone comes in contact with a liquid cryogen.

1. Promptly flush the area with large volumes of tepid water.
    - Tepid water is 105° to 111°F or 41° to 46°C.
    - For cold burns, immerse affected area in tepid water for at least 15 minutes.
2. Calm the victim.
3. Avoid aggravating the injury.
    - Do not rub or massage the affected parts of the body.
4. Cover the area with a sterile dressing.
    - You may also use a clean sheet if the exposed area is large.
    - This protects the area from further trauma.
5. Consult a physician immediately.
    - Maintain the affected area at normal body temperatures until a physician arrives.

Quick Steps: Handle Contact with Liquid Cryogens
1. Promptly flush the area with large volumes of tepid water. 2. Calm the victim. 3. Avoid aggravating the injury. 4. Cover the area with a sterile dressing. 5. Consult a physician immediately.

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Chapter 3

MR Compatibility Test Guidelines

Introduction

The document describes a set of procedures and standards that can be used to evaluate the Magnetic Resonance (MR) compatibility of hand-held, non-electronic equipment used in conjunction with the MR system.

Included are definitions of MR compatibility, a scheme for classification of equipment, and a set of tests and standards for assessing MR compatibility. This chapter also contains the step-by-step instructions to help you learn how to:

- Test MR Compatibility
- Test Magnetic Forces and Torques
- Test Heating

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What Do I Need to Know About...

This section presents the concepts necessary to successfully follow the MR compatibility guidelines and tests. Specifically, you need to understand:

- MR Compatibility Basics
  - MR Safety
  - Device Safety and Effectiveness
  - Scanner Compatibility
- Testing Approach
- Classifications
  - Device Classifications
  - Zone Classifications
- Magnetic Forces and Torques
  - Screening Test
  - Small Object Test
- Heating
  - Pre-Test Inspection
  - Heating Test
- Magnet Field Distortion

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MR Compatibility Basics

For the purposes of this document, "MR compatibility" is defined as follows:

A device shall be considered MR-compatible if, and only if:

- Its presence in the MR magnet room does not pose an increased safety risk to the patient or other personnel (MR Safety)
- It performs its intended function when used in conjunction with the MR system in a safe and effective manner (Device Safety and Effectiveness)
- Its use in conjunction with the MR scanner does not adversely impact the function of the scanner (Scanner Compatibility)

MR compatibility is dependent on field strength of the MR system. MR compatibility as tested on a specific field strength only applies to that specific field strength. For instance, items tested at 1.5T do not ensure compatibility with a 3.0T system.

A device may be MR-compatible for certain types of uses but not for others. The idea of usage classification is discussed. MR compatibility does not imply general safety. Refer to the Working Safely chapter for normal operating safety issues.

This document applies to hand-held, non-electronic devices. The degree of MR compatibility required for a device is broken into four zones according to where the device will be used during scanning. Knowledge of the correct MR Compatibility Zone allows the appropriate MR compatibility tests and standards for the device in question to be determined.

MR Safety

The "MR Safety" portion of the definition is related to whether the presence of the device in question in or near the MR system poses an additional safety risk. The primary areas of concern are:

- **Magnet forces:** The device must not be attracted to the magnet with sufficient force as to be a projectile. The device must not experience sufficient torque as to be a possible source of injury.
- **Heating:** The device or immediately surrounding tissue must not have the potential to become hot enough to cause injury during scanning.

The organization ASTM has developed several MR compatibility test standards that are used (for example by the FDA) to justify various MR safety categories.

Device Safety and Effectiveness

Device safety and effectiveness is whether the introduction of the device into the MR environment alters the way in which the device functions. In order to be MR-compatible, the device must perform its intended function effectively and in a way that does not pose a safety risk while in the MR scanner environment.

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The following are aspects of the MR environment which may impact device function:

- Large static magnetic field
- Rapidly changing magnetic field
- Radio frequency (RF) energy

NOTE: GE is not responsible for assessing the proper function of any device. The device manufacturer must make arrangements to test their device in the MR imaging environment. Written certification that the device functions as intended in the MR environment shall be provided before the device is considered MR-compatible.

Scanner Compatibility

Scanner Compatibility refers to whether the use of the device in conjunction with the scanner adversely effects scanner performance (signal to noise ratio, artifacts, image distortion, etc.). The areas of concern include:

- **Static Magnetic Field Homogeneity:** The device must not cause significant changes in the main magnetic field homogeneity.
- **RF Interference:** The device must not distort the transmit pattern of the RF energy used for imaging. The device must not generate significant RF energy in the sensitive range of the scanner.

Testing Approach

The tests used to assess MR compatibility are divided into the following areas:

- Magnetic Forces and Torques
- Heating
- Magnet Field Distortion (Static)

Each area has tests and acceptance criteria appropriate to the MR Compatibility Zones.

When a new device is considered for MR compatibility, it is first classified. Then the appropriate tests and acceptance criteria are chosen. Once the device passes the tests, and the device manufacturer has provided in writing the results of their functionality tests indicating that the device functions as intended in the MR environment, the device may be designated “MR-compatible” for the zone tested. The tests are designed such that MR safety is independent of the zone tested. For example, a Zone 4 hand-held device may be safely positioned at isocenter and scanned but may cause severe image distortions and/or artifacts.

Results of MR compatibility testing should be maintained on file at the site along with design drawings and any other documentation necessary to describe the device. Since design changes may make the device incompatible, any changes must be reviewed with and approved by the site testing personnel in order to maintain the MR compatibility of the device. Some changes may necessitate retesting.

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Classifications

In order to manage the tremendous variety of devices and applications, the following classifications are used throughout this document.

Device Classifications

The tests specified assume the device has the following properties:

- **Electrically Passive:** The device contains no electric or electronic circuitry.
- **Hand-held:** The device can be comfortably lifted and manipulated with one hand by a typical person. Such devices typically weigh less than 2 kg.

Zone Classifications

The degree of MR compatibility required depends on the intended use of the device. This measurement of compatibility is designated as Zone 1, 2, 3, or 4:

Zone 1

Zone 1 MR-compatible devices are suitable for use in the imaging volume, potentially in contact with the patient, during scanning. The highest zone of MR compatibility is Zone 1. A Zone 1 MR-compatible device may be used in the imaging volume in the region of interest without degradation of image quality or spatial accuracy. Biopsy needles and endoscopes would typically be Zone 1 devices.

Zone 2

Zone 2 MR-compatible devices are suitable for use in the imaging volume, potentially in contact with the patient, during scanning, provided that the device is at least 5.0 cm away from any anatomical feature for which spatially accurate images are required. A Zone 2 device may be used in the imaging volume but can cause spatial distortions and artifacts nearby (within 5.0 cm). Typical Zone 2 devices would include positioners and microscopes.

Zone 3

Zone 3 MR-compatible devices are suitable for use in the imaging volume, potentially in contact with the patient, when not scanning. The device should be moved outside of 1.0 m from isocenter during scanning. A Zone 3 device should be removed from the scan volume prior to scanning to avoid spatial distortion and artifacts. The majority of MR-compatible hand-held surgical instruments would be Zone 3 devices.

Zone 4

Zone 4 MR-compatible devices are suitable for use in the magnet room during scanning when kept more than 1.0 m from isocenter. A Zone 4 MR-compatible device should not cause problems only if it is no closer than 1.0 m from isocenter. Typical Zone 4 devices include furniture and carts.

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Table 3-1 Compatibility Test Selection

Zone	Magnetic Force	Heating	Field Distortion
Zone 4	4.1.2	4.2.1	N/A
Zone 3	4.1.2	4.2.1	N/A
Zone 2	4.1.2	4.2.1	4.3.1
Zone 1	4.1.2	4.2.1	4.3.1

Table 3-2 Compatibility Test Acceptance Criteria

Zone	Magnetic Force	Heating	Field Distortion
Zone 4	<30° deflection	<5°C	N/A
Zone 3	<30° deflection	<5°C	N/A
Zone 2	<30° deflection	<5°C	<100 counts above zero distortion (at 5 cm)
Zone 1	<30° deflection	<5°C	<100 counts above zero distortion

NOTE: The same type of device can be made MR-compatible to any zone depending on the material used in fabrication. For example, a scalpel can be either Zone 1, 2, or 3. A scalpel made of a non-ferrous metal might be Zone 3 MR-compatible, while one made of ceramic could be Zone 1 MR-compatible.

Magnetic Forces and Torques

The Magnetic Forces and Torques (MFT) Test Specification includes three tests:

- **MFT 1:** Screening
- **MFT 2:** Small Object Test

Screening Test

The MFT Screening test includes two parts. The results of these tests should be recorded as either pass or fail.

Part A

Part A of the MFT Screening test is designed to exclude devices that may be strongly attracted to the MR magnet. This portion of the test should be performed prior to the introduction of any device under consideration for MR compatibility into the magnet room.

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This first portion of test is to be performed outside of the screen room and requires the use of a hand-held permanent magnet. The magnet used can be an Edmund Scientific 25 lb. pull (Serial number 40847) or equivalent. During this test, a hand magnet is suspended on a string and the device is slowly brought into contact with the magnet. If the hand magnet is strongly attracted to the device, the test fails.

Part B

Part B of the MFT Screening test is designed to pass devices that experience no force or torque while in the MR magnet. The test is to be performed using the MR magnet. A portion of the test involves moving a device into and out of the magnet and detecting any translation or rotation of the device. Only devices which have passed the screening test should be tested.

The following materials are required for this test: a clear plastic box with locking lid, a plastic box-sized sheet of paper, and a marking pen. The device is placed in the clear plastic box on top of a sheet of paper, its outline is traced on the paper and the lid closed and locked. The device is then moved into the magnet bore and then removed from the bore. If the device moves or pivots as it moves into or out of the bore, it fails. The test is repeated with the device oriented 90° from its original position.

Small Object Test

The MFT Small Object test is for devices that have passed the Screening test. The test involves moving the device toward the magnet in a controlled fashion, while monitoring the force on the device. This test is to be performed using the MR magnet ramped.

A deflection meter (level - equipped protractor with a string attached) and a mesh bag are required for this test. The device is suspended from a string and slowly carried toward the MR magnet until it is positioned in a known magnetic force area. The angle of deflection of the string (and thus the force on the device) is monitored. The results of this test should be recorded as pass or fail.

Heating

The Heating tests are applicable only for devices that can physically be located inside of a 1.0 m diameter sphere centered at isocenter during scanning. Other devices may be exempt.

The Heating test is designed to detect if the device interacts with the MR system so as to create hazardous temperatures when positioned in the patient or when in contact with the patient's or operator's skin. The maximum safe internal temperature is 41°C, which represents a 4°C rise from normal internal temperature of 37°C. The maximum skin exposure temperature allowed is also 41°C. Skin temperature can reach 37°C during MR scanning, so a temperature increase of 5°C would exceed the allowed temperature. Thus, the temperature level to be detected by the heating test is 28°C, which represents a 5°C rise above normal room temperature of 23°C.

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The MR Heating test requires the use of a temperature sensitive marking material such as Omega OMEGASTIK Temperature Indicating Crayons or OMEGALAQ Temperature Indicating Lacquer. These materials melt at a specific temperature and maintain an altered appearance when cooled. The specific model used should melt at a temperature of 28°C, assuming a room temperature of 23°C.

The MR Heating test also requires the use of a tissue-simulating material. A gel phantom designed to simulate the electrical conductivity properties of tissue has been developed. The gel consists of 9.70% TX-150 (Oil Center Research, Lafayette LA), 9.06% aluminum powder (JT Baker Chemical Co., Phillipsburg, NJ), 80.97% water, and 0.27% NaCl.

Pre-Test Inspection

The pre-test inspection can be performed outside the procedure room. Only devices that have passed the required Magnetic Forces and Torques test should be tested. The pre-test inspection involves inspecting the device prior to testing for construction material. If the device is constructed entirely of a non-conductive material, such as ceramic or plastic (as indicated in information supplied by manufacturer and confirmed by inspection), it may be exempt from further Heating testing.

Heating Test

This test is used for devices that may be in surface contact with the patient or operator during scanning and for devices that may be inserted into the patient during scanning. This test is to be performed using the MR magnet ramped.

The conductive gel samples, specifically designed for the heating test, are to be used. The exposed surfaces of the device are painted with the temperature sensitive paint. The paint is water-based and undergoes a permanent change if its temperature exceeds a specific value. The device should be positioned next to or in the gel as it would be positioned in the skin. The Heating Test pulse sequence data (PSD), listed under the GE Protocols on the system, should be ran during testing. The device is then removed from the scanner, after the scan is complete and examined to determine if any surfaces exceeded the prescribed temperature. If any surfaces have exceeded the appropriate temperature threshold, the device fails. The results of this test should be recorded as pass or fail.

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Magnet Field Distortion

The Magnet Field Distortion (MFD) test is to be performed using the MR magnet ramped. Only devices which have passed the Magnetic Forces and Torques test should be tested. Zone 3 and 4 devices do not need to be tested.

MFD Test 1: Local Field Disturbance

This test is designed to evaluate local field disturbances from devices that would be in the imaging volume during scanning. A copper sulfate phantom is scanned, then the device is submerged in the solution and the phantom is re-scanned. The change in field, due to the device, is measured and the potential impacts on signal intensity and spatial accuracy are reported. Results should be recorded for Zones 1 and 2, as pass or fail.

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How Do I...

This section provides the step-by-step instructions for testing MR compatibility in a magnetic field environment. Specifically, it describes how to:

- Test MR Compatibility
- Test Magnetic Forces and Torques
  - Screening Test
  - Small Object Test
- Test Heating
  - Pre-Test Inspection
  - Devices used During Scanning

The American Society for Testing and Materials, International (ASTM) has developed the following MR compatibility standards (and are developing more):

- F 1542 Specification for the Requirements and Disclosure of Self-Closing Aneurysm Clips
- F 2052 Test Method for Measurement of Magnetically Induced Displacement Force on Medical Devices in the Magnetic Resonance Environment
- F 2119 Test Method for Evaluation of MR Image Artifacts from Passive Implants
- F 2182 Measurement of Radio Frequency Induced Heating Near Passive Implants During Magnetic Resonance Imaging
- F 2213 Measurement of Magnetically Induced Torque on Medical Devices in the Magnetic Resonance Environment
- F 2503 Practice for Marking Medical Devices and Other Items for Safety in the Magnetic Resonance Environment

These may be ordered online at http://www.astm.org.

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Test MR Compatibility

The following procedures should be used to assess whether a device is MR-compatible.

1. Make sure that all the appropriate documentation of the device has been provided.
   - The documentation should include a detailed mechanical drawing, indicating the types of materials used in fabrication.
2. Determine the appropriate device and usage classifications.
   - Device Classifications – electrically passive and hand-held
   - Zone Classifications – Zones 1, 2, 3, or 4
3. Perform the magnetic force Small Object Test.
   - If the device passes, proceed to step 4.
   - If not, the device should **not** be designated MR-compatible.
4. Confirm that the manufacturer deems the device as MR-compatible.
   - Verify the manufacturer performed a test of the device functionality in the MR system.
   - Additionally, verify the manufacturer provides satisfactory documentation, indicating that the safe function of the device is not compromised by its introduction into the MR environment.
5. Determine the appropriate tests for the device.
   - Magnetic Forces and Torques
   - Heating
   - Magnet Field Distortion
6. Perform the indicated tests and record the results.
   - If all the test results are within the acceptance criteria, the device may be considered MR-compatible to the lowest zone passed.
7. Record the results of the MR Compatibility test.
   - Document the results on the MR Compatibility Data Sheet located in the Safety Reference Forms chapter.

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Quick Steps: Test MR Compatibility
1. Make sure that all the appropriate documentation of the device has been provided. 2. Determine the appropriate device and usage classifications. 3. Perform the magnetic force Small Object Test. 4. Confirm that the manufacturer deems the device as MR-compatible. 5. Determine the appropriate tests for the device. 6. Perform the indicated tests and record the results. 7. Record the results of the MR Compatibility test.

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Test Magnetic Forces and Torques

Screening Test

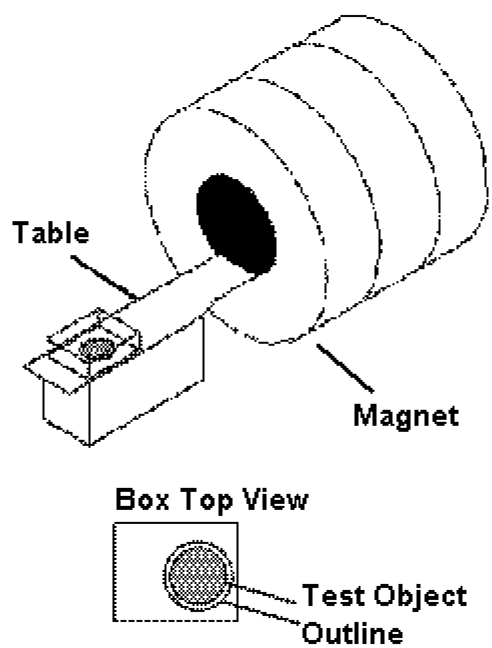
The MFT Screening Test Part A should be performed outside the magnet room. It is designed to exclude devices that may be strongly attracted to the MR magnet. Part B of the test is designed to detect any magnetic force or torque on a device. Only devices that have passed the Screening Test Part A should be tested.

Part A

1. Suspend a hand magnet with a string.
   - The hand magnet should be an Edmund Scientific 25 lb. pull or equivalent.
   - Use string as opposed to a metallic wire.
2. Slowly bring the device into contact with the magnet.
   - If the magnet moves and the string deflects from vertical before the device touches the magnet, or
   - if the device sticks to the magnet and pulls the magnet so that the string deflects from vertical, the device fails.
3. Record the results of the test as pass or fail.
   - Document the results on the MR Compatibility Data Sheet located in the Safety Reference Forms chapter.

Part B

1. While outside the magnet room, place the device to be tested into the plastic box on top of a piece of paper.
   - The paper should be taped into the bottom of the box.
   - The lid should have a lock attached.
2. Draw an outline of the device on the paper.
   - This indicates the initial position of the device.
3. Close and secure the lid.
4. Place the box on the outer end of the patient table and secure it to the table.
   - Make sure the device has not shifted after box placement on the table.



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5. Move the table, such that the box with the device is moved to magnet iso-center, then return it to the starting position.
    - If the position of the device relative to its outline has not changed (including rotation), the device passes.
    - If the position of the device relative to its outline has changed (including rotation), the device fails.
6. Repeat the test with the device oriented 90° from its original position.
7. Record the results of the test as pass or fail.
    - Document the results on the MR Compatibility Data Sheet located in the Safety Reference Forms chapter.

Quick Steps: Test Magnetic Forces and Torques – Screening Test

Part A

1. Suspend a hand magnet with a string.
2. Slowly bring the device into contact with the magnet.
3. Record the results of the test as pass or fail.

Part B

1. While outside the magnet room, place the device to be tested into the plastic box on top of a piece of paper.
2. Draw an outline of the device on the paper.
3. Close and secure the lid.
4. Place the box on the outer end of the patient table and secure it to the table.
5. Move the table, such that the box with the device is moved to magnet iso-center, then return it to the starting position.
6. Repeat the test with the device oriented 90° from its original position.
7. Record the results of the test as pass or fail.

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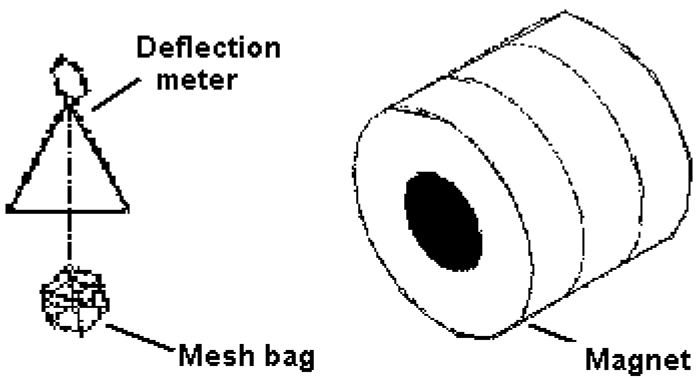
Test Magnetic Forces and Torques

Small Object Test

The MFT Small Object Test is designed to determine the magnetic force and torque on a device. Only devices which have passed the Screening Test (MFT 1) should be tested. This test requires a deflection meter and a mesh bag.

1. While outside the magnet room, place the device to be tested into a mesh bag.
   - Placing the device in a bag allows the item to freely hang from the deflection meter once the string is attached.
2. Adjust the position of the device in the bag such that its long axis is parallel to the floor.
3. Attach the string to the mesh bag and run the opposite end of the string through the center hole of the deflection meter.



4. Hold the deflection meter level out in front of you and pointing at the magnet as you enter the magnet room.
   - Walk slowly toward the magnet while observing the angle of deflection.
   - Make sure to keep the device between yourself and the magnet.
5. If the deflection angle exceeds 30°, stop the test and remove the device from the procedure room. Otherwise, continue moving toward the patient end of the magnet until the top center of the deflection meter touches the top of the magnet opening.
6. Align the deflection meter with the Z-axis of the magnet.
7. Record the maximum deflection on the MR Compatibility Log Form.
   - If the defection angle is less than 30°, the device passes.
   - If the deflection angle exceeds 30°, the device fails.
8. Record the results of the test as pass or fail.
   - Document the results on the MR Compatibility Log Form located in the Safety Reference Forms chapter.

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Quick Steps: Test Magnetic Forces and Torques – Small Object Test
1. While outside the magnet room, place the device to be tested into a mesh bag.
2. Adjust the position of the device in the bag such that its long axis is parallel to the floor.
3. Attach the string to the mesh bag and run the opposite end of the string through the center hole of the deflection meter.
4. Hold the deflection meter level out in front of you and pointing at the magnet as you enter the magnet room.
5. If the deflection angle exceeds 30°, stop the test and remove the device from the procedure room. Otherwise, continue moving toward the patient end of the magnet until the top center of the deflection meter touches the top of the magnet opening.
6. Align the deflection meter with the Z-axis of the magnet.
7. Record the maximum deflection on the MR Compatibility Log Form.
8. Record the results of the test as pass or fail.

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Test Heating

Pre-Test Inspection

The Pre-Test Inspection can be performed outside the magnet room. Only devices that have passed the required Magnetic Forces and Torques test should be tested.

1. Before testing the device, inspect it for construction material.
   - If the construction material information is supplied by the manufacturer, it should still be confirmed by your inspection.
2. Determine if the device should undergo further Heating testing.
   - If the device is constructed entirely of a non-conductive material, such as ceramic or plastic it may be exempted from further Heating testing.

Quick Steps: Test Heating – Pre-Test Inspection
1. Before testing the device, inspect it for construction material. 2. Determine if the device should undergo further Heating testing.

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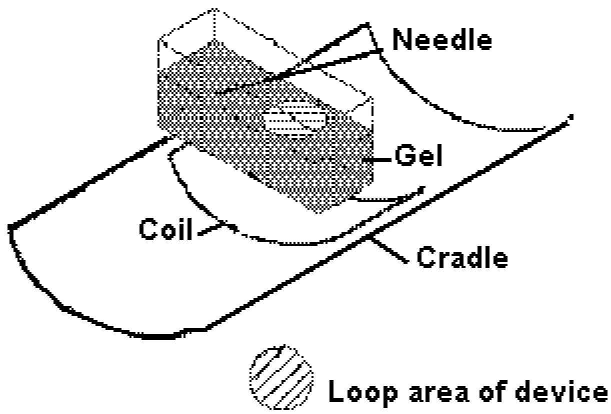
Test Heating

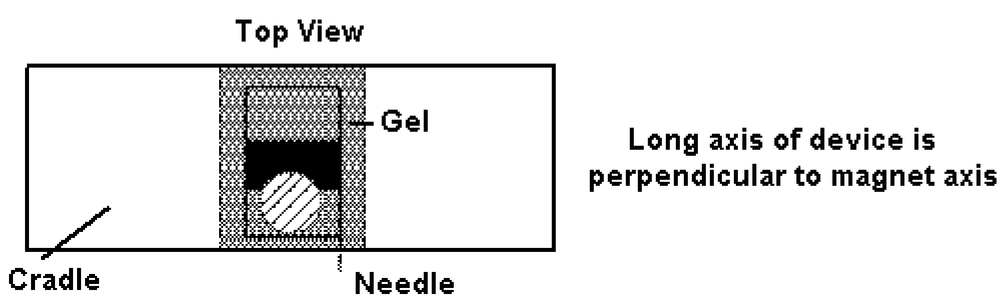
Devices used During Scanning

This portion of the Heating Test is used for devices that may be in surface contact with the patient or operator during scanning and for devices that may be inserted into the patient during scanning. This test is to be performed using the MR magnet ramped.

1. Apply the temperature indicating material to the surfaces of the device to be tested.
    - Use Omega OMEGASTIK Temperature Indicating Crayons or OMEGALAQ Temperature Indicating Lacquer.
    - These materials melt at a specific temperature and maintain an altered appearance when cooled.
2. Lay the Body Flex coil flat on the table at isocenter.
3. Position the device in the tissue simulating gel.
    - The relative position between the device and tissue should be duplicated.
    - The figure demonstrates positioning of the equipment with a needle used as the testing device.



4. Place the gel/device on top of one of the sides of the Body Flex coil.
    - The long axis of the device should be aligned perpendicular to the magnet's Z-axis as shown in the figure below.



5. Scan the device with the Heating Test protocol listed under the GE protocols.
    - The Heating Test protocol exposes the device to the maximum RF power possible on the system for a period of 15 minutes.

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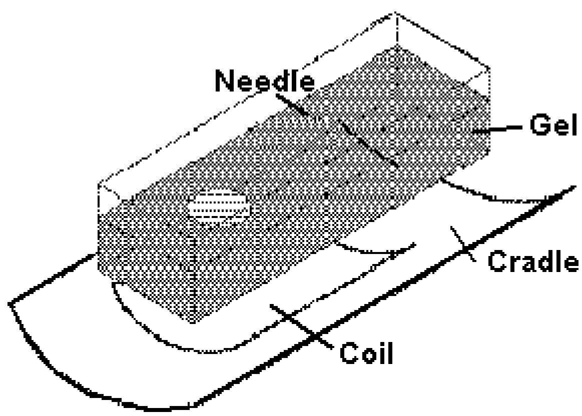
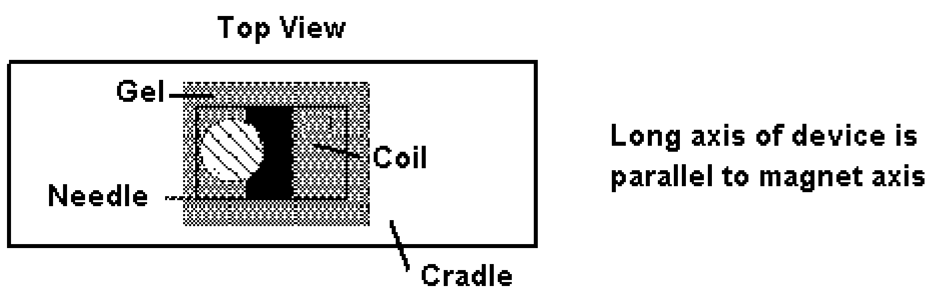
6. Observe the device via the over-patient camera during the scan.
   - If you can visually verify that the temperature sensitive material is undergoing a color change you can stop the scan before its completion to prevent damage to the device.
   - If no change is detectable during the scan, wait for the scan to finish, then remove the device and inspect for temperatures exceeding the melting point of the indicating material.
   - If the indicating material melted at any point on the device, the test fails.
7. Repeat with the long axis of the device aligned parallel to the magnet's Z-axis.



   - If the indicating material melted at any point on the device, the test fails.
   - If the device has some loop area, repeat the test with the loop area perpendicular to the magnet's Z-axis.
   - A needle is shown as the device in the figures for demonstration purposes. The needle axis represents the long axis on the device being tested.



8. Record the results of the test as pass or fail.
   - Document the results on the MR Compatibility Log Form located in the Safety Reference Forms chapter.

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Quick Steps: Test Heating – Devices used During Scanning

1. Apply the temperature indicating material to the surfaces of the device to be tested.
2. Lay the Body Flex coil flat on the table at isocenter.
3. Position the device in the tissue simulating gel.
4. Place the gel/device on top of one of the sides of the Body Flex coil.
5. Scan the device with the Heating Test protocol listed under the GE protocols.
6. Observe the device via the over-patient camera during the scan.
7. Repeat with the long axis of the device aligned parallel to the magnet's Z-axis.
8. Record the results of the test as pass or fail.

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Appendix A
Safety Reference Forms

Introduction

This appendix includes safety references to an example patient screening form and a peripheral nerve stimulation incident form. These reference forms may be duplicated for use with your Magnetic Resonance (MR) imaging system.

Specifically, this appendix includes:

- Patient Screening Form
  - Section 1: General Information
  - Section 2: Hazardous Items Checklist
  - Section 3: Magnet Room Pre-Entry Checklist
- MR Compatibility Data Sheet
- MR Compatibility Log Form

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Patient Screening Form

Section 1: General Information

Patient Screening Form – Section 1: General Information

Patient Name: ________________________________ Date: ___/___/___

Birthdate: ___/___/___ Age: _____ Height: _____ Weight: _____

Study: ______________________________________ Male ❑ Female ❑

Previous MR study: Yes ❑ No ❑

If yes, date and type of study: ____

____

Recent X-ray studies: Yes ❑ No ❑

If yes, date and type of study: ____

____

Previous surgeries: Yes ❑ No ❑

If yes, date and type of surgery:

Head ____

Neck ______________________________________

Chest ____

Abdomen ____

Extremities ____

Allergies: ____

What have you eaten in the last four hours? ____

Have you ever worked in a machine shop or similar environment where you may have been subjected to small metal slivers, particularly in the eyes? Yes ❑ No ❑

If yes, please describe: ____

____

Female patients only:

Date of last menstrual period: ___/___/___ Post menopausal: Yes ❑ No ❑

Are you or is there a chance you are pregnant? Yes ❑ No ❑

Are you currently breast feeding? Yes ❑ No ❑

Do you have an intrauterine device (IUD)? Yes ❑ No ❑

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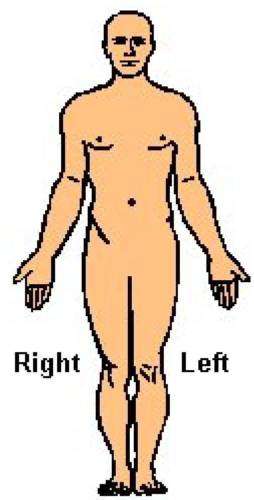
Section 2: Hazardous Items Checklist

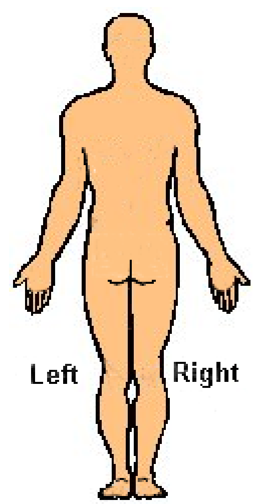
Patient Screening Form – Section 2: Hazardous Items Checklist

The following items can interfere with MR imaging and some can actually be hazardous to your safety. Please indicate if you have any these items:

Yes ❑	No ❑	Cardiac pacemaker or lead wires
Yes ❑	No ❑	Artificial heart valve
Yes ❑	No ❑	Aortic clips
Yes ❑	No ❑	Brain aneurysm clips
Yes ❑	No ❑	Neurostimulator or lead wires
Yes ❑	No ❑	Electronic implant or device
Yes ❑	No ❑	Insulin or other infusion pump
Yes ❑	No ❑	Electrodes or wires
Yes ❑	No ❑	Cochlear or other ear implant
Yes ❑	No ❑	Metallic stent, filter, or coil
Yes ❑	No ❑	Shunts
Yes ❑	No ❑	Joint replacements
Yes ❑	No ❑	Fractured bones treated with metal rods, plates, pins, screws, or nails
Yes ❑	No ❑	Any type of prosthesis
Yes ❑	No ❑	Metal mesh implant
Yes ❑	No ❑	Surgical staples or wire sutures
Yes ❑	No ❑	Shrapnel
Yes ❑	No ❑	Dentures or partial plates
Yes ❑	No ❑	Metal slivers in the eyes
Yes ❑	No ❑	Tattoo or permanent make-up
Yes ❑	No ❑	Body piercing jewelry
Yes ❑	No ❑	Any metallic fragment or foreign body
Yes ❑	No ❑	Wet diapers or incontinence products
Yes ❑	No ❑	Others (please list) _______________

Please mark the location of any metal inside or on your body on the following anatomical figures.





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Patient Screening Form – Section 2: Hazardous Items Checklist	
The following items can interfere with MR imaging and some can actually be hazardous to your safety. Please indicate if you have any these items: ______________________	**Please mark the location of any metal inside or on your body on the following anatomical figures.**

Section 3: Magnet Room Pre-Entry Checklist

Patient Screening Form – Section 3: Magnet Room Pre-Entry Checklist		
Do not enter the scan room with any metal or magnetic-sensitive items. Check the items below if you have one of the following:		
Yes ❑	No ❑	Glasses
Yes ❑	No ❑	Removable dental work
Yes ❑	No ❑	Hearing Aid
Yes ❑	No ❑	Watch
Yes ❑	No ❑	Jewelry
Yes ❑	No ❑	Wallet or money clip
Yes ❑	No ❑	Magnetic strip cards (credit cards, bank cards)
Yes ❑	No ❑	Pens or pencils
Yes ❑	No ❑	Keys
Yes ❑	No ❑	Coins
Yes ❑	No ❑	Pocket knife
Yes ❑	No ❑	Metal zippers, buttons or metal fibers/threads in clothing
Yes ❑	No ❑	Belt buckle
Yes ❑	No ❑	Shoes
Yes ❑	No ❑	Hairpins or barrettes
Yes ❑	No ❑	Bra with metal hooks
Yes ❑	No ❑	Bra and girdle underwire support
Yes ❑	No ❑	Sanitary belt
Yes ❑	No ❑	Safety pins

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Patient Screening Form – Section 3: Magnet Room Pre-Entry Checklist
Do not enter the scan room with any metal or magnetic-sensitive items. Check the items below if you have one of the following:

WARNING: **These objects must not be taken into the magnet room. Damage to the equipment, MR system, and personal injury could result.**

Signature: ____________________________________ Date: ___/___/___

Form completed by: _____________________ Relationship to patient: ______________________

Form reviewed by: ______________________ Title: _______________

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Report of Patient Peripheral Nerve Stimulation Form

Report of Patient Peripheral Nerve Stimulation	
Report Date	
Date of MR Exam	
Hospital Name	
Telephone Number	
Name of Reporting Health Care Professional	
Patient Age	
Patient Weight	
Patient Height	
Patient Pathology	
Patient Medications	
Exam Number	
Series Number	
PSD (GRE, SE, FSE, EPI, IR)	
TR (ms)	
TE (ms)	
FOV (cm)	
Slice Thickness (mm)	
Interslice Spacing (mm)	
Slew Rate (T/m/s)	
Frequency Encoding Direction (RL, AP, SI)	
Were patient's hands clasped?	
dB/dt (% peripheral nerve or T/s)	
Stimulation Severity (1=very mild, 2=mild, 3=uncomfortable, 4=very uncomfortable)	
Stimulation Description and Location	
Please Duplicate Before Use	

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MR Compatibility Data Sheet

MR Compatibility Data Sheet

Device Information

Device Name: ____________________ Model #: ______________

Manufacturer: ____________________ Serial #: ______________

Description: ____

____

Document Checklist

- ❑ A set of detailed mechanical drawings indicating the type of material used (attach copy)
- ❑ Results of the manufacturer test, indicating the device functions as intended in the MRI environment (attach copy)

Device Classification

Circle one: Hand-held Portable Stationary

Circle one: Electrically-active Electrically-passive

MR Compatibility Zone Required: 1 2 3 4

Results of Magnetic Forces Screening Test: Pass Fail

Comments: ____

List the required tests from Table 1 along with test results:

Test	Results	Status (circle lowest zone passed)				
______________	______________	4	3	2	1	none
______________	______________	4	3	2	1	none
______________	______________	4	3	2	1	none
______________	______________	4	3	2	1	none
______________	______________	4	3	2	1	none
______________	______________	4	3	2	1	none
______________	______________					
	Overall Status (circle highest of above):	4	3	2	1	none

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MR Compatibility Log Form

Instrument Description						Date:		
Manufacturer:						Completed by:		
Serial Number:								
Other:								
Distance from Instrument	Plane	Max	Min	Max Deviation from Zero	Max	Min	Max Deviation from Zero	Pass/Fail

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Appendix B

Glossary of Terms and Acronyms

Introduction

This appendix defines the MR terms necessary for a working understanding of Pulse Sequence Databases. We could in no way list all terms associated with MR as it relates to General Electric (GE) Medical Systems equipment, but choose to include a few that may prove helpful when reading the next few chapters.

This glossary also provides the identification of the various acronyms used in this guide. The phrase is written out in full, followed by the acronym enclosed in parentheses, e.g. Pulse Sequence Database (PSD).

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Glossary

3D Multi Slab: An image mode used in Time-of Flight vascular imaging for acquiring multiple overlapping 3D slabs.

90° Pulse: A pulse that rotates the magnetization vector 90° from longitudinal static magnetic field direction. This converts the longitudinal magnetization into transverse magnetization.

A/P: See Anterior/Posterior.

AFOV: See Asymmetric Field of View.

AIT: See Available Imaging Time.

Anterior/Posterior (A/P): A patient positioning selection designating the coronal plane alignment in order to ensure that the center of the region of interest is as close as possible to isocenter. The coronal plane divides the body into anterior (front) and posterior (back) sections.

Artifact: An error in the reconstructed image that does not correspond to the patient. There are three major forms of artifacts that can occur in MR imaging and contribute to poor image quality: geometric distortion, inhomogeneous signal intensity, and spurious signal.

Asymmetric Echo: An echo whose peak, at TE, is not centered in the sampling window. Also called fractional echo or partial echo.

Asymmetric Field of View (AFOV): 1. An FOV in which the vertical and horizontal dimensions are not equal. Simular to the rectangular FOV selected. 2. An imaging enhancement activated by choosing one of two FOV options: square pixel or variable FOV. Asymmetric FOV is useful for any scan which has anatomy smaller than the FOV in the phase direction. See FOV and square pixels.

Available Imaging Time (AIT): In cardiac gating, the time during which data can be collected by the MR system.

Average Flow: An Flow Analysis measurement. Summation of voxel values in a given flow region (ml/min), reflecting the volume per minute passing through the defined flow region of a specified cardiac phase or cycle.

Average Velocity: A Flow Analysis measurement. Flow Q (expressed in cm^3/sec) divided by the cross-sectional area A (expressed cm^2) of a vessel: V = Q/A (cm/sec); 1/2 V_{max} for laminar flow.

Averaging: A SNR-enhancing technique in which the same MR signal is added up, and then the sum is divided by the number of signals acquired.

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Bandwidth: A range within a band of frequencies that an MRI system is **"**tuned**"** to receive. The receive bandwidth of an image determines the number of frequencies encompassed in the image. The system's bandwidth choice depends on the TE, matrix, and FOV you select.

Beats per Minute (bpm): The average heart rate as shown by the cardiac waveform display.

Bipolar Flow-Encoding Gradients: Two gradient pulses of identical shape, but opposite polarity. Used in order to encode velocities as changes of phase, as used in Phase Contrast angiography.

BPM: See Beats per Minute.

Cardiac Phase Images: Images demonstrating different times or phases within a cardiac cycle.

Cine: Generated images for dynamic views of anatomy such as the heart. This option employs retrospective gating techniques and a Gradient Echo pulse sequence.

CNR: See Contrast-to-Noise Ratio.

Collapsed: A Maximum Intensity Projection (MIP), also called Maximum Pixel Projection (MPP) from TOF magnitude images, or PC weighted-phase images. The collapsed image is the MIP in the slice direction.

Complex Difference: A flow reconstruction type for Phase Contrast Vascular Imaging providing control of the Slab Dephasing Gradient and Phase Correction. Complex Difference reconstructions have the Dephase Gradient off and Phase Correction on.

Contrast Resolution: An image function providing the ability to differentiate anatomical density differences with respect to surrounding anatomical regions.

Contrast-to-Noise Ratio (CNR): Ratio of the absolute difference in intensities between two regions to the level of fluctuations in intensity due to noise.

Coronal: The horizontal plane along the longitudinal axis of the body dividing it into anterior (front) and posterior (back) halves.

Decubitus Position: Describes the position of a patient lying on the left or right side.

Diastole: The period between the end of the T-wave and the beginning of the R-wave in the cardiac cycle. Also called ventricular filling.

Dynamic-Range Compression: A method of enhancing Phase Contrast image quality by applying a projection Dephasing Gradient to suppress signal from stationary tissues.

Echo Rephasing: Re-establishment of spin phase coherence, accomplished via a 180 degree RF pulse or gradient switching. See Refocusing.

Echo Time: See Time to Echo (TE).

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Echo Train Length (ETL): The number of 180° refocusing pulses played out during one TR period.

EDR: See Extended Dynamic Range.

Effective R-R interval (RR): The inverse of bpm (Beats per Minute) measured in msec: RR = 60,000 divided by bpm.

Effective TR: The "average" repetition time, or TR, in cardiac gating. Measured as the number of RR intervals between successive excitations of a particular slice location - e.g., RR, 2xRR, 3xRR, 4xRR.

Effective value: A typical or average value - for example, effective TR. Since you can not control your patients heart rate, you can not control true TR in a gated study. You can control the effective TR by telling the system not to trigger at every beat.

ETL: See Echo Train Length.

Even-Echo Rephasing: Rephasing of moving spins on symmetric, even echoes (e.g., 2, 4, or 6) in Multi-Echo sequences.

Extended Dynamic Range (EDR): An imaging enhancement that uses 32-bit processing instead of the conventional 16-bits to improve SNR.

F/W: See Fat/Water Suppression.

Fast Cardiac Gating (FastCard): A 2D Time-of Flight, gradient recalled, single-breath PSD for acquiring multiple phases of the cardiac cycle at a single slice location.

Fat/Water Suppression (F/W): An imaging enhancement technique that suppresses signal within the imaging volume from either fat or water by applying a frequency-selective saturation pulse.

FID: See Free Induction Decay.

Field of View (Acquisition FOV): The area of the anatomy being imaged, usually expressed in centimeters. FOV image size is a function of the acquisition matrix times the pixel size.

First-Order Phase Correction: Phase errors in a Phase Contrast image can be modeled as a linear shading across the image in the x and y direction. In first order Phase Correction, the slopes of the x and y shading are determined to reduce the shading.

Flip Angle: Flip angle is the rotational angle of the magnetization vector produced by a RF pulse relative to the longitudinal axis of the static magnetic field. Flip angle adjusts contrast.

Flow Analysis: A flow reconstruction type for Cine-PC and 2D PC providing control of the Slab Dephasing Gradient and Phase Correction. Flow Analysis reconstructions have the Dephase Gradient off and Phase Correction off.

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Flow Axis: The orthogonal axis (S/I, R/L, A/P) for which flow has been encoded in a flow image.

Flow Compensation (Flow Comp): An imaging enhancement using the system's gradients to put flowing protons into phase with stationary protons, thereby reducing flow artifacts. Applied in the slice and frequency directions.

Flow Encoding: A technique used in MR to measure or display motion such as blood flow within vessels.

Flow Image Set: An image type produced by Phase Contrast scans. Flow images are phase images that may or may not be magnitude weighted. If magnitude weighted, a multiplicative magnitude mask for noise suppression is applied to the phase image. Phase correction and scaling for velocity encoding may also be applied to the image. The default is for correction to take place, but correction can be turned off by selecting the Flow Analysis reconstruction mode on the vascular options screen.

Flow Recon Type: A user-selectable option for selecting a Slab Dephasing Gradient and a Phase Correction technique. See Phase Difference, Complex Difference, and Flow Analysis.

Flow-Related Enhancement: A process by which the signal intensity of moving fluids, like blood or CSF, can be increased compared with the signal of stationary tissue. Occurs when unsaturated, fully magnetized spins replace saturated spins between RF pulses.

FOV Center: The center of a scan image, which is ideally located at the magnet's isocenter.

Fractional Echo: A feature instructing the system to collect just part of the data it normally would. Reduces susceptibility and flow artifacts.

Fractional NEX: A feature instructing the system to use about half or exactly three-quarters of the phase encoding acquired in conventional imaging. Decreases scan time significantly.

Free Induction Decay (FID): The measurable magnetic resonance signal that occurs as the transverse magnetism, produced by the application of the 90° pulse, decays toward zero.

Frequency: The scanning direction associated with the frequency gradient. Usually corresponds to the image's long axis.

Gating: An MR technique for imaging rapidly moving anatomy such as the heart. Uses equipment such as a standard electrocardiograph to trigger data acquisition.

GMN: See Gradient Moment Nulling.

Gradient Echo Imaging: A pulse sequence that reverses gradient polarity to rephase protons and form echoes. Permits short TRs and flip angles of less than 90° to excite only a portion of the longitudinal magnetization.

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Gradient Echo: A basic Fast Scan pulse sequence that uses pulses of 1° to 180° to excite the protons of interest and rephase them. Gradient Echo uses gradients rather than conventional RF pulses.

Gradient Moment Nulling (GMN): The application of gradients to correct phase errors caused by velocity, acceleration or other motion. First-order gradient nulling is the same as Flow Compensation.

Gradient Moment: In MR angiography, the first moment describes a gradient's effect on the phase of a spin with constant velocity; the second moment, its effect on spins experiencing acceleration; the third moment, its effect on spins experiencing jerk.

Gradient-Recalled Acquisition in the Steady State (GRASS): See Gradient Echo and MPGR Gradient Echo.

GRASS: See Gradient-Recalled Acquisition in the Steady State.

G_x, G_y, G_z: Symbols for MR gradients. Subscripts indicate the spatial direction of each gradient.

Intersequence Delay: The time between each image in the cardiac cycle.

Intravoxel Spin-Phase Dispersion: A loss of phase coherence and therefore, signal intensity that can result when a wide spectrum of flow velocities exist, when higher orders of motion like acceleration are present, or when there are minor variations in magnetic field homogeneity.

Inversion Recovery (IR): A pulse sequence that inverts the magnetization and then measures the recovery rate as the nuclei return to equilibrium. This rate of recovery depends on T1.

Inversion Time (TI): The time between the center of the first (180°) inverting pulse and the beginning of the second (90°) refocusing pulse in an IR pulse sequence.

IR: See Inversion Recovery.

Isocenter: The point at which the three gradient planes cross.

Isochromats: Spins sharing the same phase and frequency at a given point in time.

Isometric Contraction: The time immediately after the R-wave when the heart prepares for contraction but does not change in volume.

J-Coupling: Also called Spin-Spin Coupling. The interaction between multiple lines and nuclei. When this interaction takes places the nuclei split their energy levels according to J (the spin-spin coupling constant).

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Magnetic Field Gradient: A device for varying the strength of the static magnetic field at different spatial locations. This is used for slice selection and determining the spatial locations of protons being imaged. Also used for Velocity Encoding, Flow Comp, and in place of RF pulses during Gradient Echo acquisitions to rephase spins. Commonly measured in gauss per centimeter.

Magnetic Resonance (MR): The absorption or emission of electromagnetic energy by nuclei in a static magnetic field after excitation by a suitable RF pulse.

Magnetic Resonance Imaging (MRI): The creation of images using the magnetic resonance phenomenon. The current application involves imaging the distribution of hydrogen nuclei (protons) in the body. The image brightness in a given region usually depends jointly on the spin density and the relaxation times. Image brightness is also affected by motion such as blood flow.

Magnetic Resonance Signal: The electromagnetic signal (in the radio frequency range) produced by the precession of the transverse magnetization of the spins. The rotation of the transverse magnetization induces a voltage in the coil. This voltage is amplified by the receiver.

Magnetic susceptibility: In electromagnetism, the magnetic susceptibility is the degree of magnetization of a material in response to an applied magnetic field. The intensity of magnetization, I, is related to the strength of the inducing magnetic field, H, through a constant of proportionality,k, known as the magnetic susceptibility. $I = kH$.

Magnetization Transfer: A technique that improves contrast by saturating the short T2 component of tissue such as gray/white matter and skeletal muscle.

Maximum Intensity Projection (MIP): A technique for producing multiple projection images from a volume of image data (i.e., 3D volume or a stack of 2D slices). The volume of image data is processed along a selected angle and the pixel with the highest signal intensity is projected onto a two-dimensional image.

MIP: See Maximum Intensity Projection.

MPGR: See Multi-Planer Gradient Echo.

MR: See Magnetic Resonance.

MRI: See Magnetic Resonance Imaging.

MSMP: See Multi-Slice, Multi-Phase Imaging.

MSSP: See Multi-Slice, Single-Phase Imaging.

Multi-Planer Gradient Echo (MPGR): A pulse sequence that represents a combination of Gradient Echo and Spin Echo sequences. Acquires data sequentially rather than slice-by-slice.

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Multi-Slice, Multi-Phase (MSMP) Imaging: Multi-slice, multi-phase cardiac gating pulse sequence that produces images at multiple heart locations and several different cardiac phases at each location.

Multi-Slice, Single-Phase (MSSP) Imaging: Multi-slice, single-phase cardiac gating pulse sequence that produces images at multiple heart locations, each at a different phase of the cardiac cycle.

NEX: See Number of Excitations.

Number of Excitations (NEX): The number of times a pulse sequence is repeated in a given acquisition.

PD-Weighted: See Proton Density-Weighted.

Phase Difference: A flow reconstruction type for Phase Contrast Vascular imaging providing control of the Slab Dephasing Gradient and Phase Correction. Phase difference reconstructions have the Dephase Gradient off and Phase Correction on.

Phase Encoding: The act of localizing an MR signal by applying a gradient pulse to alter the phase of spins before signal readout.

Phase FOV: The Phase Field of View option provides faster scans by scaling down the size of the field of view in the phase direction by 3/4 or 1/2. The phase FOV option is not compatible with some PSD and imaging options.

Projection Dephasing Gradient: A gradient applied to diminish signal from stationary tissues in thick slab 2D Phase Contrast angiography.

Proton Density-Weighted (PD-weighted): PD-weighted images have contrast that is primarily due to the number of protons in the structures. PD-weighted images result when scan timing parameters are selected that minimize the T1 (long TRs) and the T2 (short TEs) contrast effects.

PSD: See Pulse Sequence Database.

Pulse Length or Width: The duration of a pulse, expressed in milliseconds.

Pulse Sequence Database (PSD): A series of RF and gradient pulses and the intervals between them used in conjunction with gradient magnetic fields to produce magnetic resonance images.

Radio Frequency (RF): The frequency (intermediate between audio and infrared frequencies) used in magnetic resonance systems to excite nuclei to resonance.

Radiofrequency Pulse (RF Pulse): A burst of RF energy which, if it is at the correct Larmor frequency, will rotate the macroscopic magnetization vector by a specific angle, dependent on the amplitude and duration of the pulse.

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Ramp Pulse: An RF excitation pulse that has smaller flip angles for spins flowing into the slab. As spins penetrate deeper into the slab, the flip angle increases. For example, a 2:1 Ramp Pulse, with a nominal flip angle of 20°, provides the entry slices with a flip angle of about 13°, and the exit slices with a flip of about 27°.

Readout Gradient: A gradient pulse, applied when an MR signal is collected, used for frequency encoding.

Refocusing: The re-establishment of phase coherence via gradient or RF pulse. See Echo Rephasing, Gradient Echo, and Gradient Moment Nulling.

Relaxation Time: The time required for 63% of the nuclei to revert to their original state in the magnetic field after the RF pulse is turned off.

Repetition Time (TR): The time between successive excitations of a slice. That is, the time from the beginning of one pulse sequence to the beginning of the next. In conventional imaging, it is a fixed value equal to a user-selected value. In cardiac-gated studies, however, it can vary from beat to beat depending on the patient's heart rate.

Rephasing Gradient: A gradient applied in the opposite direction of a recent selective excitation pulse, in order to correct for gradient-induced phase shifts.

RF: See Radio Frequency.

R-R Interval: That part of an ECG waveform representing the heart's electrical activity showing the time between the peak of one R-wave and the peak of the next one. Each R-R interval represents the length of one cardiac cycle.

SAR: Specific Absorption Rate refers to the Radio Frequency power absorbed per unit of mass of an object (Watts/kg). Absorption of RF energy may result in increased tissue temperature.

Saturation Pulse: A slice-selective RF pulse applied, often followed by a Dephasing Gradient, to saturate spins and therefore minimize their signal. Used, for example, to minimize signal from flowing blood in the slice direction.

Saturation: Repeated application of radio frequency pulses in a time that is short compared to the T1 of the tissue, producing incomplete realignment of the net magnetization with the static magnetic field.

Scan Time: The amount of time needed to acquire data.

Sequential IR: An Inversion Recovery sequence in which the system applies the 180°/90°/180° pulses a slice at a time. With the alternative, non-sequential, the system applies the initial 180° pulse to all slices, then returns to each slice to complete the sequence.

Signal-to-Noise Ratio (SNR): The ratio of signal amplitude to noise - i.e., the amplitude of signal emitted by the patient's protons, divided by the amount of patient noises and electronic noise inherent in any electronic instrument.

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Slice Select: The scanning direction associated with the slice-select gradient. Usually corresponds to the direction of the scanning range.

SNR: See Signal-to-Noise Ratio.

Spatial Encoding: A method by which data is collected in order to formulate a three-dimensional image in a two-dimensional plane.

SPGR: See Spoiled Gradient Echo.

Spin Echo imaging: A magnetic resonance imaging technique in which the Spin Echo magnetic resonance signal rather than the Free Induction Decay is used.

Spin-Spin Coupling: See J-Coupling.

Spoiled Gradient Echo (SPGR): A Gradient Echo pulse sequence designed for acquiring T1-weighted images in 2D or 3D mode.

Spoiler Pulse: A gradient pulse applied to dephase spins and to minimize or eliminate residual signal.

SSFP: See Steady State Free Precession.

Steady State Free Precession (SSFP): 1. A Gradient Echo pulse sequence designed for acquiring T2-weighted images in 3D mode. 2. A condition achieved by repeatedly exciting an MR sample with phase-coherent RF pulses at a repetition rate (TR) which is shorter than T2.

T1: The characteristic time constant for the magnetization's return to the longitudinal axis after being excited by an RF pulse. Also called Spin Lattice or Longitudinal Relaxation Time.

T1-Weighted: Scan protocols that allow the T1 effects to predominate over the other relaxation effects.

T2*: The characteristic time constant for loss of transverse magnetization and MR signal due to T2 and local field inhomogeneties. Since such inhomogeneities are not compensated for by gradient reversal, contrast in gradient-echo images depends on T2*.

T2*-Weighted: Scan protocols that allow the T2* effects to predominate over the other contrast effects. There are three primary gradient echo pulse sequences that can be used to produce varying T2*-weighted images: Gradient Echo, SPGR, and SSFP.

T2: The characteristic time constant for loss of phase coherence among spins, caused by their interaction, and the resulting loss in the transverse-magnetization MR signal. Also referred to as Spin-Spin or Transverse Relaxation Time.

T2-Weighted: Scan protocols that allow the T2 effects to predominate over the other contrast effects.

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TE Min: The shortest possible TE time for a given prescription, used to minimize flow dephasing and T2 effects.

TE: See Time to Echo.

TE1: The time from the middle of the first excitation pulse to the middle of the first readout in an Asysmmetrical Spin Echo pulse sequence.

TE2: The time between the middle of the first excitation pulse and the middle of the second readout in an Asysmmetrical Spin Echo pulse sequence.

Threshold: A technique for setting the desired pixel signal intensity values the system uses to process an image.

Throughplane: A flow-encoding direction which is perpendicular to the imaging plane.

TI: See Inversion Time.

Time to Echo (TE): The time between the center of the excitation pulse and the peak of the echo, which usually occurs at the center of the readout.

Time-of-Flight (TOF) Angiography: A 2D or 3D imaging technique that relies primarily on flow-related enhancement to distinguish moving from stationary spins in creating MR angiograms. Blood that has flowed into the slice will not have experienced RF pulses and will therefore appears brighter than stationary tissue.

TOF Angiography: See Time-of-Flight Angiography.

TR: See Repetition Time.

Trigger Delay: The time between the occurrence of the triggering pulse and the actual onset of imaging.

Trigger Window (TW): In cardiac gating, a period during which no further data can be acquired. During this period, the system waits for the next R-wave trigger, which initiates a new sequence of data acquisition.

Trigger: In cardiac/respiratory gating, signal sent by the cardiac/respiratory monitor to activate data acquisition.

TW: See Trigger Window.

Variable Bandwidth (VB): An imaging option that lets you narrow the system's receiver bandwidth to increase SNR. Narrowing the bandwidth forces the system to detect signals from a small range of frequencies. This means the system discards more random electronic noise, improving SNR. The system narrows the Variable Bandwidth only as much as the selected TE allows.

VB: See Variable Bandwidth.

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Velocity Encoding (VENC): A value entered to prescribe the highest velocities to be encoded without aliasing in Phase Contrast angiography.

VENC: See Velocity Encoding.

Very Selective Saturation (VSS): Used in spectroscopy.

Volume Imaging: An acquisition technique in which signal is collected from an entire volume rather than individual slices. Permits reconstruction of extremely thin slices, and usually enhances SNR.

VSS: See Very Selective Saturation.

Water Suppression: The suppression of the water signal in a MR spectrum, usually by a specialized excitation sequence.

Weighted-Phase Images: Images that present flow data. Directional-flow images demonstrate flow along a single axis; speed-flow images combine all flow information into a single presentation.

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Appendix C

Maintenance Service Schedules

Introduction

This appendix includes the planned maintenance (PM) service schedules, which represent the manufacturer's recommendations. These schedules are necessary for a qualified service personnel to perform the PM checks on your Magnetic Resonance (MR) imaging system.

Specifically, this appendix includes:

- 0.7T Service Schedule
- 1.5T Service Schedule for HDx and HDx prior systems
- 3.0T Service Schedule for HDx and HDx prior systems
- HDxt 1.5T and 3.0T Service Schedule for HDxt
- Discovery MR750 3.0T and Discovery MR450 1.5T and Optima MR450w 1.5T Service Schedule
- Optima MR360 and Brivo MR355 System Service Schedule

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PM Service Schedules

Specific customer requirements and/or your site environment may necessitate more or less frequent intervals for PM service. An agreement to perform PMs less frequently than these recommendations can be made with the understanding that a reduction of system performance may result.

To further enhance productivity, the PM procedures have been divided into four types:

- Type 1 – Safety and Regulatory
- Type 2 – Image Quality
- Type 3 – Other (System not available for patient scans)
- Type 4 – Other (System available for patient scans)

The PM matrices list all the PM procedures and the frequency they should be completed, according to schedules A - D (Table APX C-1) or A - F (Table APX C-2). The schedules indicate the procedures that are performed during each visit. They also show the type (1 - 4) for each procedure. The services should be completed at the indicated intervals and should be performed only by qualified service personnel.

Table APX C-1 4/year PM schedule

A	B	C	D
0 to 3 months	4 to 6 months	7 to 9 months	10 to 12 months

Table APX C-2 6/year PM schedule

A	B	C	D	E	F
0 to 2 months	3 to 4 months	5 to 6 months	7 to 8 months	9 to 10 months	11 to 12 months

There are different service schedules for each system type:

- 0.7T Service Schedule
- 1.5T Service Schedule for HDx and HDx prior systems
- 3.0T Service Schedule for HDx and HDx prior systems
- HDxt 1.5T and 3.0T Service Schedule
- Discovery MR750 3.0T and MR450 1.5T and Optima MR450w Service Schedule
- Optima MR360 and Brivo MR355 System Service Schedule

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0.7T Service Schedule

The 0.7T PM service schedule (Table APX C-3) lists all the PM procedures and the frequency they should be completed within a 3-month time frame, according to schedules A - D (Table APX C-1). This results in services completed four times per year.

CAUTION: Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

Table APX C-3 0.7T System PM Schedule

Item	Type	A	B	C	D
Magnet Room					
Check Oxygen Monitor operation/installation date	1	X			
Perform Physical Acquisition Controller (PAC) Leakage Current test	1			X	
Check magnet room radio frequency (RF) integrity with correlated noise	3	X	X	X	X
Check cardiac gating cable	1	X		X	
Check patient blower and filter	3	X	X	X	X
Check pneumatic Patient Alert System	1	X			X
RF					
Check RF cabinet fans and filters	4	X	X	X	X
Perform Power Monitor functional checks	1		X		
Check RF output power	1	X	X	X	X
Patient Handling					
Check table operation	1	X	X	X	X
Check emergency release of cradle	1		X		X
Check patient transport casters and armboard set screws	1	X	X	X	X
Check brake pedal cable and brake operation	1		X		X
Check table gas shocks	1		X		X
Check table/cradle level	3			X	
Clean lightweight cradle wheels	3			X	X

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Item (Continued)	Type	A	B	C	D
Replacement of gas shocks (every 2 years)	3		X		
Gradients					
Check Gradient Cabinet fans and filters	4	X	X	X	X
Check eddy current compensation	2	X	X	X	X
Check gradient cables connection and support	3		X		
Check gradient calibration	2	X	X	X	X
PDU					
Check PDU Module in GRFD Cabinet emergency off and stop circuits	1			X	
Inspect PDU power connections	3			X	
Operator Workspace for OpenSpeed					
Check computer fans and clean air intake grills	3	X	X	X	X
Clean workstation mouse and videocam lens	3		X		X
Set Silicon Graphics, Inc. (SGI) system clock	4	X	X	X	X
Operator Workspace for OpenSpeed Excite					
Clean disk space by entering the following in a c-shell: **storelog**	4	X	X	X	X
Adjust the time from Guided Install	4	X	X	X	X
Clean dust of global operator console	3	X	X	X	X
General System					
Check system cabinet fans and clean filters	4	X	X	X	X
Check and delete error/message log/T-test files	3	X	X	X	X
Check shim	2	X	X	X	X
Check SNR	2	X	X	X	X
Check cabinet inlet air temperature	4	X	X	X	X
Check PM supplies	4			X	
Update Configuration File in Site Log Book	4		X		X

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Item (Continued)	Type	A	B	C	D
Perform Site Restoration - Check the Daily Quality Assurance (DQA) phantom, remove GE test scans, and check cabinet doors and covers	3	X	X	X	X
Check laser alignment lights	1	X	X	X	X

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Item (Continued)	Type	A	B	C	D
3M Laser Camera					
Check laser camera fans	3	X	X	X	X
Clean and vacuum laser camera interior	3		X		
Clean laser camera suction cups	3	X	X	X	X
Clean laser camera transport plate/docking unit	3	X	X	X	X
Run laser camera cleaning film	3	X	X	X	X
Clean laser camera exterior	4		X		
Check laser camera air shock pressure (Mobile systems only)	4		X		
Replace laser camera external docking unit switches	3	X			X
Magnet and Cryogens					
Check Magnet Monitor Connectivity	4	X	X	X	X
Calculate cryogen boil-off rates	4	X	X	X	X
Monitor Magnet Thermal Performance	4	X	X	X	X
Evaluate cryogen delivery schedule (If applicable)	4		X		X
Evaluate helium transfill efficiency	4		X		X
Verify cryogen meter calibration	4	X			
Inspect cryogen vent	1		X		
Test GE Magnet Rundown Unit (MRU)	1	X	X	X	X
Inspect GE MRU	1		X		
Check drip pan for fluid build-up	3		X		X
Inspect Sumitomo system (Replace adsorber every 20,000 hours)	3			X	
Replace Desiccant Pack in water flow meter	3	X	X	X	
Inspect flow meter for Magnet Monitor	3	X	X	X	X
SCC Cabinet					
Check air inlets, water temperature, and water flow	3	X	X	X	X

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1.5T Service Schedule for HDx and HDx prior systems

The 1.5T PM service schedule (Table APX C-4) lists all the PM procedures and the frequency they should be completed within a 2-month time frame, according to schedules A - F (Table APX C-2). This results in services completed six times per year.

CAUTION: Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

NOTE: An asterisk (*) indicates the procedure should be performed during every site visit.

Table APX C-4 1.5T System PM Schedule

Item	Type	A	B	C	D	E	F
Magnet Room							
Check Oxygen Monitor operation/installation date	1	X					
Inspect front cover and cable take-up	3						X
Inspect for discoloration the inside of the bore and the front and back of the magnet	1	X					
Perform Physical Acquisition Controller (PAC) Leakage Current test	1			X			
Check magnet room radio frequency (RF) integrity with correlated noise	3	X		X		X	
Check cardiac gating cable	1	X		X	X		X
Check patient blower and filter	3	X	X	X	X	X	X
Check pneumatic Patient Alert System	1	X			X		
RF							
Check RF cabinet fans and filters	4	X	X	X	X	X	X
Check Erbtec blower output	3	X	X	X	X		
Perform Power Monitor functional checks –RF/Penetration cabinet	1					X	
Perform Power Monitor functional checks – RF/Penetration II cabinet	1					X	
Perform Power Monitor functional checks – RF/Power Distribution Unit (PDU) cabinet	1					X	
Check RF output power	1	X	X	X	X	X	X

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Item (Continued)	Type	A	B	C	D	E	F
Patient Handling							
Check emergency release of cradle and patient transport	1	X	X	X	X	X	X
Check patient transport docking and alignment	3		X		X		X
Check patient transport casters, armboard set screws, and bumper strips	1	X	X	X	X	X	X
Check patient transport caster locks	1	X			X		
Check cradle longitudinal drive clutch	1		X		X		X
Check patient transport hydraulic filter	3			X			X
Clean lightweight cradle wheels	3	X			X		
Check foot pedal spring installation date	3	X					
Gradients							
Check Gradient Cabinet fans and filters	4	X	X	X	X	X	X
Check eddy current compensation	2	X	X	X	X	X	X
Check gradient cables connection and support	3		X				
Check gradient calibration	2	X	X	X	X	X	X
Check fluid level and valve of heat exchanger	3	X		X		X	
Check water chiller for gradient coil cooling	3	X		X		X	
Check pump motor lubrication	4	X					
PDU							
Check Standard PDU fans and filters	4		X		X		X
Check Standard PDU emergency off and stop circuits/indicator lights	1			X			
Inspect Standard PDU and power connections	3			X			
Check PowerTech and/or Transtector	3	X			X		
Check Compact PDU fans and filters	4		X		X		X
Check Compact PDU emergency off and stop circuits/indicator lights	1			X			
Inspect Compact PDU and power connections	3			X			
Check PDU Module in RF and Advance Control Gradient Driver (ACGD)/PDU Cabinet emergency off and stop circuits	1			X			
Inspect ACGD Cabinet PDU power connections	3			X			
Computer (Horizon 5.X)							

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Item (Continued)	Type	A	B	C	D	E	F
Check Operator Console (OC) and Independent Console (IC) computer fans and clean air intake grills	3	X	X	X	X	X	X
Clean OC/IC DAT drive	4	X	X	X	X	X	X
Check OC/IC console power supply fan	3	X	X	X	X	X	X
Clean and vacuum OC/IC console interior	3	X					
Operator Workspace (8.X and 9.X)							
Check computer fans and clean air intake grills	3	X	X	X	X	X	X
Clean workstation mouse and videocam lens	3	X			X		
Set Silicon Graphics, Inc. (SGI) system clock	4	X	X	X	X	X	X
Operator Workspace (Excite, HD, HDx)							
Clean disk space by entering the following in a c-shell: **storelog**	4	X	X	X	X	X	X
Adjust the time from Guided Install	4	X		X		X	
Clean dust of global operator console	3	X		X		X	
General System							
Check system cabinet fans and clean filters	4	X	X	X	X	X	X
Check and delete error/message log/T-test files/save info	3	X	X	X	X	X	X
Check shim	2	X	X	X	X	X	X
Check SNR	2	X	X	X	X	X	X
Check cabinet inlet air temperature	4		X		X		X
Check PM supplies	4		X				
Update Configuration File in Site Log Book	4		X			X	
Verify PM completion on van equipment (Mobile systems only)	4	X			X		
Perform Site Restoration - Check the Daily Quality Assurance (DQA) phantom, remove GE test scans, and check cabinet doors and covers	3	X	X	X	X	X	X
Check modem (United States sites only)	3	X	X	X	X	X	X
Check laser alignment lights	1	X	X	X	X	X	X
Review System Health Report	4	X	X	X	X	X	X

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Item (Continued)	Type	A	B	C	D	E	F
3M Laser Camera							
Check laser camera fans	3	X	X	X	X	X	X
Clean and vacuum laser camera interior	3		X			X	
Clean laser camera suction cups	3	X	X	X	X	X	X
Clean laser camera transport plate/docking unit	3	X	X	X	X	X	X
Run laser camera cleaning film	3	X	X	X	X	X	X
Clean laser camera exterior	4		X			X	
Check laser camera air shock pressure (Mobile systems only)	4		X			X	
Replace laser camera external docking unit switches	3	X			X		
Twin Accessory Cabinet (TAC)							
Filter replacement	4	X	X	X	X	X	X
Tip seal replacement	3						X
Clean inlet screen	3						X
Solenoid valve replacement	3						X

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Item (Continued)	Type	A	B	C	D	E	F
Magnet and Cryogens							
Check cryogen levels (Phone site for information)	4	*	*	*	*	*	*
Calculate cryogen boil-off rates/record compressor run times (Phone Site for information)	4	*	*	*	*	*	*
Evaluate cryogen delivery schedule (If applicable)	4	X	X	X	X	X	X
Evaluate helium transfill efficiency	4	X	X	X	X	X	X
Verify cryogen meter calibration - GE Magnets	4	X					
Verify cryogen meter calibration - Oxford Magnets	4	X					
Inspect cryogen vent	1		X				
Test Magnet Emergency Rundown Unit (ERU)	1	X	X	X	X	X	X
Inspect Magnet ERU	1		X				
Test GE Magnet Rundown Unit (MRU)	1	X	X	X	X	X	X
Inspect GE MRU	1		X				
Check and empty collection bottles (GE S-One and Oxford magnets only)	3	X	X	X	X	X	X
Inspect Cryogenics Technics, Inc. (CTI) system (Oxford magnets only)	3		X			X	
Record cryostat pressure and flow rates (GE magnets only)	3	X	X	X	X	X	X
Inspect Leybold system (Replace adsorber every 24,000 hours) (GE magnets only)	3	X		X		X	
Inspect Balzers system (Replace adsorber every 26,000 hours) (GE magnets only)	3	X		X		X	
Inspect Sumitomo system (Replace adsorber every 20,000 hours)	3	X		X		X	
Perform Oxford mobile specific inspections	3			X			X
Inspect Oxford water cooled power supply valve	3	X					
Check/replace Equipment Diagnostic Monitor (EDM) battery	3	X					
Change Desiccant Pack water flow meter	3	X		X		X	

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3.0T Service Schedule for HDx and HDx prior systems

The 3.0T PM service schedule (Table APX C-5) lists all the PM procedures and the frequency they should be completed within a 2-month time frame, according to schedules A - F (Table APX C-2). This results in services completed six times per year.

CAUTION: Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

NOTE: An asterisk (*) indicates the procedure should be performed during every site visit.

Table APX C-5 3.0T System PM Schedule

Item	Type	A	B	C	D	E	F
Magnet Room							
Inspect front cover and cable take-up	3						X
Inspect for discoloration the inside of the bore and the front and back of the magnet	1	X					
Perform Physical Acquisition Controller (PAC) Leakage Current test	1			X			
Check magnet room radio frequency (RF) integrity with correlated noise	3	X		X		X	
Check cardiac gating cable	1	X		X	X		X
Check patient blower and filter	3	X	X	X	X	X	X
Check pneumatic Patient Alert System	1	X			X		
RF							
Check RF Cabinet fans and filters	4	X	X	X	X	X	X
Check RF output power	1	X	X	X	X	X	X
Patient Handling							
Check emergency release of cradle and patient transport	1	X	X	X	X	X	X
Check patient transport docking and alignment	3		X		X		X
Check patient transport casters and armboard set screws and bumper strips	1	X	X	X	X	X	X
Check patient transport caster locks	1	X			X		

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Item (Continued)	Type	A	B	C	D	E	F
Check cradle longitudinal drive clutch	1		X		X		X
Check patient transport hydraulic filter	3			X			X
Clean lightweight cradle wheels	3	X			X		
Check foot pedal spring installation date	3	X					
Gradients							
Check Gradient Cabinet fans and filters	4	X	X	X	X	X	X
Check eddy current compensation	2	X	X	X	X	X	X
Check gradient cables connection and support	3		X				
Check gradient calibration	2	X	X	X	X	X	X
Check fluid level and valve of heat exchanger	3	X		X		X	
Check water chiller for gradient coil cooling	3	X		X		X	
Check pump motor lubrication	4	X					
PDU							
Check PDU module in ACGD/PDU Cabinet emergency off and stop circuits	1			X			
Inspect PDU power connections	3			X			
Operator Workspace (with Octane computer)							
Check computer fans and clean air intake grills	3	X	X	X	X	X	X
Clean workstation mouse and videocam lens	3	X			X		
Set Silicon Graphics, Inc. (SGI) system clock	4	X	X	X	X	X	X
Operator Workspace (HD and HDx)							
Clean disk space by entering the following in a c-shell: **storelog**	4	X	X	X	X	X	X
Adjust the time from Guided Install	4		X		X		X
Clean dust of global operator console	3		X		X		X
General System							
Check system cabinet fans and clean filters	4	X	X	X	X	X	X

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Item (Continued)	Type	A	B	C	D	E	F
Check and delete error/message log/T-test files	3	X	X	X	X	X	X
Check shim	2	X	X	X	X	X	X
Check SNR	2	X	X	X	X	X	X
Check cabinet inlet air temperature	4		X		X		X
Check PM supplies	4		X				
Update configuration file in Site Log Book	4		X			X	
Perform site restoration - Check Daily Quality Assurance (DQA) phantom, remove GE test scans, and check cabinet doors and covers	3	X	X	X	X	X	X
Check modem	3	X	X	X	X	X	X
Check laser alignment lights Not applicable for 3.0T VH/i	1	X	X	X	X	X	X
Review System Health Report	4	X	X	X	X	X	X
Accessory Cabinet (ACC)							
Clean filter	4	X	X	X	X	X	X
Perform Power Monitor functional check	1						X
Magnet and Cryogens							
Check cryogen levels	4	*	*	*	*	*	*
Calculate cryogen boil-off rates/record compressor run times	4	*	*	*	*	*	*
Evaluate cryogen delivery schedule	4	X	X	X	X	X	X
Evaluate helium transfill efficiency	4	X	X	X	X	X	X
Verify cryogen meter calibration (GE magnets only)	4	X					
Inspect cryogen vent	1		X				
Test GE Magnet Rundown Unit (MRU)	1	X	X	X	X	X	X
Inspect GE (MRU)	1		X				
Record cryostat pressure and flow rates (GE magnets only)	3	X	X	X	X	X	X
Inspect Sumitomo Heavy Industry (SHI) system	3	X		X		X	

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HDxt 1.5T and 3.0T Service Schedule

The HDxt PM service schedule (Table) lists all the PM procedures and the frequency they should be completed. There are four PMs per year.

CAUTION: **Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.**

Table APX C-6 HDxt 1.5T and 3.0T PM Schedule

Subsystem	Type	A	B	C	D
Image Quality					
Alignment Light Check	IQ	X	X	X	X
Patch Update		As applicable	As applicable	As applicable	As applicable
DQA-II Calibration	IQ	X			
LVShim	IQ	X	X	X	X
Eddy Current Check	IQ	X	X	X	X
Coherent Noise	IQ	X	X	X	X
Signal to Noise Check	IQ	X	X	X	X
Spike Noise Check	IQ	X	X	X	X
16-channel switch diagnostic	IQ	X	X	X	X
Perform Save Info	N/A	X	X	X	X
PM Check	IQ	X	X	X	X
Magnet Room					
Oxygen Monitor Operation	Safety	X			
Cardiac Gating Cable	Safety	X	X	X	X
Patient Blower and Filter	Check	X	X	X	X
Pneumatic Patient Alarm System	Safety		X		X

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Subsystem	Type	A	B	C	D
Inspect Front Cover Cable Take-up	Check				X
PAC Leakage Current Test	Safety			X	
RF/System Cabinet					
Fan and Filter for SRFD	Clean	X	X	X	X
Fan and Filter for MKS	Clean	X	X	X	X
Check RF Output Power	Check	X	X	X	X
Check/Clean Filters in RFS Cabinet	Clean		X		
Power Monitor Check	Check				X
Patient Handling					
Patient Table Checks	Safety	X	X	X	X
Lite Patient Transport	Safety	X	X	X	X
Signa OR - compatible table	Safety	X	X	X	X
Signa Oncology Patient Transport Table	Safety	X	X	X	X
Gradient					
Fans and Filters	Clean	X	X	X	X
Check Fluid Levels and valve of Heat Exchanger	Clean	X		X	
CFT-150 System maintenance	Check		X		X
Lytron System maintenance	Check		X		X
Mobile Hx System maintenance	Check		X		X

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Subsystem	Type	A	B	C	D
Gradient cable / PDU Connections & Support	Check			X	
PDU					
Emergency Off & Stop Circuits/Indicator Lights	Safety	X		X	
Clean PDU Air intake	Clean			X	
TAC Cabinet - TRM Systems only					
Filter Replacement	Check	X	X	X	X
Clean Inlet Screen	Clean			X	
Solenoid Valve Replacement	Replace			X	
Vacuum Check	Check				X
Computer					
Storelog	Check	X	X	X	X
Clean dust, set time	Clean		X		X
BrainWave cabinet					
Clean the Filters on Brainwave Lite PC	Clean		X		
Integrated Patient Comfort Module (ICPM) Chilled Air Blower (CAB) (3.0T Option)					
Check Cabinet Filter and Condenser	Clean		X		X
Magnet					
Verify Magnet Monitor Calibration	Check		X		
Test GE Magnet Rundown Unit (MRU)	Safety	X	X	X	X
Inspect GE MRU	Safety		X		
Inspect cryocooler systems	Check	X		X	

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Subsystem	Type	A	B	C	D
Inspect Cryogen Vent	Check	X			

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Discovery MR750 3.0T and MR450 1.5T and Optima MR450w 1.5T Service Schedule

The Discovery MR750 3.0T and MR450 1.5T and Optima 450w PM service schedule (Table APX C-7) lists all the PM procedures and the frequency they should be completed according to schedules A -D (Table APX C-7).

CAUTION: **Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.**

Table APX C-7 Discovery MR750 3.0T and MR 450 1.5T and Optima 450w PM Schedule

Subsystem	PM check performed	Test Purpose	A	B	C	D
Image Quality						
Patient Handling	Laser Light Alignment	Check	X	X	X	X
Image Quality	DQAII Calibration	Adjust		X		X
IBIS	Patch Update	Adjust	X	X	X	X
Image Quality	SPT (PM Mode)	Check	X	X	X	X
Image Quality	LVShim (PM Mode)	Check	X	X	X	X
Image Quality	EPI White Pixel (PM Mode)	Check	X	X	X	X
IBIS	PM Check	Check	X	X	X	X
Magnet Room						
Magnet Room	Oxygen Monitor Operation (Option)	Safety			X	
Magnet Room	PAC Leakage Current Test	Safety			X	
Magnet Room	Cardiac Gating Cable	Safety		X		X
Magnet Room	Patient Blower and Filter	Check	X	X	X	X
Magnet Room	Pneumatic Patient Alert System	Safety		X		X
PGR Cabinet						
PGR Cabinet	Inlet Filters	Clean	X	X	X	X
RF Power	UPM Functional Check	Safety				X
Patient Handling						
Patient Handling	General Checks	Safety		X		X

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Subsystem	PM check performed	Test Purpose	A	B	C	D
Patient Handling	Hydraulic Fluid Check	Check				X
Heat Exchanger						
Heat Exchanger	Check fluid levels and top off	Check	X	X	X	X
Heat Exchanger	HEC Coolant Deionization	Replace			X	
Heat Exchanger	HEC Filter Check	Check		X		X
PDU						
Teal PDU	Emergency Off & Stop Circuits/Indicator	Safety			X	
Teal PDU	LightsLeak Sensor Functionality Check	Safety			X	
Computer						
GOC	Set time	Check	X		X	
GOC	Clean dust	Clean	X		X	
Computer System	T-File Cleanup	Clean		X		
Computer System	Storelog/File Clean up	Clean		X		
Computer System	Review Overnight Diag Results	Check	X	X	X	X
Computer System	One-Wire Diagnostic	Check		X		
Options Cabinet						
BrainWave	Clean filters on BrainWave Lite PC (Option)	Clean		x		x
Magnet						
Magnet & Cryogens	Inspect Cryogen Vent	Check			X	
Magnet & Cryogens	Test GE Magnet Rundown Unit (MRU)	Safety	X	X	X	X
Magnet & Cryogens	Inspect SH1 System (Replace Adsorber every 20K hours)	Replace		X		X
Magnet covers	Inspect for discoloration the inside of the bore and the front and back of the magnet	Check	X			
Secondary PEN Wall						
Penetration Panel	Body Coil Air Filter	Clean		X		X

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Optima MR360 and Brivo MR355 System Service Schedule

Optima MR360 and Brivo MR355 System Service Schedule (Table APX C-8) lists all the PM procedures and the frequency they should be completed according to schedule A - D (Table APX C-1).

CAUTION: Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

Table APX C-8 Optima MR360 and Brivo MR355 System Service Schedule

Subsystem	PM check performed	Test Purpose	A	B	C	D
Image Quality						
Alignment Light Check	IQ	2	X	X	X	X
Patch update (Service contract customers only)	Software	3	X	X	X	X
DQA Tool II	IQ	2	X	X	X	X
LVShim	IQ	2	X	X	X	X
Grafidy3	IQ	2	X	X	X	X
Coherent Noise	IQ	2	X	X	X	X
Signal to Noise Check	IQ	2	X	X	X	X
Spike Noise Check	IQ	2	X	X	X	X
PM Check (Service Contract Customers Only)	IQ	2	X	X	X	X
Magnet Room						
Oxygen Monitor Operation	Safety	1	X			
Patient Blower & Filter	Check	3	X	X	X	X
Pneumatic Patient Alert System	Safety	1		X		X
System Cabinet						
Power Monitor Check	Check	3				X
Cabinet FAN & Water Line	Check	3	X	X	X	X
Driver Module Lite Functional Check	Check	3		X	X	X
Cabinet Monitor Functional Check	Check	3		X		X

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Subsystem	PM check performed	Test Purpose	A	B	C	D
Clean Inlet Filter	Check	4	X	X	X	X
Gradient Cable Connection Check	Check	3			X	
DC Power Check	Check	3		X		X
PHPS Lite Functional Check	Check	3		X		X
MCS/LCS Coolant Level Check	Check	4	X	X	X	X
Patient Handling						
Patient Table Checks	Safety	1	X	X	X	X
System Cooling Unit and BRM Chiller						
Check Fluid levels and Valve of Heat Exchanger	Clean	3		X		X
Replace Lytron Pump for 4KW	Clean	3				X
PDU						
Emergency Off & Stop Circuits/Indicator Lights	Safety	1	X	X	X	X
Computer						
Storelog	Check	4	X	X	X	X
Clean Dust. Set time	Check	3		X		X
Magnet						
Test GE Magnet Rundown Unit (MRU)	Safety	1	X	X	X	X
PAC Leakage Current Test	Safety	1			X	
Inspect Cryocooler Systems	Check	3	X		X	
Inspect Cryogen Vent	Check	3	X			
Verify Cryogen Meter Calibration	Check	4		X		

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Appendix D

1.5T Spatial Magnetic Field Data

Magnet Information

This information is provided to comply with IEC 60601-2-33 clause 6.8.3 bb. The peak magnetic field, peak gradient of the magnetic field, and the peak force product (magnetic field times gradient) and their spatial locations are provided. Note that these values typically occur on the magnet covers.

To find the magnet type used with your system, contact your field engineer

Spatial Magnetic Field

Maximum forces and torques on ferromagnetic objects depends on the force product $\left(|B|\left|\frac{\partial B}{\partial z}\right| = |B||grad(B)|\right)$.

The maximum static magnetic fringe field gradient $\left(\left|\frac{\partial B}{\partial z}\right| = |grad(B)|\right)$ has been listed by MR compatibility investigators in the past.

For each maximum $|B|, \left|\frac{\partial B}{\partial z}\right|, |B|\left|\frac{\partial B}{\partial z}\right|$ the maximum values, the spatial locations (cylindrical coordinates (Z,R)), and the values of the other (typically non-maximum) parameters are given below for GE 1.5T magnets. Note that 1 T/m = 100 G/cm. In addition, 1 T^2/m = 1000000 G^2/cm.

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Figure APX D-1 Magnet location of fringe-field maximum

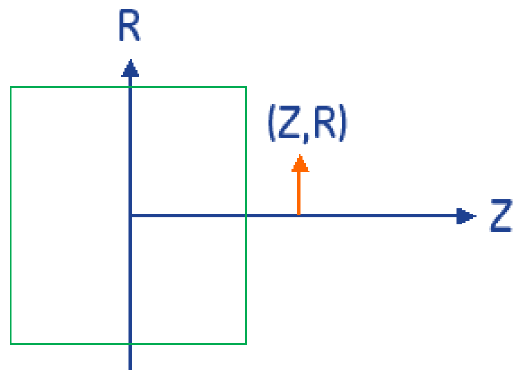

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Table APX D-1 Maximum $|B|,|gradB|,and|B|*|gradB|$ for 1.5T magnet

1.5T Systems	Field	R(m)	Z(m)	B Field (T)	Gradient (T/m)	Product (T^2/m)
Optima MR450W 1.5T (DVw 1.5T)	peak B	0.426	0.688	2.4	14.4	35.0
	Peak Gradient	0.561	0.746	1.9	18.8	36.3
	Peak product (BxGradient)	0.450	0.712	2.4	17.1	40.4
Discovery MR450 1.5 T (1.5T LCC HDv)	peak B	0.306	0.598	1.9	2.8	5.2
	Peak Gradient	0.371	0.837	1.2	5.8	7.1
	Peak product (BxGradient)	0.322	0.755	1.5	5.2	8.1
Signa 1.5 T HDx Series (1.5T LCC HDx-BRM, Hde)	peak B	0.319	0.627	1.9	3.4	6.5
	Peak Gradient	0.391	0.807	1.4	7.4	10.5
	Peak product (BxGradient)	0.369	0.766	1.6	6.9	11.3
Signa 1.5 T TwinSpeed (1.5T LCC HDx-TRM)	peak B	0.306	0.608	1.9	2.9	5.4
	Peak Gradient	0.384	0.828	1.3	6.5	8.3
	Peak product (BxGradient)	0.344	0.757	1.6	5.9	9.5
1.5T LCC HD*	peak B	0.327	0.615	1.9	3.6	6.9
	Peak Gradient	0.399	0.812	1.4	7.3	10.3
	Peak product of BxGradient	0.378	0.784	1.5	7.0	10.8
Cx1.5T w. CRM enclosure*	peak B	0.317	0.670	1.8	4.2	7.6
	Peak Gradient	0.413	0.827	1.3	7.6	10.0
	Peak product (BxGradient)	0.378	0.771	1.6	7.3	11.9
SV 1.5T*	peak B	0.276	0.596	1.7	1.3	2.3
	Peak Gradient	0.276	0.852	1.3	3.5	4.4
	Peak product (BxGradient)	0.276	0.788	1.5	3.2	4.7
SIV 1.5T*	peak B	0.276	0.623	1.7	1.4	2.4
	Peak Gradient	0.276	0.889	1.2	3.5	4.2
	Peak product (BxGradient)	0.276	0.807	1.5	3.3	4.8
SIII 1.5T*	peak B	0.276	0.642	1.7	0.9	1.6
	Peak Gradient	0.276	0.898	1.3	2.7	3.6
	Peak product (BxGradient)	0.276	0.834	1.5	2.5	3.7

*For older magnet designs (<1996) the data is provided independent of the gradient performance.

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1.5T Magnetic Field Plots

Check with your service engineer for specific field plot information. Magnetic field plots (assumes no proximate ferromagnetic materials) may be found at:

http://www.gehealthcare.com/company/docs/siteplanning.html#mr

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Appendix E

3.0T Spatial Magnetic Field Data

Magnet Information

This information is provided to comply with IEC 60601-2-33 clause 6.8.3 bb. The peak magnetic field, peak gradient of the magnetic field, and the peak force product (magnetic field times gradient) and their spatial locations are provided. Note that these values typically occur on the magnet covers.

To find the magnet type used with your system, contact your field engineer.

Spatial Magnetic Field

Maximum forces and torques on ferromagnetic objects depends on the force product $\left(|B|\left|\frac{\partial B}{\partial z}\right| = |B||grad(B)|\right)$.

The maximum static magnetic fringe field gradient $\left(\left|\frac{\partial B}{\partial z}\right| = |grad(B)|\right)$ has been listed by MR compatibility investigators in the past.

For each maximum $|B|, \left|\frac{\partial B}{\partial z}\right|, |B|\left|\frac{\partial B}{\partial z}\right|$ the maximum values, the spatial locations (cylindrical coordinates (Z,R)), and the values of the other (typically non-maximum) parameters are given below for GE magnets. Note that 1 T/m = 100 G/cm. In addition, 1 T^2/m = 1000000 G^2/cm.

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Figure APX E-1 Magnet location of fringe-field maximum

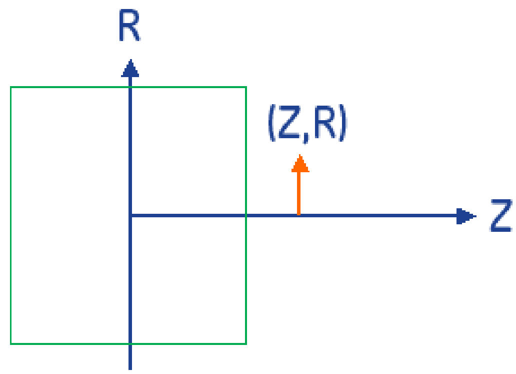

Table APX E-1 Maximum $|B|, |gradB|, and |B| * |gradB|$ for 3.0T magnets

3.0 T Systems	ID	R(m)	Z(m)	B Field (T)	Gradient (T/m)	Product (T^2/m)
Discovery MR750 Series	peak B	0.306	0.624	3.6	3.6	12.8
	Peak Gradient	0.548	0.922	1.8	13.9	25.0
	Peak product (BxGradient)	0.389	0.859	2.6	12.1	31.8
3T HDx Signa 3.0T Series	peak B	0.306	0.608	3.6	3.6	12.8
	Peak Gradient	0.495	0.910	2.1	14.9	30.7
	Peak product (BxGradient)	0.351	0.770	3.3	11.9	39.3
3T-Maestro	peak B	0.321	0.626	3.7	4.8	17.6
	Peak Gradient	0.494	0.911	2.1	12.6	26.1
	Peak product of BxGradient	0.382	0.869	2.5	10.8	27.0
3T94 Series	peak B	0.306	0.713	3.4	1.9	6.5
	Peak Gradient	0.416	1.071	2.4	9.1	21.9
	Peak product (BxGradient)	0.412	1.026	2.7	8.3	22.7

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Appendix F

IEC EMC Compliance

Per IEC 60601-1-2 Edition 2.1 Medical Electrical Equipment needs special precautions regarding Electro Magnetic Compatibility (EMC) and needs to be installed and put into service according to the EMC information provided in the following tables. Table APX F-1, Table APX F-2, Table APX F-3 and Table APX F-4 provide details about the level of compliance and provide information about potential interactions between devices.

Portable and mobile RF communications equipment can affect Medical Electrical Equipment.

WARNING: **The MR System may be interfered with by other equipment, even if that other equipment complies with CISPR EMISSION requirements.**

WARNING: **The MR System should not be used adjacent to or stacked with other equipment; if adjacent or stacked use is necessary, the MR System should be observed in order to verify normal operation in the configuration in which it will be used.**

WARNING: **The MR System should be used only in a shielded location named as the Magnet Room. Magnetic and RF Shielded Room requirements defined in Pre-Installation Manual.**

WARNING: **The use of accessories, transducers and cables other than those specified, with the exception of transducers and cables sold by GE Healthcare or replacement parts for internal components, may result in increased emissions or decreased immunity of the MR system.**

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Adhering to the recommendations provided herein for the interaction of the MR System with other electrical devices within the electromagnetic environment may not eliminate all the disturbances, however, the system will maintain its essential performance by continuing to acquire, display, and store quality diagnostic quality images safely.

Table APX F-1 Guidance And Manufacturer's Declaration – Electromagnetic Emissions

<table>
<tr><th colspan="3">The MR System is intended for use in the electromagnetic environment specified below. The customer or the user of the MR System should assure that it is used in such an environment.</th></tr>
<tr><th>Emissions Test</th><th>Compliance Level</th><th>Electromagnetic Environment - Guidance</th></tr>
<tr><td>RF emissions
CISPR 11</td><td>Group 2</td><td>The MR system must emit electromagnetic energy in order to perform its intended function. Nearby electronic equipment may be affected.</td></tr>
<tr><td>RF emissions
CISPR 11</td><td>Class A</td><td rowspan="3">The MR system is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.</td></tr>
<tr><td>Harmonic emissions
IEC 61000-3-2</td><td>Not Applicable</td></tr>
<tr><td>Voltage fluctuations/flicker emissions
IEC 61000-3-3</td><td>Not Applicable</td></tr>
</table>

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Table APX F-2 Guidance And Manufacturer's Declaration – Electromagnetic Immunity

The MR System is intended for use in the electromagnetic environment specified below. The customer or the user of the MR System should assure that it is used in such an environment.			
Immunity test	**IEC 60601 test level**	**Compliance Level**	**Electromagnetic environment – guidance**
Electrostatic discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	±6 kV contact ±8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	±2 kV for power supply lines ±1 kV for input/output lines	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV line(s) to line(s) ±2 kV line(s) to earth	±1 kV differential mode ±2 kV common mode	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5% UT (>95% dip in UT) for 0,5 cycle 40% UT (60% dip in UT) for 5 cycles 70% UT (30% dip in UT) for 25 cycles	Not applicable	Mains power quality should be that of a typical commercial or hospital environment. If the user of the MR System requires continued operation during power mains interruptions, it is recommended that the MR System be powered from an uninterruptible power supply or a battery.
	<5% UT (>95% dip in UT) for 5 sec.	<5% UT (>95% dip in UT) for 5 sec.	
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
NOTE UT is the a.c. mains voltage prior to application of the test level.			

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Table APX F-3 Guidance And Manufacturer's Declaration – Electromagnetic Immunity (cont'd)

The MR System is intended for use in the electromagnetic environment specified below. The customer or the user of the MR System should assure that it is used in such an environment.

Immunity test	IEC 60601 test level	Compliance Level	Electromagnetic environment – guidance
			Portable and mobile RF communications equipment should be used no closer to any part of the MR System, including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter. Recommended separation distance
Conducted RF IEC 61000-4-6	3 Vrms 150 kHz to 80 MHz	3 Vrms	$d = 1,2\sqrt{P}$
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2,5 GHz	3 V/m	$d = 1,2\sqrt{P}$ 80 MHz to 800 MHz $d = 2,3\sqrt{P}$ 800 MHz to 2.5GHz where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey[a], should be less than the compliance level in each frequency range[b] Interference may occur in the vicinity of equipment marked with the following symbol: 

NOTE 1 At 80 MHz and 800 MHz, the higher frequency range applies.
NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.
a Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the MR System issued exceeds the applicable RF compliance level above, the MR System should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the MR System.
b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

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Table APX F-4 Recommended Separation Distances

The MR System is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the MR System can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the MR System as recommended below, according to the maximum output power of the communications equipment.			
Rated maximum output power of transmitter W	**Separation distance according to frequency of transmitter m**		
	150 kHz to 80 MHz $d = 1,2\sqrt{P}$	**80 MHz to 800 MHz** $d = 1,2\sqrt{P}$	**800 MHz to 2.5GHz** $d = 2,3\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.37	0.37	0.74
1	1.17	1.17	2.33
10	3.69	3.69	7.38
100	11.67	11.67	23.33
For transmitters rated at a maximum output power not listed above, the recommended separation distance d in metres (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer. NOTE 1 At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies. NOTE 2 These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

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Temperature and Humidity Specifications

System Suite

Use the specifications listed in Table APX F-5 and Table APX F-6 for designing your HVAC (heating, ventilation, and air conditioning) system. Proper insulation and moisture barrier should be installed within the environmental controlled space (e.g. area above drop ceiling) for humidity, condensation, and temperature control.

NOTE: To help prevent a patient from feeling uncomfortably warm during a scan, make sure the magnet room temperature does not exceed 69.8.F (21.C) maximum. For Discovery MR750 3.0T systems, if the scan room temperature exceeds 75.2F (24C), then the SAR is automatically derated, which means that the current scan parameters may trip the SAR monitor.

Table APX F-5 Temperature And Humidity Specifications

Area	Temperature		Humidity		Max. Room Gradient °F (°C)
	Range °F (°C)	Change °F/Hr (°C/Hr)	Range%	Change%/Hr	
Equipment Room at Inlet to Equipment	59-89.6* (15-32)*	5 (3)	30-75*	5	5 (3)**
Magnet Room	59-69.8 (15-21)	5 (3)	30-60*	5	5 (3)
Operator's Control Room	59-89.6* (15-32)*	5 (3)	30-75*	5	5 (3)
Note * Non-condensing humidity with 50% nominal at 65.F (18.3.C). ** Room temperature gradient specification applies from floor to height of top discharge of equipment cabinets.					

Table APX F-6 Discovery MR750 3.0T Temperature And Humidity Specifications

Area	Temperature		Humidity	
	Range °F (°C)	Change °F/Hr[1] (°C/Hr)	Range%	Change%/Hr[2]
Equipment Room at Inlet to Equipment	59-89.6 (15-32)	5 (3)	30-70	5
Magnet Room	59-69.8 (15-21)	5 (3)	30-60	5
Operator's Control Room	59-89.6 (15-32)	5 (3)	30-70	5
Note [1] Operating temperature gradient limits shall be between -5F (-3C) degrees C/hour and 5F (3C) degrees C/hour when averaged over 1 hour. [2] Operating humidity gradient limits shall be between -5% RH/hour and 5% RH/hour, when averaged over 1 hour.				

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MRCC Operating Environment

The MR Common Chiller (MRCC) operating environment specifications do not apply to Discovery MR750 3.0T systems.

The MRCC is designed to be located external to the building and may be used or be transported in environments meeting the following specifications.

- Ambient Temperature: -22 °F (-30°C) to 110°F (43°C)
- Humidity: 5-100%

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Appendix G

China RoHS Labeling Directive

Signa 1.5T & 3.0T Systems

The following product pollution control information is provided according to SJ/T11364-2006 *Marking for Control of Pollution caused by Electronic Information Products*.

This symbol indicates the product contains hazardous materials in excess of the limits established by the Chinese standard SJ/T11363-2006 *Requirements for Concentration Limits for Certain Hazardous Substances in Electronic Information Products*. The number in the symbol is the Environment-friendly Use Period (EFUP), which indicates the period during which the toxic or hazardous substances or elements contained in electronic information products will not leak or mutate under normal operating conditions so that the use of such electronic information products will not result in any severe environmental pollution, any bodily injury or damage to any assets. The unit of the period is "Year".

In order to maintain the declared EFUP, the product shall be operated normally according to the instructions and environmental conditions as defined in the product manual, and periodic maintenance schedules specified in Product Maintenance Procedures shall be followed strictly.

Consumables or certain parts may have their own label with an EFUP value less than the product. Periodic replacement of those consumables or parts to maintain the declared EFUP shall be done in accordance with the Product Maintenance Procedures.

This product must not be disposed of as unsorted municipal waste, and must be collected separately and handled properly after decommissioning.

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Signa HD & HDx 1.5T & 3.0T Systems, Signa HDi 1.5T System, Discovery MR750 3.0T, Discovery MR450 1.5T and Optima MR450w
Table of hazardous substances' name and concentration.

Component Name	Hazardous substances' name					
	(Pb)	(Hg)	(Cd)	(Cr(VI))	(PBB)	(PBDE)
Magnet	X	O	O	X	O	O
Patient Table	X	O	O	X	O	O
Systems Cabinet	X	O	X	X	O	O
Coils	X	O	O	X	O	O
Global Operating Console (GOC)	X	O	X	X	O	O
Chiller	O	O	O	O	O	O
Accessories (Uninterruptable Power Supply)	X	O	X	X	X	X
LCD Monitor	O	X	O	O	O	O

O: Indicates that this toxic or hazardous substance contained in all of the homogeneous materials for this part is below the limit requirement in SJ/T11363-2006.
X: Indicates that this toxic or hazardous substance contained in at least one of the homogeneous materials used for this part is above the limit requirement in SJ/T11363-2006
Data listed in the table represents best information available at the time of publication.
Applications of hazardous substances in this medical device are required to achieve its intended clinical uses, and/or to provide better protection to human beings and/or to environment, due to lack of reasonably (economically or technically) available substitutes.

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Signa HDe 1.5T, Optima MR360 and Brivo MR355 Systems
Table of hazardous substances' name and concentration.

Component Name	Hazardous substances' name					
	(Pb)	(Hg)	(Cd)	(Cr(VI))	(PBB)	(PBDE)
Magnet	X	O	O	X	O	O
Patient Table	X	O	O	X	O	O
Systems Cabinet	X	O	X	X	X	X
Coils	X	O	O	X	O	O
Global Operating Console (GOC)	X	O	X	X	O	O
Chiller	O	O	O	O	O	O
LCD Monitor	O	X	O	O	O	O

O: Indicates that this toxic or hazardous substance contained in all of the homogeneous materials for this part is below the limit requirement in SJ/T11363-2006.
X: Indicates that this toxic or hazardous substance contained in at least one of the homogeneous materials used for this part is above the limit requirement in SJ/T11363-2006
Data listed in the table represents best information available at the time of publication.
Applications of hazardous substances in this medical device are required to achieve its intended clinical uses, and/or to provide better protection to human beings and/or to environment, due to lack of reasonably (economically or technically) available substitutes.

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